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MSc in Quantitative Finance

Rethinking Token Valuation for Digital Platforms

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Abstract

Tokenized economies have attracted growing attention from both researchers and practitioners, yet many valuation frameworks continue to rely on assumptions that are difficult to reconcile with real-world digital platforms. This thesis investigates how token valuation changes when these assumptions are progressively relaxed. Building on the Cong–Li–Wang framework, the analysis examines the effects of finite access duration and external asset competition on equilibrium token prices, adoption incentives, and dominance thresholds. It then considers a distinct class of reserve-backed token ecosystems, motivated by a real-world case involving RIV Technologies, where the central problem becomes the optimal management of token issuance rather than valuation alone. The results show that token value depends not only on platform fundamentals, but also on the institutional design of the ecosystem itself. Horizon structure, competitive conditions, and issuance policy emerge as key determinants of sustainable value creation, highlighting the close relationship between token valuation, protocol design, and digital asset economics.

Keywords: Token Valuation; Digital Assets; Platform Economics; Continuous-Time Finance; Optimal Issuance; Reserve-Backed Tokens; Fintech.

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Introduction

Financial theory has traditionally focused on assets whose value derives from cash flows, consumption opportunities, or contractual claims. Tokenized digital economies challenge this separation. The rapid growth of digital platforms has created new forms of economic organization in which value is generated, exchanged, and accumulated through mechanisms that often fall outside the traditional boundaries of financial theory.

Among these innovations, tokenized ecosystems occupy a particularly interesting position. They combine elements of asset pricing, platform economics, incentive design, and corporate finance within a single economic architecture. Despite their growing relevance, a fundamental question remains largely open: how should the value of a token be determined when its economic role extends beyond that of a conventional financial asset?

Addressing this question requires a framework that jointly accounts for participation dynamics, access duration, competitive pressure, and supply policy — features that standard asset pricing models do not address simultaneously. Tokenized systems are characterized by properties that rarely coexist in traditional markets, including endogenous participation incentives, network effects, finite access rights, protocol-controlled supply, and reserve-backed structures. Understanding how these mechanisms interact is essential not only for theoretical valuation, but also for the practical design of sustainable digital economies.

This thesis studies token valuation from the perspective of quantitative finance, with the objective of bringing theoretical token-pricing models closer to the economic constraints faced by real digital platforms. The starting point is the framework developed by Cong, Li, and Wang in *Tokenomics: Dynamic Adoption and Valuation* [6], which provides a rigorous foundation for understanding how token prices may internalize expected platform growth and, by doing so, reduce the effective cost of participation for users. Their contribution is central because it formalizes a mechanism through which tokenization can affect adoption endogenously: when future platform growth is capitalized into current token prices, holding the token

becomes less costly, and participation incentives are strengthened.

The purpose is not to replace that insight, but to extend it. The benchmark model is analytically powerful, yet it relies on assumptions that abstract from several features that are relevant in practical tokenized ecosystems. In particular, the original setting treats access rights as perpetual and studies token value within a closed platform environment. Real-world access rights, however, are often finite, renewable, cancellable, or exposed to termination risk. Moreover, users and investors rarely evaluate a token in isolation: they compare it with alternative uses of capital. Finally, in many token ecosystems, supply is not purely exogenous. Protocols, treasuries, or issuing entities may actively decide how many tokens to release, creating a direct link between valuation, reserve management, and issuance policy.

The main contribution of the thesis is therefore to investigate how token valuation changes when some of these simplifying assumptions are progressively relaxed. The guiding idea is that token valuation and token design are inseparable problems. The value of a token depends not only on expected platform growth, but also on the institutional structure through which access is granted, terminated, renewed, backed, and supplied.

Chapter 1 extends the Cong–Li–Wang framework by introducing finite access duration. Two specifications are considered: a deterministic case, in which access expires after a known maturity date, and a stochastic case, in which termination occurs according to a Poisson process with constant intensity. The analysis shows that finite duration does not eliminate the adoption-enhancing mechanism of the benchmark, but that the way duration is modeled is decisive: deterministic expiration significantly depresses token values and adoption incentives, while stochastic termination preserves the qualitative structure of the perpetual benchmark through probabilistic survival rather than hard truncation. Chapter 2 embeds the finite-horizon token in a competitive environment by comparing its equilibrium value against an external asset priced under the same risk-neutral measure. This comparison leads to the characterization of a unique dominance threshold in platform activity, whose comparative statics reveal how token competitiveness depends jointly on platform fundamentals, the effective horizon of access rights, and the value of the outside opportunity. Chapter 3 shifts the focus from equilibrium valuation to protocol design. Motivated by direct engagement with the token architecture developed by RIV Technologies [16], a fintech company building a reserve-backed token ecosystem in which issuance is tied to the growth of productive underlying assets, the chapter formulates the protocol's issuance decision as a static optimization problem in which the protocol balances intrinsic value creation, reserve accumulation, premium stabilization, and issuance

costs. The analysis delivers explicit conditions for the existence and uniqueness of the optimal issuance rate and an approximate closed-form solution that makes the underlying trade-offs directly visible.

Taken together, the three chapters develop a unified view of token valuation as a problem of economic design under uncertainty. The first chapter asks whether the capitalization mechanism survives once access rights are made finite. The second asks when a token remains competitive against an external asset. The third asks how a reserve-backed protocol should manage issuance when supply is endogenous. These questions are distinct, but they are connected by the same central insight: token value is not an intrinsic property of the token itself, but an equilibrium outcome shaped by the rules governing access, competition, and supply.

Chapter 1

Token Valuation Beyond Perpetuity

The theoretical foundation of token-based platform economies has been rigorously formalized by Cong, Li and Wang in *Token Economics: Dynamic Adoption and Valuation* [6], building on standard continuous-time asset pricing frameworks Duffie [9], Cochrane [5].

Their framework demonstrates how tokenization can generate user-side incentives through a capitalization mechanism: by internalizing expected platform growth, token prices reduce the effective carry cost of participation and stimulate adoption in equilibrium, consistent with dynamic optimization frameworks Merton [12]. A central simplifying assumption of their model is that access rights are perpetual, allowing token values to reflect an infinite horizon of expected utility flows.

While analytically powerful, the perpetual benchmark abstracts from a key feature of most real-world access rights: expiration. In practice, platform participation is rarely indefinite (Ascarza and Hardie [1]). Subscription contracts have predetermined renewal dates, usage rights are finite, and many services are subject to termination risk. As a result, the horizon over which benefits can be capitalized is either contractually bounded or probabilistically truncated. Introducing maturity therefore alters the economic environment in a structurally meaningful way by constraining the intertemporal capitalization channel that underlies token valuation.

This chapter extends the Cong-Li-Wang (CWL) framework by incorporating finite access duration into the tokenized platform economy. Two specifications are considered. First, access expires deterministically after a known maturity date, introducing a mechanical valuation decay as the remaining horizon shortens. Second, access termination occurs stochastically according to a known hazard rate, embedding expiration risk directly into equilibrium pricing. In both cases, expiration risk is

anticipated and priced ex ante, preserving rational expectations and the internal consistency of the dynamic equilibrium.

Importantly, the introduction of maturity modifies the valuation structure without altering the fundamental equilibrium mechanism. Preferences, technology, and the Markov structure of the model remain unchanged. Finite duration operates through the pricing equation, reducing the capitalization horizon and introducing a valuation drag. The central question is therefore whether the adoption-enhancing channel identified in the perpetual benchmark survives once access rights are made economically realistic.

The analysis shows that while finite maturity attenuates token values relative to the perpetual case, it does not eliminate the user-side incentive mechanism. The effective carry cost channel remains operative, and equilibrium adoption incentives persist under both deterministic and stochastic termination. The perpetual specification emerges as a limiting benchmark corresponding to the case of infinite horizon and vanishing expiration risk.

By stress-testing the core mechanism under realistic maturity constraints, this chapter establishes the structural robustness of token-based capitalization. It provides the theoretical foundation for the subsequent comparative analysis with alternative access structures and for the applied case study developed in later chapters.

1.1 Basic Economic Intuition: Tokens, Access Rights, and Maturity

Before introducing the formal model, it is useful to clarify the economic objects studied in this chapter. The terminology of tokenized platforms is often used in different ways, but the analysis below relies on one specific interpretation: a token is not treated simply as a speculative crypto-asset, but as a digital instrument that gives access to an economic environment.

A **token** is a digital object that can be held, transferred, and priced. In this thesis, the token is studied primarily as an economic claim that allows the holder to interact with a platform. The user holds the token not only because its market price may increase, but also because holding it gives access to services, transactions, or interactions inside the platform.

A **tokenized access right** is therefore similar to an entry ticket, a usage claim, a membership pass, or a subscription-like right. For example, a token may allow a user to trade in a digital marketplace, access premium functions in a software platform, use a payment network, enter a digital community, consume content, obtain

cloud-computing capacity, or use services inside a fintech ecosystem.

The token is valuable because the platform is useful. A token that gives access to an empty digital marketplace is not very valuable, because there are few buyers, sellers, and transactions. The same token becomes more valuable if the marketplace grows. Similarly, a payment-network token is more valuable when many merchants and users accept it; a content-platform token is more valuable when the platform hosts more content and more interactions; and a service-ecosystem token is more valuable when the ecosystem offers more useful services.

This is what the model summarizes through **platform activity**. Platform activity is an aggregate measure of how economically useful, intense, or productive the platform is. Depending on the application, it may represent transaction volume, number of active users, matching intensity, service usage, content production, payment flows, or the overall scale of interactions among users. The model keeps this variable abstract because different platforms generate value in different ways. What matters for the theory is the direction of the effect: higher activity makes access more useful, and therefore makes the access token more valuable.

This intuition also explains the capitalization mechanism in the CWL framework. If users expect the platform to grow in the future, the token price today may already reflect part of that future growth. Expected appreciation then reduces the effective cost of holding the token and can encourage adoption. In simple terms, users are more willing to hold the token when they expect that the platform will become more useful and that the access right will become more valuable over time.

The reason for introducing **maturity** comes from a practical observation: in real digital platforms, access is rarely an unconditional right to benefit from the platform forever. Users often buy subscriptions, licenses, memberships, usage credits, service passes, or other rights that expire, need to be renewed, or depend on the continuation of the relationship with the platform. Hence, the relevant economic question is not only how valuable the platform may become in the future, but also whether the token holder will still have the right to benefit from that future value when it arrives.

Maturity measures exactly this time dimension. It is the length of time over which the token continues to generate access value. A one-month access token, a one-year subscription token, and a perpetual access token are economically different even if they refer to the same platform. The shorter the remaining access horizon, the less future platform growth can be incorporated into the current token value.

This is not a minor technical detail. It changes the economic meaning of the token. Consider two tokens that give access to the same platform and are exposed

to the same platform activity. If one token expires tomorrow and the other expires in ten years, they should not have the same value. The long-lived token allows the holder to benefit from future platform growth for a much longer period; the short-lived token does not. Treating both as perpetual would overstate the value of the short-lived access right.

Introducing maturity is therefore a way to make the CWL mechanism closer to real digital platforms. If token prices are supposed to reflect future platform benefits, the model must specify who is entitled to receive those benefits and for how long. Maturity answers exactly this question.

Two forms of finite duration are considered. Under **deterministic maturity**, the expiration date is known in advance. This is similar to a subscription that ends after a fixed number of months, a pass that expires at the end of the year, or a prepaid access right that is valid only until a specified date. In this case, the value of the token mechanically declines as the expiration date approaches, because the remaining window of useful access becomes shorter.

Under **stochastic maturity**, expiration is uncertain. The token does not have a known final date, but access may terminate randomly. This can represent cancellation risk, user exit, platform discontinuation, loss of eligibility, or another event that unexpectedly ends the access relationship. In this case, the holder does not know exactly when access will stop, but prices the probability that the token survives long enough to generate future utility.

The innovation of this chapter is to introduce these finite-duration structures into the CWL framework and ask whether the tokenization mechanism survives once access rights are no longer perpetual. The central economic question is simple: if tokens are valuable because they capitalize future platform activity, what happens when the right to benefit from that future activity has an expiration date or may terminate randomly?

1.2 Model Environment

The economy consists of a platform that issues tokens and a continuum of heterogeneous users who decide whether to participate by holding them. Participation is endogenous and depends on individual valuations as well as on aggregate platform conditions, which evolve dynamically over time in a continuous-time setting and shape the benefits of access (see Duffie [9]). The modeling framework builds on the benchmark setting of Cong, Li, and Wang (CWL)([6]), while introducing a modified interpretation of access rights. The key building blocks of the model are described

below.

Users and Preferences

The economy is populated by a continuum of heterogeneous users who can participate in the platform by holding tokens. Users differ in the utility they derive from accessing platform services, reflecting heterogeneity in preferences, usage intensity, or platform valuation.

Each user i is characterized by an individual type u_i, which captures the user's intrinsic valuation of participation, reflecting the utility derived from accessing services conditional on the platform's activity level. Preferences are modeled in exponential form, so that the utility flow derived from participation is proportional to e^{u_i}. This specification guarantees strictly positive utility flows, preserves tractability when aggregating heterogeneous users, and allows differences in user types to translate into economically meaningful dispersion in platform valuations. Moreover, the exponential formulation is standard in continuous-time models of platform participation and token-based economies, ensuring consistency with the existing literature.

User types are distributed according to a continuous distribution with cumulative distribution function $G(u)$ and density $g(u)$, both defined on a compact support $[\underline{U}, \overline{U}]$. We assume $G \in C^1$ and g continuous on $[\underline{U}, \overline{U}]$, ensuring sufficient regularity to apply threshold-based participation arguments and envelope conditions in the equilibrium analysis that follows. The individual participation decision can be interpreted within a standard continuous-time optimization framework, in the spirit of Merton [12].

Platform Environment

The platform is populated by a continuum of users who decide whether to participate by holding tokens. Let N_t denote the size of the platform's active user base at time t. In the continuum-of-users framework, N_t represents the normalized mass of participating users and takes values in the interval $[0, 1]$.

Formally, participation is determined endogenously through users' optimal decisions. At each point in time, only users whose type exceeds a participation threshold choose to hold tokens and access the platform. As a result, the user base can be expressed as the measure of participating types:

$$N_t = \int_{\mathcal{U}_t} dG(u),$$

where $\mathcal{U}_t \subseteq [\underline{U}, \overline{U}]$ denotes the set of user types that optimally choose to participate

at time t. This corresponds to the cumulative distribution of user types above the endogenous participation cutoff. This representation highlights that the size of the user base is not fixed, but responds to economic incentives and platform conditions.

Platform activity is summarized by an aggregate state variable A_t, which captures the overall level of economic activity generated on the platform. The variable A_t is intentionally kept abstract, as it may represent different notions depending on the application, such as transaction volume, matching efficiency, content creation, or the intensity of the interactions among users. Higher values of A_t reflect a more active and productive platform environment, increasing the benefits of participation for users.

The evolution of platform activity is assumed to follow a stochastic process under the physical probability measure $\mathbb{P}$. Specifically, A_t evolves according to a geometric diffusion:

$$\frac{dA_t}{A_t} = \hat{\mu}^A dt + \sigma^A d\hat{Z}_t, \tag{1.1}$$

where $\hat{\mu}^A$ denotes the expected growth rate of platform activity, σ^A its volatility, and $\hat{Z}_t$ a standard Brownian motion under $\mathbb{P}$.

This specification captures aggregate shocks affecting the platform's activity level, such as technological innovations, demand fluctuations, or changes in the economic environment.

Participation and Holding Costs

Participation in the platform entails a per-period cost $\phi > 0$ for users who hold tokens and access platform services. This cost captures generic frictions associated with participation, such as effort, time costs, or platform-related fees. The parameter ϕ is assumed to be constant across users and over time, implying that participation decisions are driven by heterogeneity in users' valuations of platform access. In addition, holding tokens involves a per-period carrying cost, capturing opportunity costs, liquidity frictions, or balance-sheet constraints associated with maintaining a token position over time. These costs ensure that token holdings are not frictionless. While they are not modeled explicitly at this stage, they will be incorporated later when characterizing equilibrium token demand.

1.3 Tokenized Access with Deterministic Expiration

We begin by characterizing active access rights and their maturity structure, and then derive the corresponding utility flow and individual optimization problem. This framework isolates the implications of finite token lifetime before extending the analysis to stochastic maturity.

Active Access Rights and Token Maturity

Let $q_{i,\tau}$ denote the number of tokens purchased by user i at time τ. Tokens are interpreted as subscription-like access rights with finite maturity $T > 0$. In this setting, a token purchased at time τ grants access to the platform only over the interval $[\tau, \tau + T]$.

The economic interpretation is straightforward. If a user buys an access token today and the token expires in one year, the token can only capitalize the benefits generated during that year. Future platform growth after the expiration date no longer belongs to the holder of that specific access right. The deterministic maturity case therefore captures a hard contractual endpoint: as the expiration date approaches, the time window over which access can generate utility mechanically shrinks.

At any time $t \geq 0$, an access right is active only if the token has not expired, that is, if $t - \tau \in [0, T]$. Tokens outside this interval no longer generate access or utility.

To capture the activity status of the token, define

$$q_{i,t} := q_{i,\tau} \mathbf{1}_{\{0 \leq t-\tau \leq T\}}, \tag{1.2}$$

where the indicator function ensures that only non-expired access rights are active at time t. Note that $q_{i,t}$ does not represent a new purchase decision at time t, but rather the active component at time t of a token purchase executed at time τ.

Once a token has been purchased, the issuance quantity $q_{i,\tau}$ is predetermined and no longer enters individual decision-making. What matters for incentives at time t is whether access rights are still active and the economic value they represent. For this reason, individual optimization is formulated in terms of the market value of active access rights. Let

$$x_{i,t} = P_t q_{i,t}, \tag{1.3}$$

where P_t denotes the equilibrium price of a token at time t. P_t reflects the market

value of one unit of active access, incorporating the expected stream of future utility generated over the remaining lifetime of the access right. Accordingly, $x_{i,t}$ represents the total value of access services held by user i at time t.

Recall that in Section 1.2, N_t was defined as the platform user base, representing the mass of users participating in the platform at time t. In the present setting, participation is tied to the possession of active access rights. Access rights are finite-lived but automatically renewed upon expiration, so that the aggregate number of active passes M remains constant over time. The user base N_t is not cohort-specific; rather, it is determined instantaneously by the participation cutoff, exactly as in the perpetual benchmark of CWL. That is, N_t measures the mass of users whose access rights remain active at time t, rather than the total platform population across all issuance vintages. As access rights expire deterministically after time T, the size of this effective user base evolves mechanically over time through contract expiration.

Utility Flow from Tokenized Platform Access

Having defined active access rights and the associated effective user base, we now characterize the instantaneous utility flow generated by tokenized platform participation. The instantaneous utility increment of user i from platform participation is given by:

$$dv_{i,t} = (x_{i,t})^{1-\alpha}(N_t A_t e^{u_i})^{\alpha} dt - \phi dt - r x_{i,t} dt. \tag{1.4}$$

The first term represents the gross utility flow generated by platform participation. Utility is produced through the interaction between the value of active token holdings $x_{i,t}$, which generate utility only while access rights are active, the effective user base N_t, aggregate platform activity A_t, and a user-specific preference parameter. An increase in the mass of active users raises the value of access for each participant, while higher platform activity amplifies the utility generated by a given level of access and participation. The parameter $\alpha \in (0, 1)$ governs the relative importance of platform-side versus access-side inputs in utility generation.

The remaining terms represent a per-period participation cost and a carrying cost. The parameter r denotes the instantaneous risk-free rate and captures the opportunity cost of holding token value, reflecting the return foregone by not investing in the risk-free asset.

Remark 1.1. **Cobb-Douglas** The gross utility flow is generated by a Cobb-Douglas aggregator of the form:

$$f(x, N, A, e^u) = x^{1-\alpha}(NAe^u)^{\alpha}.$$

A Cobb-Douglas aggregator is characterized by a constant elasticity of substitution equal to one and by diminishing marginal returns in each input. In this context, it implies that active token holdings and platform-side factors are imperfect but essential complements in the generation of utility. The choice of this function follows the benchmark specification in the CWL model.

Unlike the benchmark CWL framework, access rights in the present model are finite-lived and generate utility only while they remain active. As a result, the flow payoff $dv_{i,t}$ no longer represents utility from a perpetual access position, but rather utility generated by active, finite-maturity access rights that cease to contribute upon expiration. This is the first departure from CWL: the same tokenized participation mechanism is retained, but the access right now has a finite economic life.

Stochastic Discount Factor

To price token payoffs in continuous time, we introduce a stochastic discount factor (SDF), denoted by Λ_t. The SDF is defined as the state-price deflator that converts future payoffs into time-t present values under the physical probability measure $\mathbb{P}$, following the standard asset pricing framework (see, Cochrane [5]). Economically, Λ_t captures both time discounting and compensation for risk. We assume that Λ_t follows a diffusion process, consistent with the standard continuous-time asset pricing framework. Let r denote the instantaneous risk-free rate, consistent with its interpretation as the opportunity cost of holding token value in the utility specification, and η, which is assumed to be constant, the market price of risk associated with the Brownian motion driving aggregate uncertainty. The dynamics of the SDF are given by:

$$\frac{d\Lambda_t}{\Lambda_t} = -rdt - \eta d\hat{Z}_t^{\Lambda}, \tag{1.5}$$

where $\hat{Z}_t^{\Lambda}$ denotes a standard Brownian motion under the physical probability measure $\mathbb{P}$.

Token Valuation with Finite Maturity

Up to this point, all economic quantities and dynamics have been defined under the physical probability measure $\mathbb{P}$, which governs agents' beliefs and the evolution of uncertainty. While this representation is natural for modeling preferences and incentives, asset pricing requires working under a risk-neutral measure. To price token payoffs, we therefore introduce an equivalent probability measure $\mathbb{Q}$, under which discounted asset prices are martingales. The change of measure from $\mathbb{P}$ to $\mathbb{Q}$ is induced

by the stochastic discount factor Λ_t defined above. Under $\mathbb{P}$, uncertainty is driven by a Brownian motion $\hat{Z}_t$. Under the risk-neutral measure $\mathbb{Q}$, the corresponding Brownian motion is denoted by Z_t, with the drift adjustment determined by the market price of risk η.

Remark 1.2. **Change of Measure with Girsanov Theorem**

Considera a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$. The Girsanov theorem is used to change the probability measure from the physical measure $\mathbb{P}$ to an equivalent risk-neutral measure $\mathbb{Q}$. Under $\mathbb{P}$, suppose a generic diffusion process X_t follows:

$$dX_t = \mu dt + \sigma d\hat{W}_t,$$

where $\hat{W}_t$ is a Brownian motion. Define under the new measure $\mathbb{Q}$, the process W_t as a Brownian motion:

$$W_t = \hat{W}_t + \int_0^t \theta_s ds,$$

where θ_t is an adapted process satisfying standard integrability conditions. As a consequence, the dynamics of X_t under $\mathbb{Q}$ becomes:

$$dX_t = (\mu - \sigma\theta_t)dt + \sigma dW_t.$$

so that the drift term is adjusted to reflect risk-neutral pricing.

In our model, aggregate platform activity evolves under the physical measure $\mathbb{P}$ according to (1.1). Since aggregate platform activity risk is priced through its exposure to the stochastic discount factor, the risk adjustment under the pricing measure is determined by the correlation between the Brownian motions driving the SDF and aggregate activity. Assume that under $\mathbb{P}$, the Brownian motions driving the pricing kernel and platform activity have instantaneous correlation ρ. Then:

$$dZ_t = \rho d\hat{Z}_t^{\Lambda} + \sqrt{1-\rho^2} dW_t,$$

where W_t is a Brownian motion independent of $\hat{Z}_t^{\Lambda}$. After the change of measure, using $d\hat{Z}_t^{\Lambda} = dZ_t^{\Lambda} - \eta dt$, we obtain:

$$d\hat{Z}_t = \rho(dZ_t^{\Lambda} - \eta dt) + \sqrt{1-\rho^2} dW_t = dZ_t - \rho\eta dt,$$

where Z_t is a Brownian motion under $\mathbb{Q}$. Substituting the adjusted Brownian motion

into (1.1), we obtain:

$$\frac{dA_t}{A_t} = (\hat{\mu}^A - \rho\sigma^A\eta)dt + \sigma^A dZ_t,$$
$$\frac{dA_t}{A_t} = \mu^A dt + \sigma^A dZ_t. \tag{1.6}$$

The term $\rho\sigma^A\eta$ represents the risk adjustment induced by the exposure of aggregate activity to the stochastic discount factor.

Recall that under $\mathbb{P}$, the stochastic discount factor satisfies (1.5).

The solution of this SDE is

$$\Lambda_t = \Lambda_0 \exp(-rt - \eta\hat{Z}_t^\Lambda - \frac{1}{2}\eta^2 t).$$

Define the exponential martingale

$$\xi_t = \exp(-\eta\hat{Z}_t^\Lambda - \frac{1}{2}\eta^2 t).$$

By standard results on exponential martingales, ξ_t is a $\mathbb{P}$-martingale with unit expectation. Consequently, $\mathbb{E}^{\mathbb{P}}[\xi_t] = 1$ for all t, and the process $\Lambda_t B_t$, where $B_t = e^{rt}$, is a martingale.

This guarantees that the Radon-Nikodym derivative defining the change of measure is well-defined and that the risk-neutral measure $\mathbb{Q}$ is equivalent to $\mathbb{P}$.

Let $y_{i,t}$ denote agent i's undiscounted cumulative payoff from platform participation up to time t. The process $y_{i,t}$ is assumed to be of finite variation and its increments are driven by the instantaneous flow payoff $dv_{i,t}$ defined in (1.4).

Let $\mathbb{E}^{\mathbb{P}}$ and $\mathbb{E}^{\mathbb{Q}}$ denote expectations under the physical and risk-neutral probability measures, respectively. Agent i evaluates her lifetime payoff under the physical measure $\mathbb{P}$ using the stochastic discount factor Λ_t:

$$\mathbb{E}^{\mathbb{P}}\left[\int_0^\infty \frac{\Lambda_t}{\Lambda_0} dy_{i,t}\right].$$

Using the change-of-measure relation implied by the pricing kernel, this expression can be equivalently written under the equivalent martingale measure $\mathbb{Q}$ as

$$\mathbb{E}^{\mathbb{Q}}\left[\int_0^\infty e^{-rt} dy_{i,t}\right].$$

The equivalence between SDF-discounted payoffs under the physical measure and discounted expectations under the risk-neutral measure follows from the martingale

pricing approach (see Duffie [9]). Hence, throughout the remainder of the analysis we work under the risk-neutral measure $\mathbb{Q}$, unless stated otherwise.

In the present framework, tokens do not grant perpetual access to the platform. Instead, access rights expire deterministically after a finite maturity T. As a consequence, utility flows generated by a given token purchase cease once access rights expire. To formalize this feature, let m_t denote the remaining maturity of a token held at time t. Given a purchase time τ, the remaining maturity at time $t \geq \tau$ is defined as:

$$m_t := T - (t - \tau). \tag{1.7}$$

For $m_t \leq 0$, the token has expired and no further utility is generated.

Accordingly, although the lifetime payoff is formally written over an infinite horizon, the effective contribution of a given token purchase is confined to the interval during which $m_t > 0$. Utility flows satisfy $dy_{i,s} = 0$ for all s such that $m_s \leq 0$.

The token price satisfies $P_t = 0$ for $t > \tau + T$. For $t \leq \tau + T$, the token price equals the expected discounted payoff over the remaining lifetime:

$$P_t = \mathbb{E}_t^{\mathbb{Q}}\left[\int_t^{\tau+T} e^{-r(s-t)} dy_{i,s}\right] = \mathbb{E}_t^{\mathbb{Q}}\left[\int_0^{m_t} e^{-ru} dy_{i,t+u}\right].$$

where $\mathbb{E}_t^{\mathbb{Q}}$ denotes expectation under $\mathbb{Q}$ conditional on the information available at time t.

Markov Equilibrium

Definition 1.3. ***Markov Equilibrium*** *A Markov equilibrium with state variable (A_t, m_t) is characterized by an equilibrium token price P_t and agents' decisions such that:*

- *Optimality: given the equilibrium price process P_t, agents optimally choose token purchases at issuance dates so as to maximize their expected lifetime payoff.*
- *Market Clearing: the token market clears at all times, that is, aggregate active token values are consistent with the total supply of passes M.*
- *Markov Property: equilibrium prices and agents' decisions depend only on the current state variable A_t and the remaining maturity m_t, and not on the past history of the economy.*

Let M denote the total number of passes issued by the platform. At any time t, the total number of outstanding passes is given by aggregating individual pass

holdings across agents:

$$M = \int_{i\in[0,1]} q_{i,t} di.$$

Assume a unit mass of agents indexed by $i \in [0,1]$. Using the definition of individual token value (1.3), the aggregate supply of passes can be equivalently expressed as:

$$M = \int_{i\in[0,1]} \frac{x_{i,t}}{P_t} di$$

Rearranging, total active token value in the economy satisfies:

$$\int_{i\in[0,1]} x_{i,t} di = P_t M. \tag{1.8}$$

Remark 1.4. **Interpretation of M under Finite Maturity** In contrast to the CWL framework, individual access rights in the present model have finite maturity. However, to isolate the effect of maturity on pricing from issuance dynamics, we assume automatic renewal: each expired pass is instantaneously replaced by a new pass with identical remaining maturity m_t. As a result, the aggregate number of active passes M remains constant over time. Importantly, the user base N_t is not the size of a specific cohort, but the instantaneous mass of users who choose to hold active passes at time t, given the current cutoff $\underline{u}_t$. The renewal mechanism ensures that the platform always has M passes outstanding, but which users hold them is determined endogenously at each instant.

This aggregation result plays a central role in characterizing equilibrium, as it links individual token values to the market-clearing condition in the token market.

Individual Payoff Structure and Optimization Problem

In a tokenized platform, the token price is determined in equilibrium by market clearing together with agents' optimal token demand.

In our setting, the equilibrium token price cannot depend solely on platform activity A_t, but must also depend on the remaining time to expiration. We therefore conjecture a Markov representation $P_t = P(A_t, m_t)$. Under the risk-neutral measure $\mathbb{Q}$, we assume that the token price follows a diffusion process of the form

$$dP_t = \mu_t^P P_t dt + \sigma_t^P P_t dZ_t, \tag{1.9}$$

where μ_t^P and σ_t^P denote, respectively, the drift and volatility of the token price, and Z_t is the Brownian motion driving aggregate platform uncertainty. Since the equilibrium price P_t is a function of aggregate platform activity, the diffusion term of

P_t is entirely inherited from that of A_t. At this stage, μ_t^P and σ_t^P are left unspecified.

Agent i derives instantaneous utility from platform participation through access rights granted by token holdings. Under the risk-neutral measure, expected capital gains are reflected in the opportunity cost of holding tokens. In particular, holding one unit of the token entails an effective per-period cost equal to $r - \mu_t^P$. Here, μ_t^P denotes the drift of the token price under the risk-neutral measure and therefore captures the effective expected appreciation relevant for pricing and carry costs. At each time t, agent i chooses whether to participate in the platform and, conditional on participation, chooses her active token holdings $x_{i,t}$ to maximize expected lifetime payoff under the risk-neutral measure. In flow terms, the agent's instantaneous decision solves:

$$dy_{i,t} = \max\left\{0, \max_{x_{i,t}}[(x_{i,t})^{1-\alpha}(N_t A_t e^{u_i})^{\alpha} - \phi - (r - \mu_t^P)x_{i,t}]\right\}dt \tag{1.10}$$

For a given state (A_t, N_t) and given the risk-neutral drift μ_t^P, the objective is concave in $x_{i,t}$ (since $\alpha \in (0,1)$), so an interior optimum is characterized by the first-order condition. We assume $r - \mu_t^P > 0$ to ensure an interior solution and to rule out arbitrage opportunities. Differentiating with respect to $x_{i,t}$ yields:

$$\frac{\partial}{\partial x_{i,t}}[(x_{i,t})^{1-\alpha}(N_t A_t e^{u_i})^{\alpha} - \phi - (r - \mu_t^P)x_{i,t}] = 0.$$
$$(1-\alpha)\left(\frac{N_t A_t e^{u_i}}{x_{i,t}}\right)^{\alpha} - (r - \mu_t^P) = 0$$

Solving for $x_{i,t}$

$$x_{i,t}^* = N_t A_t e^{u_i}\left(\frac{1-\alpha}{r - \mu_t^P}\right)^{\frac{1}{\alpha}} \tag{1.11}$$

We determined the agent's optimal active holdings $x_{i,t}^*$ as a function of the current state variables and μ_t^P.

Substituting $x_{i,t}^*$ back into the payoff delivers the maximized instantaneous net benefit from participation. The participation decision is governed by the condition that this optimized payoff is non-negative. This pins down a cutoff type $\underline{u}_t$ such that the agents with $u_i \geq \underline{u}_t$ participate, while the others stay out. Imposing the indifference condition at the cutoff type yields:

$$\underline{u}_t = -\ln(N_t) + \ln\left(\frac{\phi}{A_t \alpha}\right) - \left(\frac{1-\alpha}{\alpha}\right)\ln\left(\frac{1-\alpha}{r - \mu^P(A_t, m_t)}\right).$$

Given the distribution $G(\cdot)$ of types, the user base is therefore

$$N_t = 1 - G(\underline{u}_t)$$

Market Clearing and Equilibrium Token Price

Consider $x^*_{i,t}$ the optimal holdings implied by (1.11). Market clearing requires that the aggregate value of active token holdings equals the market value of the outstanding pass supply M, i.e.

$$\int_{i\in[0,1]} x^*_{i,t} di = P(A_t, m_t)M,$$

where M is the (constant) mass of passes outstanding under the automatic renewal mechanism. This condition links individual optimal demand to the equilibrium pricing function $P(\cdot,\cdot)$. To close the model, we must characterize the drift μ^P_t under $\mathbb{Q}$, which enters agent's effective carry cost via the term $r - \mu^P_t$. This requires deriving the risk-neutral dynamics of P_t as a function of the state variables (A_t, m_t). Aggregating (1.11) over participating agents ($u_i \geq \underline{u}_t$), aggregate demand can be written as

$$\int_{i\in[0,1]} x^*_{i,t}\, di = N_t A_t \left(\frac{1-\alpha}{r - \mu^P(A_t, m_t)}\right)^{\frac{1}{\alpha}} \int_{\underline{u}_t}^{\overline{U}} e^u\, dG(u).$$

Define the aggregate transaction-needs shifter:

$$S_t = S(\underline{u}_t) := \int_{\underline{u}_t}^{\overline{U}} e^u dG(u), \tag{1.12}$$

so that market clearing becomes

$$P(A_t, m_t) = \frac{N_t S_t A_t}{M}\left(\frac{1-\alpha}{r - \mu^P(A_t, m_t)}\right)^{\frac{1}{\alpha}}. \tag{1.13}$$

This is a static equilibrium relation: it links $P(A_t, m_t)$ to contemporaneous aggregates and the implied expected appreciation $\mu^P(A_t, m_t)$. The static equilibrium relation derived above does not pin down the expected appreciation of the token. To close the model, the equilibrium price must be consistent with its own risk-neutral dynamics. We next determine $\mu^P(A_t, m_t)$ from the risk-neutral dynamics of P_t via Itô's lemma applied to $P(A_t, m_t)$. Since m_t decreases deterministically at unit rate, $dm_t = -dt$, and Itô's lemma must account for the maturity derivative P_m.

Token Price Dynamics

The token dynamics must be derived using Itô's lemma for functions of two state variables. Under $\mathbb{Q}$, platform activity follows (1.6), while remaining maturity evolves deterministically as $dm_t = -dt$, for $m_t > 0$, with expiration occurring at $m_t = 0$. Since the equilibrium token price is a function of aggregate platform activity, both processes are driven by the same Brownian motion under $\mathbb{Q}$. Applying Itô's lemma to $P(A_t, m_t)$ (see Oksendal [15]), we obtain:

$$dP_t = \frac{\partial P_t}{\partial m_t} dm_t + \frac{\partial P_t}{\partial A_t} dA_t + \frac{1}{2}\frac{\partial^2 P_t}{\partial m_t^2}(dm_t)^2 + \frac{1}{2}\frac{\partial^2 P_t}{\partial A_t^2}(dA_t)^2 + \frac{\partial^2 P_t}{\partial m_t \partial A_t}(dm_t)(dA_t).$$

Since $dA_t \cdot dm_t = 0$ and $(dm_t)^2 = 0$ and $(dA_t)^2 = (A_t\sigma_t^A)^2 dt$ we obtain:

$$\begin{aligned} dP_t &= P_A\, dA_t + P_m\, dm_t + \frac{1}{2} P_{AA}\, (dA_t)^2 \\ &= P_A\Big(A_t\mu^A\, dt + A_t\sigma^A\, dZ_t\Big) + P_m(-dt) + \frac{1}{2} P_{AA}\, (A_t\sigma^A)^2 dt \\ &= \Big(P_A A_t\mu^A - P_m + \frac{1}{2} P_{AA} A_t^2 (\sigma^A)^2\Big) dt + P_A A_t \sigma^A\, dZ_t. \end{aligned} \tag{1.14}$$

Comparing this expression with the generic risk-neutral representation (1.9) we obtain the implied drift and diffusion as functions of the state variables $\mu_t^P := \mu^P(A_t, m_t)$ and $\sigma_t^P := \sigma^P(A_t, m_t)$:

$$\mu_t^P P(A_t, m_t) = P_A A_t \mu^A - P_m + \frac{1}{2} P_{AA} (A_t\sigma^A)^2 \tag{1.15}$$

and

$$\sigma_t^P P(A_t, m_t) = P_A A_t \sigma^A.$$

The key implication is that, relative to the perpetual-access benchmark, the finite-maturity setting introduces the additional term $-P_m$ in the drift, which plays the role of a theta-like effect analogous to that arising in finite-maturity derivative pricing. Economically, this term captures the mechanical erosion of token value due to the shrinking horizon over which access rights generate utility. This term is precisely what makes the equilibrium pricing problem fundamentally different: the token pricing equation will now be a partial differential equation in (A, m), rather than an ordinary differential equation in A alone. This characterization is standard in continuous-time asset pricing models with Markov state variables.

Remark 1.5. A natural concern in the finite-maturity setting is that the additional term $-P_m$ may fully offset the capitalization benefits of tokenization. However, that

the maturity decay term operates through the pricing equation, whereas adoption incentives depends on the effective carry cost $r-\mu^P(A_t, m_t)$ in individual optimization.

Even in the presence of deterministic expiration, tokenization continues to enhance adoption incentives whenever the risk-neutral appreciation component $\mu^P(A_t, m_t)$ remains positive. Finite maturity therefore disciplines valuation by limiting the capitalization horizon, but does not mechanically eliminate the growth-driven pricing channel of the tokenized economy.

Absence of Arbitrage and Equilibrium Consistency

We conclude the analysis by verifying that the equilibrium token price derived above is consistent with the absence of arbitrage. Throughout this section, we make explicit the dependence of aggregate adoption and transaction intensity on the state variables and write $N_t = N(A_t, m_t)$ and $S_t = S(A_t, m_t)$. Under the fundamental theorem of asset pricing, the absence of arbitrage implies that the value of any traded asset, augmented by its cumulative payoff stream and discounted by the stochastic discount factor, must be a martingale under the physical measure. In the present setting, this condition requires that

$$\Lambda_t P_t + \int_0^t \Lambda_s D_s ds$$

is a martingale, where D_t denotes the aggregate flow payoff (dividend) associated with holding the token. Equivalently, the drift of this process must be zero.

Using Itô's lemma, we have

$$d(\Lambda_t P_t) = \Lambda_t dP_t + P_t d\Lambda_t + d\Lambda_t dP_t.$$

Using the dynamics of the stochastic discount factor under $\mathbb{P}$ and the corresponding no-arbitrage restriction implied by risk-neutral pricing, the drift condition yields

$$rP_t = \mu_t^P P_t + D_t$$

In equilibrium, the aggregate dividend reflects the marginal contribution of token holdings to platform surplus, evaluated at optimal demand. Using the optimal demand expression and the envelope condition, the aggregate flow dividend can be written as

$$D_t = (1-\alpha)\left(\frac{N_t S_t A_t}{P_t M}\right)^{\alpha} P_t.$$

Substituting both the drift of the token price obtained from Itô's lemma applied to $P(A_t, m_t)$ and the expression for D_t into the no-arbitrage condition yields the

equilibrium pricing equation:

$$rP(A_t, m_t) = P_A A_t \mu^A + \frac{1}{2} P_{AA} A_t^2 (\sigma^A)^2 - P_m + \\ + (1-\alpha)\left(\frac{N(A_t, m_t) S(A_t, m_t) A_t}{P(A_t, m_t) M}\right)^{\alpha} P(A_t, m_t). \tag{1.16}$$

This equation characterizes the equilibrium token price as the solution to a partial differential equation in the state variables (A_t, m_t). The additional term $-P_m$ reflects the deterministic decay of remaining maturity and constitutes the key difference relative to the perpetual-access benchmark of CWL.

Together with the endogenous adoption conditions determining N_t and S_t, this equation ensures that the equilibrium token price is both internally consistent with agents' optimal behavior and free of arbitrage opportunities.

Markov Equilibrium with Finite-Maturity Access Rights

The equilibrium construction follows the continuous-time Markov equilibrium logic of Cong, Li, and Wang [6], and can be interpreted more broadly within equilibrium frameworks where aggregate states and valuation interact endogenously, as in Brunnermeier and Sannikov [3]. However, finite token maturity introduces an additional state variable, which fundamentally alters the equilibrium pricing problem.

Because access rights expire deterministically, the token price depends not only on aggregate activity A_t but also on the remaining maturity m_t. As a result, the equilibrium pricing equation becomes a partial differential equation in (A, m) rather than an ordinary differential equation in A alone.

The equilibrium pricing equation for the token must be complemented by appropriate boundary conditions. As in the benchmark CWL framework, aggregate platform activity admits a natural lower bound. When aggregate activity vanishes, the platform generates no utility and participation is not profitable for any agent. Accordingly, the token price must satisfy the lower boundary conditions

$$P(0, m) = 0 \qquad \text{for all } m \geq 0.$$

Access rights expire deterministically once the remaining maturity reaches zero, after which no further utility flows are generated. This implies a terminal boundary condition in the maturity dimension:

$$P(A, 0) = 0 \qquad \text{for all } A \geq 0.$$

This terminal condition assumes that access rights do not embed any renewal option or continuation value. Upon expiration, the token delivers no further utility and has zero scrap value. Unlike perpetual tokens, whose value is sustained by indefinitely long future access, finite-maturity tokens naturally lose all value upon expiration.

Proposition 1.6. ***Finite-Maturity Tokenized Equilibrium*** *Under standard regularity conditions, a Markov equilibrium with state variables (A_t, m_t) is characterized by the following objects:*

1. *An equilibrium token price $P(A_t, m_t)$ solving the equilibrium pricing partial differential equation subject to the boundary conditions $P(0, m) = 0$ and $P(A, 0) = 0$;*

2. *A participation cutoff $\underline{u}_t = \underline{u}(A_t, m_t)$, such that agents with $u_i \geq \underline{u}_t$ participate, and endogenous user base given by $N_t = 1 - G(\underline{u}_t)$;*

3. *Optimal active token holdings $x^*(A_t, m_t)$ satisfying the intra-temporal optimality condition equating marginal transaction benefits to the effective carry cost $r - \mu^P(A_t, m_t)$;*

Together, these objects jointly determine equilibrium prices, participation, and token demand in the finite-maturity tokenized economy.

Remark 1.7. From the equilibrium relation (1.13), the token price scales linearly with platform activity A up to the growth-adjusted carry term $(r - \mu^P(A, m))^{-1/\alpha}$. Hence, whenever $r - \mu^P(A, m) > 0$ and the risk-neutral appreciation component $\mu^P(A, m)$ does not decrease in A, the equilibrium price function $P(A, m)$ is strictly increasing in A for any fixed remaining maturity m.

Economically, higher platform activity raises the marginal productivity of access rights and therefore increases the capitalization of platform-generated surplus into token prices. This monotonicity property will play a key role in Chapter 2, where it ensures that dominance regions relative to an external asset can be characterized through well-defined activity thresholds.

As in the perpetual-access benchmark, tokens capitalize expected future platform growth into current prices, thereby reducing the effective carry cost of participation and fostering user adoption. However, finite token maturity fundamentally alters this mechanism. Because access rights expire deterministically, expected productivity growth is capitalized only over a finite horizon. As a result, the feedback loop between token prices and user adoption remains operative but is attenuated by the decay of remaining maturity. In this sense, finite-maturity tokens retain the

adoption-accelerating role emphasized in Cong et al. [6], while introducing a natural discipline on token valuation. The equilibrium token price reflects not only expected platform growth and network effects, but also the shrinking time window over which access rights generate utility.

1.4 Non-Tokenized Access Rights under Deterministic Maturity

We consider a non-tokenized economy in which platform access is provided through deterministic, finite-maturity subscriptions that are not traded. This benchmark allows us to separate the effects of access duration from those of asset pricing and capitalization.

Active Access Rights and Deterministic Maturity and Utility Flow (Non-Tokenized)

In the non-tokenized economy, platform access is provided through deterministic, finite-maturity subscription contracts. These access rights grant participation over a fixed time interval but are not represented by a tradable asset and therefore do not carry a market price.

As in the tokenized economy, platform access is defined over a deterministic interval of length T, and access rights are active only while this interval has not expired. Accordingly, the activity status of access rights is defined as in (1.2).

For consistency with the tokenized framework, we continue to denote individual decisions in terms of an effective access variable $x_{i,t}$. In the non-tokenized economy, however, access rights are not priced and are not represented by a traded asset. As a result, effective access coincides directly with active access rights, and no additional transformation is involved:

$$x_{i,t} = q_{i,t}.$$

Accordingly, introducing $x_{i,t}$ does not add a new economic object, but merely preserves notational consistency with the tokenized case.

Let N_t denote the mass of users whose access rights are still active at time t. Under deterministic maturity, the user base evolves mechanically over time through contract expiration.

The instantaneous utility increment from platform participation coincides with equation (1.4).

In contrast to the tokenized economy, $x_{i,t}$ represents access services rather than the market value of a traded asset. Accordingly, the term $rx_{i,t}$ reflects a static opportunity cost and does not embed capital gains. Utility flows are generated only while access rights remain active and cease deterministically upon expiration. Aggregate platform activity A_t evolves stochastically as specified in the model setup (see (1.1)). In the non-tokenized economy, aggregate risk affects utility directly but does not give rise to asset valuation or risk-adjusted pricing.

Individual Payoff Structure and Optimization Problem

In the non-tokenized economy, access rights are not represented by a traded asset. Agents therefore do not face capital gains or losses, and the only relevant cost of access intensity is the static opportunity cost r.

At each time t, agent i chooses whether to participate in the platform and, conditional on participation, selects her level of access $x_{i,t}$ to maximize instantaneous utility. Individual decisions are taken as given with respect to aggregate state variables.

The instantaneous decision problem is

$$dy_{i,t} = \max\left\{0, \max_{x_{i,t}}\left[(x_{i,t})^{1-\alpha}(N_t A_t e^{u_i})^{\alpha} - \phi - rx_{i,t}\right]\right\}dt. \tag{1.17}$$

Conditional on participation, the objective function is concave in $x_{i,t}$ for $\alpha \in (0,1)$. An interior solution is characterized by the first-order condition

$$(1-\alpha)\left(\frac{N_t A_t e^{u_i}}{x_{i,t}}\right)^{\alpha} - r = 0, \tag{1.18}$$

which yields the optimal access choice

$$x^*_{i,t} = N_t A_t e^{u_i}\left(\frac{1-\alpha}{r}\right)^{\frac{1}{\alpha}}. \tag{1.19}$$

Substituting the optimal choice into the payoff function delivers the maximized instantaneous net benefit from participation. Participation is optimal whenever this value is non-negative. This defines a cutoff type $\underline{u}_t$ such that agents with $u_i \geq \underline{u}_t$ participate.

The cutoff type satisfies

$$\underline{u}_t = -\ln(N_t) + \ln\left(\frac{\phi}{A_t\alpha}\right) - \left(\frac{1-\alpha}{\alpha}\right)\ln\left(\frac{1-\alpha}{r}\right), \tag{1.20}$$

and the endogenous user base is given by

$$N_t = 1 - G(\underline{u}_t). \tag{1.21}$$

Remark 1.8. Although access rights have finite deterministic maturity, remaining time to expiration does not constitute an economic state variable in the non-tokenized economy. As in the tokenized framework, access becomes inactive once the remaining maturity reaches zero. However, in the absence of asset pricing, maturity affects payoffs only mechanically through the activity condition $m_t > 0$ and does not enter individual optimization or participation thresholds.

Proposition 1.9. ***Non-Tokenized Equilibrium with Deterministic Subscription Maturity*** *In the non-tokenized economy with deterministic finite-maturity access rights, a Markov equilibrium with state variable A_t is characterized by a participation cutoff $\underline{u}_t^{NT}$ and an endogenous user base N_t^{NT} satisfying*

$$\begin{aligned}
\underline{u}_t^{NT} &= -\ln(N_t^{NT}) + \ln\left(\frac{\phi}{A_t\alpha}\right) - \left(\frac{1-\alpha}{\alpha}\right)\ln\left(\frac{1-\alpha}{r}\right), \\
N_t^{NT} &= 1 - G(\underline{u}_t^{NT}).
\end{aligned}$$

1.5 Token Pricing under Stochastic Access Duration

In the deterministic framework, access rights expire at a known and common horizon, and remaining maturity constitutes an explicit state variable of the economy. We now introduce a stochastic specification in which access termination occurs randomly over time.

Rather than expiring at a fixed date, tokenized access rights are subject to a Poisson-based termination mechanism with constant intensity. This modification replaces the deterministic countdown of remaining maturity with a survival structure governed by a hazard rate. As a consequence, expiration risk no longer enters the equilibrium through an additional state variable, but through an effective killing term in the valuation equation.

Economically, this specification captures environments in which access duration is uncertain and termination risk is idiosyncratic, while preserving tractability and a Markov pricing structure in aggregate platform activity.

An example is a platform access token that does not have a fixed expiration date,

but may stop generating access because the service is discontinued, the user exits, or the protocol terminates the access relationship. In such a case, the relevant question is not how many months are left until a known deadline, but how likely it is that the access right survives into the future. The stochastic specification therefore replaces a hard maturity date with a survival probability.

For instance, a user may hold a token that grants access as long as the platform relationship remains active, but access may unexpectedly end because eligibility conditions are no longer satisfied, the account is closed, the service is suspended, or the platform changes the terms under which access is granted. In all these cases, there is no predetermined expiration date. What matters for valuation is the probability that the token remains active long enough for the holder to benefit from future platform activity.

Poisson Process

We formalize stochastic access termination through a Poisson process with constant intensity. The process defines the random time at which tokenized access rights are irreversibly terminated.

Definition 1.10. ***Poisson Process*** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration satisfying the usual conditions. A Poisson process $(J_t)_{t\geq 0}$ with intensity $\delta > 0$ is a counting process such that:*

1. *$J_0 = 0$,*
2. *J_t has independent increments,*
3. *for any $s < t$, the increment $J_t - J_s$ follows a Poisson distribution with parameter $\delta(t-s)$,*
4. *almost surely, J_t has jumps of size one.*

The intensity parameter δ governs the arrival rate of jump events and is assumed to be constant over time.

Definition 1.11. ***Stochastic Termination Time*** *For each user i, define the termination time τ_i^* as the first jump time of an idiosyncratic Poisson process $(J_{i,t})_{t\geq 0}$ with intensity δ:*

$$\tau_i^* := \inf\{t \geq 0 : J_{i,t} = 1\}. \tag{1.22}$$

The random time τ_i^* represents the moment at which the access right associated with the token held by user i is terminated. Termination is interpreted as an absorbing event: once it occurs, access rights permanently cease to generate utility.

Remark 1.12. The termination time τ_i^* is exponentially distributed with parameter δ, i.e.

$$\mathbb{P}(\tau_i^* > t) = e^{-\delta t}, \qquad t \geq 0.$$

Moreover, conditional on survival up to time t, the expected remaining lifetime satisfies $\mathbb{E}_t[\tau_i^* - t] = \frac{1}{\delta}$.

The parameter δ can be read as the instantaneous intensity of termination. A low value of δ describes an access right that is expected to survive for a long time, while a high value of δ describes a fragile access relationship. Relative to deterministic maturity, the holder never knows the exact expiration date in advance. What is priced instead is the probability that the token remains active long enough to deliver future utility.

The exponential survival structure implied by the Poisson termination time will enter the valuation problem through an effective killing rate in the risk-neutral pricing equation.

Active Access Rights and Stochastic Token Maturity

We now characterize active access rights in the presence of stochastic termination. Throughout the analysis, we normalize the issuance date of access tokens to $\tau = 0$.

Let $q_{i,0}$ denote the number of tokens purchased by user i at issuance.

Under stochastic termination, access rights remain active until the random termination time τ_i^*. Let

$$\chi_{i,t} := \mathbf{1}_{\{t<\tau_i^*\}} \tag{1.23}$$

The number of active access rights held by user i at time t is therefore

$$q_{i,t} = q_{i,0}\chi_{i,t}. \tag{1.24}$$

Relative to the deterministic maturity case, the only modification concerns the activation mechanism. Accordingly, the definitions of active access rights, their market value, instantaneous utility flows and the mass of users whose access rights remain active at time t coincide with those in the deterministic framework (see equations (1.3)-(1.4)), with the deterministic activity indicator replaced by $\chi_{i,t}$. Relative to CWL, this is the second maturity extension considered in the chapter: access is still finite in an economic sense, but finiteness operates through random termination rather than through a known calendar date.

Token Valuation with Stochastic Termination

Valuation is conducted using the stochastic discount factor Λ_t introduced in Section 1.3. As in the deterministic framework, we work under the equivalent risk-neutral measure $\mathbb{Q}$ induced by Λ_t, so that discounted asset prices are martingales.

Under stochastic termination, the pricing kernel remains unchanged (see (1.5)). The only modification arises from the survival structure of access rights, which affects the expected stream of future utility flows.

Let $y_{i,t}$ denote agent i's undiscounted cumulative payoff from platform participation up to time t, with increments driven by the instantaneous flow payoff $dv_{i,t}$ (see (1.10)).

The token price at time t is given by the expected discounted value of future utility flows that accrue only while access rights remain active:

$$P_t = \mathbb{E}_t^{\mathbb{Q}} \left[\int_t^{\infty} e^{-r(s-t)} \mathbf{1}_{\{s<\tau_i^*\}} \, dy_{i,s} \right]. \tag{1.25}$$

Using the change of variables $s = t + u$, this expression can be rewritten as

$$P_t = \mathbb{E}_t^{\mathbb{Q}} \left[\int_0^{\infty} e^{-ru} \mathbf{1}_{\{t+u<\tau_i^*\}} \, dy_{i,t+u} \right].$$

Under the assumption of a constant termination intensity δ, the survival probability satisfies

$$\mathbb{E}_t^{\mathbb{Q}} \left[\mathbf{1}_{\{t+u<\tau_i^*\}} \right] = \mathbb{P}(\tau_i^* > t + u \mid \mathcal{F}_t) = e^{-\delta u}.$$

Therefore, stochastic termination effectively modifies the discounting of future payoffs, and the token price can be expressed as

$$P_t = \mathbb{E}_t^{\mathbb{Q}} \left[\int_0^{\infty} e^{-(r+\delta)u} \, dy_{i,t+u} \right]. \tag{1.26}$$

Relative to the deterministic maturity framework, stochastic termination replaces the fixed remaining lifetime by an exponential survival adjustment. Future utility flows are discounted not only by the risk-free rate, but also by the termination intensity, reflecting the possibility that access rights may cease before utility is realized.

Markov Equilibrium with Stochastic Termination

We now characterize the Markov equilibrium in the presence of stochastic access termination. Relative to the deterministic maturity framework, the structure of agents' optimization, participation and market clearing remains unchanged.

Under stochastic termination, aggregate platform activity A_t is the only state variable of the economy as in the perpetual CWL benchmark ([6]). Termination risk enters the pricing equation through an effective killing rate, rather than through an additional state dimension.

Agents take prices as given and choose active token holdings to maximize instantaneous net benefits.

In particular, conditional on participation, optimal active holdings satisfy

$$x^*_{i,t} = N_t A_t e^{u_i} \left(\frac{1-\alpha}{r+\delta-\mu^P(A_t)} \right)^{\frac{1}{\alpha}},$$

where μ^P_t denotes the risk-neutral drift of the token price.

An agent participates if and only if the maximized instantaneous net benefit from holding access rights is non-negative. This defines a cutoff type $\underline{u}_t$ such that agents with $u_i \geq \underline{u}_t$ participate in the platform.

Imposing the indifference condition at the cutoff yields

$$\underline{u}_t = -\ln N_t + \ln\left(\frac{\phi}{A_t \alpha}\right) - \frac{1-\alpha}{\alpha} \ln\left(\frac{1-\alpha}{r+\delta-\mu^P(A_t)}\right). \tag{1.27}$$

Given the distribution of user types $G(\cdot)$, the endogenous user base is therefore

$$N_t = 1 - G(\underline{u}_t). \tag{1.28}$$

Market clearing continues to require that the aggregate value of active token holdings equals the market value of the outstanding pass supply. Aggregating individual demand over participating agents implies the same static equilibrium relation between prices, aggregates, and expected appreciation:

$$P(A_t) = \frac{N_t S_t A_t}{M} \left(\frac{1-\alpha}{r+\delta-\mu^P(A_t)} \right)^{\frac{1}{\alpha}}, \tag{1.29}$$

where S_t captures the distribution of transaction needs among participating users (see (1.12)).

The Markov structure of the equilibrium is therefore preserved.

While the static equilibrium relation retains the same functional form, the equilibrium drift $\mu^P(A_t)$ must now be consistent with a pricing equation that incorporates stochastic termination. This distinction will be reflected in the risk-neutral dynamics of the equilibrium price and in the resulting pricing partial differential equation.

Token Price Dynamics with Stochastic Termination

The equilibrium token price depends only on aggregate platform activity and admits a Markov representation of the form

$$P_t = P(A_t).$$

Given (1.6) and applying Itô's lemma to the pricing function $P(A_t)$ yields

$$\begin{aligned} dP_t &= P_A(A_t)\, dA_t + \frac{1}{2} P_{AA}(A_t)\, (dA_t)^2 \\ &= \left(P_A A_t \mu^A + \frac{1}{2} P_{AA} A_t^2 (\sigma^A)^2 \right) dt + P_A A_t \sigma^A \, dZ_t. \end{aligned} \tag{1.30}$$

Comparing this expression with the generic risk-neutral price dynamics (1.9) the implied drift and volatility are given by

$$\mu_t^P P(A_t) = P_A A_t \mu^A + \frac{1}{2} P_{AA} A_t^2 (\sigma^A)^2, \qquad \sigma_t^P P(A_t) = P_A A_t \sigma^A. \tag{1.31}$$

Relative to the deterministic maturity framework, the maturity-derivative term $-P_m$ is absent. This reflects the fact that access expiration no longer occurs through the mechanical passage of time, but through stochastic termination. As shown in the previous section, termination risk enters the valuation problem through an effective killing rate that modifies the intertemporal discounting of payoffs, rather than through an additional state variable.

As a consequence, while the pricing equation reverts to a one-dimensional Markov structure, stochastic termination continues to affect equilibrium prices through the equilibrium drift, which must be consistent with the modified valuation of future utility flows.

Absence of Arbitrage and Equilibrium Consistency under Stochastic Termination

We conclude the stochastic analysis by verifying that the equilibrium token price derived above is consistent with the absence of arbitrage.

In equilibrium, the absence of arbitrage requires that the discounted token price augmented by its cumulative payoff stream is a martingale. In the presence of stochastic termination of access rights, the standard no-arbitrage condition must be adjusted to account for the expected loss of continuation value. Because access rights may terminate at any instant with constant intensity δ, token holders face an

additional expected capital loss proportional to the current token value.

Accordingly, the absence-of-arbitrage condition becomes

$$(r+\delta)P_t = \mu_t^P P_t + D_t, \tag{1.32}$$

or, equivalently,

$$rP_t = \mu_t^P P_t + D_t - \delta P_t. \tag{1.33}$$

The term δP_t captures the expected loss in token value due to stochastic termination. Importantly, this effect does not arise from the local dynamics of the token price, but from valuation. While the token price continues to evolve as a diffusion driven by aggregate platform activity, the presence of termination risk shortens the effective horizon over which future payoffs can be realized.

In this sense, stochastic termination enters the pricing problem through an effective killing rate that modifies the intertemporal discounting of payoffs, rather than through an additional state variable. This mechanism is directly analogous to default intensity in reduced-form credit risk models. As in the deterministic case, aggregate dividends reflect the marginal contribution of access rights to platform surplus, evaluated at optimal demand. Using the optimal holdings and the envelope condition, the aggregate flow dividend takes the same functional form as before:

$$D_t = (1-\alpha)\left(\frac{N_t S_t A_t}{P_t M}\right)^{\alpha} P_t.$$

Substituting the risk-neutral drift obtained from Itô's lemma applied to $P(A_t)$ (1.30) yields the equilibrium pricing equation:

$$rP(A_t) = P_A A_t \mu^A + \frac{1}{2} P_{AA} A_t^2 (\sigma^A)^2 - \delta P(A_t) + (1-\alpha)\left(\frac{N(A_t)S(A_t)A_t}{P(A_t)M}\right)^{\alpha} P(A_t). \tag{1.34}$$

Relative to the deterministic maturity framework, the pricing equation no longer features a maturity-derivative term. Instead, stochastic termination enters valuation through the killing term $-\delta P(A_t)$. This term captures the expected loss of future access value due to random termination and plays a role analogous to maturity decay in the deterministic setting.

The equilibrium pricing equation is now an ordinary differential equation in aggregate platform activity. Nevertheless, stochastic termination continues to discipline token valuation by limiting the effective capitalization horizon. Expected platform growth and network effects are capitalized into prices only to the extent that access rights are expected to survive.

Together with the endogenous adoption conditions determining N_t and S_t, the pricing equation ensures that the equilibrium token price is internally consistent with agents' optimal behavior and free of arbitrage opportunities.

Proposition 1.13. ***Tokenized Equilibrium with Stochastic Termination*** *Under standard regularity conditions, a Markov equilibrium with stochastic access termination is characterized by the following objects:*

1. *An equilibrium token price $P(A_t)$ solving the equilibrium pricing equation with killing term $-\delta P(A_t)$, subject to the boundary condition $P(0) = 0$;*

2. *A participation cutoff $\underline{u}_t = \underline{u}(A_t)$, such that agents with $u_i \geq \underline{u}_t$ participate, and an endogenous user base given by $N_t = 1 - G(\underline{u}_t)$;*

3. *Optimal active token holdings $x^*(A_t)$ satisfying the intra-temporal optimality condition equating marginal transaction benefits to the effective carry cost $r + \delta - \mu^P(A_t)$.*

Together, these objects jointly determine equilibrium prices, participation, and token demand in the tokenized economy with stochastic termination.

As in the deterministic maturity and perpetual-access benchmarks, tokens capitalize expected future platform growth into current prices, thereby reducing the effective carry cost of participation and fostering user adoption. However, stochastic termination fundamentally modifies the capitalization mechanism. Because access rights may terminate at any time with constant intensity, expected productivity growth is discounted not only by time preference and risk, but also by survival risk.

As a result, the feedback loop between token prices and user adoption remains operative but is disciplined by the termination intensity. Unlike deterministic maturity, which truncates capitalization at a known horizon, stochastic termination limits token valuation through an endogenous effective discounting of future utility flows. The equilibrium token price therefore reflects expected platform growth and network effects, adjusted for the probability that access rights cease before utility is realized.

1.6 Non-Tokenized Platform with Stochastic Access Termination

Stochastic User Participation and Utility Flow (Non-Tokenized)

We characterize user participation and utility generation in the non-tokenized economy under stochastic exit. Normalizing the subscription issuance date to $\tau = 0$, access rights are governed by an idiosyncratic Poisson process with intensity δ, with termination time τ_i^* that governs the duration of platform participation and survival indicator $\chi_{i,t}$ (see (1.23)).

We denote $q_{i,0}$ the quantity of access rights acquired by user i at issuance. The number of active access rights held by user i at time t is therefore $q_{i,t}$ defined as in (1.24)

In the non-tokenized economy, access rights are not represented by a tradable asset and do not carry a market price. Accordingly, effective access coincides directly with active access rights as in the deterministic non-tokenized benchmark. Thus, unlike the tokenized framework, no valuation transformation is involved, and access intensity reflects purely physical participation rather than the market value of a traded claim. The instantaneous utility increment from platform participation coincides with the previous frameworks (see (1.4)), with stochastic termination affecting participation exclusively through the survival indicator.

Stochastic exit truncates the utility stream at a random time but does not otherwise affect the structure of instantaneous payoffs. Aggregate platform activity A_t evolves stochastically as specified in the model setup (see (1.1)) and affects utility directly through network interactions.

In the absence of tokenization, no asset pricing problem arises. Accordingly, the stochastic discount factor plays no operational role in this environment, and aggregate risk is not priced but only affects welfare through utility flows.

Individual Payoff Structure and Optimization under Stochastic Exit

In the non-tokenized economy with stochastic exit, access rights are not represented by a traded asset. As a result, agents do not face capital gains or losses, and the only relevant cost of access intensity is the static opportunity cost r. Stochastic exit truncates participation at a random time but does not otherwise affect instantaneous incentives (see (1.17)).

Conditional on remaining active, agents choose access intensity to maximize instantaneous utility (see (1.18)). As in the deterministic non-tokenized economy, the optimization problem is static and individual decisions are taken as given with respect to aggregate variables.

Accordingly, the optimal access choice, participation threshold, and endogenous user base coincide with those derived in the deterministic non-tokenized benchmark (see (1.19)-(1.20)-(1.21)). Stochastic exit affects participation exclusively through the survival indicator $\chi_{i,t}$ and does not alter optimal access intensity or cutoff conditions.

Proposition 1.14. ***Non-Tokenized Equilibrium under Stochastic Exit*** *In the non-tokenized economy with stochastic exit, a Markov equilibrium with state variable A_t is characterized by a participation threshold $\underline{u}_t^{NT}$ and an endogenous user base N_t^{NT} satisfying*

$$\underline{u}_t^{NT} = -\ln(N_t^{NT}) + \ln\left(\frac{\phi}{A_t\alpha}\right) - \left(\frac{1-\alpha}{\alpha}\right)\ln\left(\frac{1-\alpha}{r}\right), \tag{1.35}$$

$$N_t^{NT} = 1 - G(\underline{u}_t^{NT}). \tag{1.36}$$

Moreover, if the cross-sectional distribution of agent types $G(u)$ exhibits an increasing hazard rate, the equilibrium user base N_t^{NT} is increasing in aggregate platform activity A_t.

1.7 Numerical Implementation and Equilibrium Implications

This section studies the quantitative implications of the equilibrium structures developed in Sections 1.2–1.5. The goal is not merely to illustrate the model numerically, but to assess how the introduction of finite access duration reshapes token valuation, expected appreciation, user adoption, and platform stability relative to the perpetual benchmark of Cong, Li and Wang (CWL). In particular, the numerical analysis is designed to evaluate whether the capitalization mechanism emphasized in the perpetual-access economy survives once access rights are made economically realistic.

The numerical exercise should therefore be read as a direct test of the chapter's main innovation. Relative to CWL, the only conceptual change is the introduction of finite access duration, either as a known expiration date or as a stochastic termination event. Differences in prices, expected appreciation, and adoption can thus be interpreted as the economic effect of maturity.

The comparison is conducted across three environments. The first is the perpetual benchmark, corresponding to the original CWL economy, in which access rights are infinitely lived. The second is the deterministic finite-maturity specification developed in Section 1.2, where tokenized access expires after a known horizon and valuation is disciplined by the maturity-decay term $-P_m$ in the pricing equation. The third is the stochastic termination specification of Section 1.4, in which access duration is random and expiration risk enters valuation through the killing term $-\delta P(A)$. For the deterministic case, the numerical exercise considers three representative maturity configurations, corresponding approximately to short, intermediate, and long access horizons. For the stochastic case, three alternative hazard rates are examined, ranging from low to high termination intensity.

The numerical evidence reveals a sharp asymmetry between deterministic and stochastic expiration. Deterministic maturity generates a substantial downward shift in token values, weakens expected appreciation, delays adoption, and increases implied adoption volatility, especially when the remaining maturity is short. By contrast, stochastic termination preserves the qualitative structure of the perpetual benchmark to a remarkable extent: token values remain high, expected appreciation stays positive, adoption curves remain close to CWL, and adoption volatility is only mildly affected. Quantitatively, this confirms the theoretical message of Chapter 1: introducing realistic access duration does not mechanically eliminate the tokenization mechanism, but the way duration is modeled is economically decisive.

All numerical results and graphical representations are obtained through a dedicated Python implementation of the equilibrium system. The computational procedure, including the numerical solution methods, parameter choices, and robustness checks, is described in detail in Appendix A (3.6).

1.7.1 Price Dynamics over Adoption Stages

A first useful perspective is to study equilibrium token prices as functions of the endogenous adoption level $N(A)$. This representation makes it possible to compare how much valuation is generated at comparable stages of platform diffusion, abstracting from the fact that different models reach those stages at different productivity levels.

The results show that the perpetual CWL benchmark always delivers the highest price schedule. This is the natural consequence of perpetual capitalization: in the absence of expiration, token value reflects an infinite stream of expected utility flows. Once access becomes finite, the capitalization horizon contracts and the equilibrium price falls. The reduction is particularly severe in the deterministic specification. Short deterministic maturities generate the lowest price schedule across all adoption

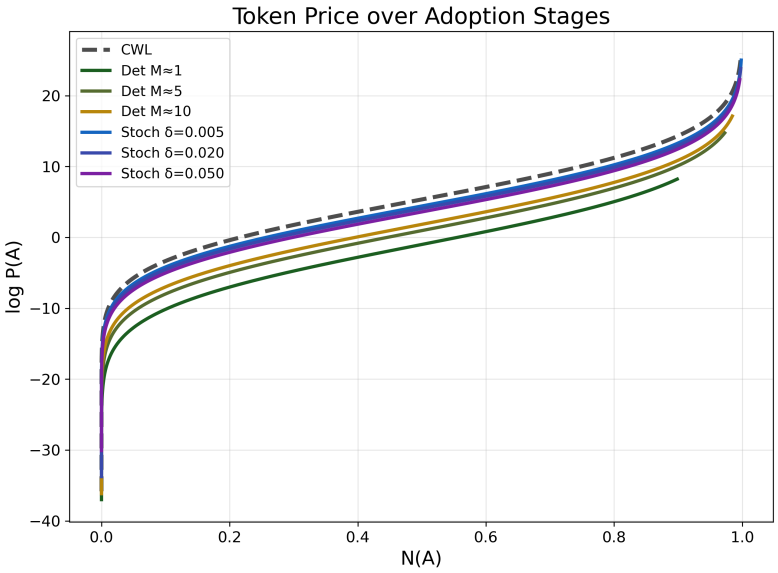


Figure 1.1: Log token price as a function of adoption across models. The perpetual CWL benchmark delivers the highest valuation at all adoption levels, reflecting full capitalization of future utility flows. Deterministic maturity introduces a sharp downward shift, especially for short horizons, while stochastic termination generates more moderate discounts and preserves a price schedule close to the benchmark.

stages, while longer maturities partially recover valuation but remain substantially below the perpetual benchmark.

The stochastic specification behaves very differently. Although token prices remain below CWL for all hazard rates, the gap is comparatively modest. Even when termination risk is non-negligible, the stochastic curves remain close to the perpetual benchmark and dominate the deterministic ones. This confirms that a random access horizon does not destroy the capitalization channel. Instead, it acts as a survival-based discounting mechanism: future utility flows are attenuated, but not abruptly truncated.

From an economic standpoint, the figure highlights the difference between two conceptually distinct forms of finite duration. Deterministic maturity introduces a hard truncation of the future, so that the token can capitalize platform growth only over a strictly bounded horizon. Stochastic termination instead induces a soft discounting of continuation value, because the token retains the possibility of surviving into future productive states.

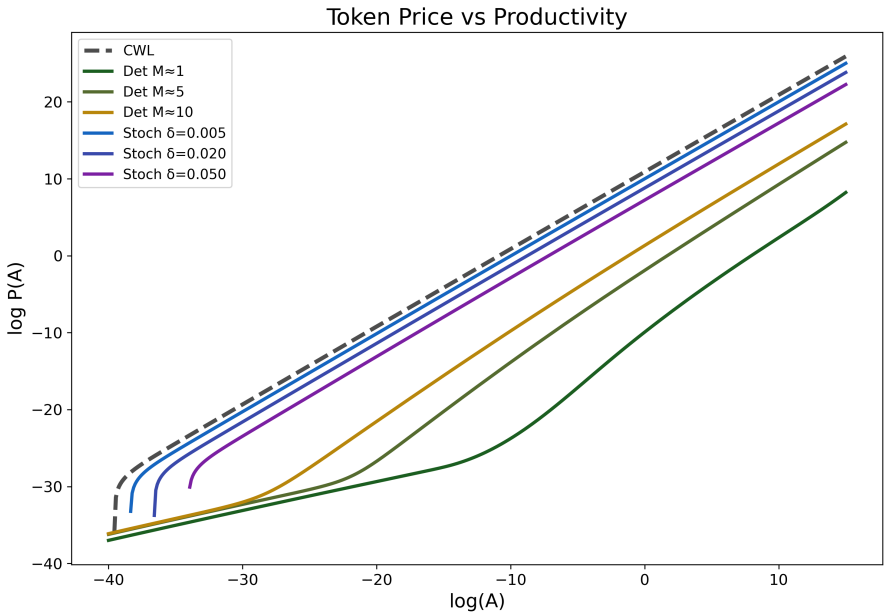


Figure 1.2: Log token price as a function of platform activity. Prices increase monotonically with activity in all specifications, consistent with the theoretical predictions. Finite maturity primarily reduces the level of valuation rather than altering its slope. Deterministic expiration produces the largest downward shift, while stochastic termination preserves a pricing schedule close to the perpetual benchmark.

1.7.2 Price Scaling with Platform Activity

The next step is to study token prices directly as functions of platform activity. Since the equilibrium pricing equations are written in terms of aggregate activity, this representation is the most immediate numerical counterpart of the theoretical objects characterized in Propositions 1.6 and 1.13.

Several features emerge clearly. First, for all specifications, token prices are increasing in platform activity. This is fully consistent with the equilibrium relations derived in the theory: higher platform activity raises the productivity of access rights, strengthens aggregate participation incentives, and increases the capitalization of future platform surplus into current token values. Second, the main quantitative effect of maturity is a downward shift in the level of the pricing function rather than a change in its overall slope. In particular, the stochastic specifications preserve a pricing schedule that is nearly parallel to the CWL benchmark, whereas deterministic maturity shifts the entire function downward, especially for short horizons.

This pattern is important because it shows that finite maturity does not overturn the fundamental monotonic relation between platform productivity and token valuation established in the theoretical chapter. Instead, maturity acts primarily as a valuation discipline. In other words, token prices continue to respond positively to



Figure 1.3: Expected token appreciation μ^P as a function of platform activity. Stochastic termination preserves positive expected returns and converges toward the perpetual benchmark at high productivity levels. Deterministic maturity significantly weakens expected appreciation and, for short horizons, generates negative drift over a large region, undermining participation incentives.

improvements in platform fundamentals, but the extent to which future improvements are capitalized depends critically on the effective horizon of access rights.

The distinction between deterministic and stochastic duration remains decisive here as well. Under deterministic expiration, the pricing function incorporates a direct maturity-decay component through the term $-P_m$, which mechanically erodes value as the remaining horizon shortens. Under stochastic termination, the same disciplining effect operates through the killing term $-\delta P(A)$, but does so in a smoother way. As a result, stochastic termination reduces valuation without fundamentally altering the equilibrium pricing structure inherited from the perpetual benchmark.

1.7.3 Expected Appreciation and the Effective Carry Cost

The core mechanism of the CWL framework is not driven by token prices per se, but by the effect of expected token appreciation on the effective carry cost of participation. For this reason, the behavior of the risk-neutral drift μ^P is particularly informative. In the perpetual and deterministic specifications, a positive μ^P lowers the effective carry cost $r - \mu^P$ and therefore supports adoption incentives. In the stochastic specification, the relevant carry becomes $r + \delta - \mu^P$, but the same logic continues to apply: higher expected appreciation mitigates the effective cost of holding the token and sustains participation.

The numerical evidence reveals a stark difference across specifications. In the stochastic models, expected token appreciation remains positive throughout the relevant range of productivity and converges toward the perpetual benchmark at high productivity levels. Indeed, for larger hazard rates, expected returns are even higher

at low productivity levels before gradually declining. This pattern is economically intuitive. When the platform is still in an early stage of development, the possibility of future adoption growth remains substantial, so expected appreciation continues to compensate users for holding the token despite the presence of survival risk.

By contrast, the deterministic specification produces a much weaker, and in the short-maturity case sharply negative, expected appreciation profile. For very short maturities, the drift becomes strongly negative over a wide region of the state space. This implies that the effective carry cost $r - \mu^P$ rises dramatically rather than falling, so tokenization ceases to enhance participation and may even become a value-destroying access technology within this specification. Longer deterministic horizons mitigate this effect, but the expected appreciation remains significantly below both the stochastic specifications and the CWL benchmark.

This result is central for the economic interpretation of Chapter 1. It shows that the mere presence of finite duration is not what threatens tokenization. What matters is whether access duration truncates future value mechanically or probabilistically. Deterministic expiration can overturn the sign of expected appreciation and thereby undermine the very mechanism that justifies token issuance. Stochastic termination, instead, preserves positive expected appreciation and therefore leaves the logic of token-based capitalization largely intact, reflecting the fact that it modifies both the pricing equation and the effective carry cost, ensuring internal consistency between valuation and user incentives.

1.7.4 Adoption Dynamics and Threshold Effects

The effect of maturity on participation is most directly visible in the adoption curves. Since user participation is endogenous and pinned down by the cutoff condition derived in the theoretical section, the numerical adoption schedules summarize how each specification transforms the mapping from platform productivity into platform diffusion.

The perpetual CWL benchmark exhibits the familiar S-shaped adoption pattern: participation starts slowly, accelerates in the intermediate region, and eventually converges toward full adoption. The stochastic extensions remain very close to this benchmark. Low hazard rates produce adoption curves that are almost indistinguishable from CWL, while higher hazard rates delay adoption only moderately. Hence, even when access is not perpetual, the tokenized platform continues to generate strong diffusion incentives as long as expiration is modeled stochastically.

The deterministic case is again much more fragile. All deterministic maturities shift the adoption curve to the right, implying that much higher productivity is

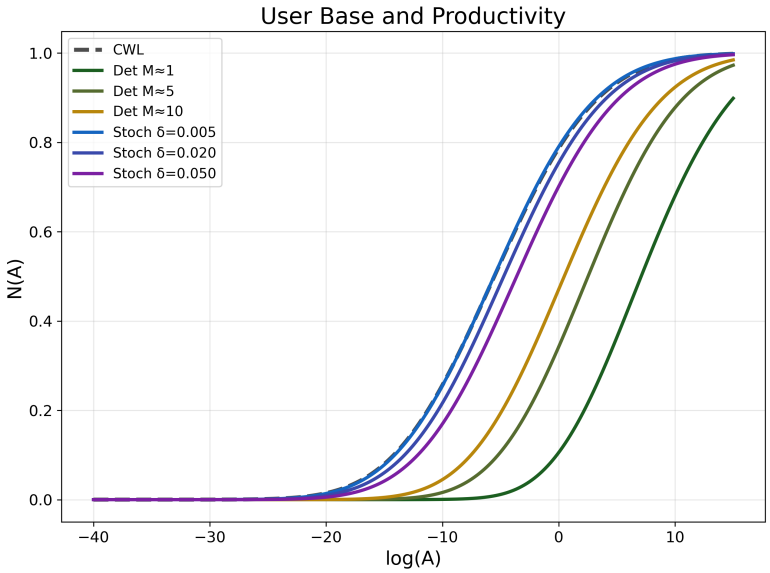


Figure 1.4: Adoption as a function of platform activity across models. The CWL benchmark exhibits the standard S-shaped diffusion pattern. Stochastic termination produces adoption curves close to the benchmark, while deterministic maturity shifts adoption to the right, requiring substantially higher productivity to sustain comparable participation levels.

required to sustain the same level of user participation. This effect is especially severe for short maturities, where adoption remains negligible for a very large range of productivity values and only rises once platform activity becomes extremely high. Intermediate and long maturities perform better, but still remain far below both the stochastic specifications and the CWL benchmark.

The threshold interpretation is made explicit in Table 1.1, which reports the productivity levels required to reach intermediate and high adoption.

Model	Parameter	$\log A$ at $N = 0.50$	$\log A$ at $N = 0.90$
CWL	–	-5.539	3.426
Deterministic	$m \approx 1.06$	7.088	15.049
Deterministic	$m \approx 4.99$	2.523	10.769
Deterministic	$m \approx 9.76$	0.470	8.997
Stochastic	$\delta = 0.005$	-5.565	3.309
Stochastic	$\delta = 0.020$	-4.727	4.142
Stochastic	$\delta = 0.050$	-3.619	5.240

Table 1.1: Adoption thresholds across models. The reported values are the productivity levels required to reach intermediate and high adoption.

The table makes the economic magnitude of the effect transparent. In the CWL benchmark, intermediate adoption is reached at a relatively low activity level, while

deterministic maturity shifts the required activity sharply upward. For the shortest deterministic horizon, reaching $N = 0.50$ requires $\log A = 7.088$, compared with -5.539 in the perpetual benchmark. This large difference means that known expiration does not merely reduce the token price; it changes the scale of platform fundamentals required for the token to support broad participation. By contrast, stochastic termination keeps the thresholds much closer to CWL. Even when the hazard rate increases, the required activity levels move gradually rather than collapsing the adoption mechanism altogether.

Deterministic maturity creates a severe adoption barrier because users know that access value will disappear after a finite horizon, reducing both equilibrium prices and expected appreciation. Stochastic termination also limits duration, but by preserving continuation value in expectation it allows tokenization to retain most of its adoption-enhancing power.

1.7.5 Comparative Gaps Relative to the Perpetual Benchmark

To make the quantitative deviations from CWL more transparent, it is useful to examine directly the differences in adoption and pricing between each finite-duration model and the perpetual benchmark.

The adoption-gap figure shows that all finite-duration specifications imply lower adoption than CWL, but the magnitude of the gap varies enormously across models. The deterministic specification generates large negative deviations, especially around the intermediate region of platform activity where the adoption curve is steepest. This is precisely the region in which the capitalization channel matters most, because adoption responds strongly to incentives and future platform growth is still economically important. By contrast, the stochastic deviations are small and remain close to zero throughout most of the state space.

The same message emerges even more strongly for prices. Deterministic maturity generates very large negative price gaps relative to CWL, with the short-maturity case exhibiting the most dramatic collapse. Stochastic termination instead produces moderate and relatively stable valuation discounts. Hence, the numerical evidence clearly indicates that finite duration does not have a uniform effect on the token economy. Rather, the economic consequences depend critically on whether finite access is represented by a deterministic cutoff or by probabilistic survival.

This distinction is economically crucial for the broader interpretation of the model. If one looked only at the deterministic specification, one might conclude that introducing realistic maturity destroys the token mechanism. The stochastic

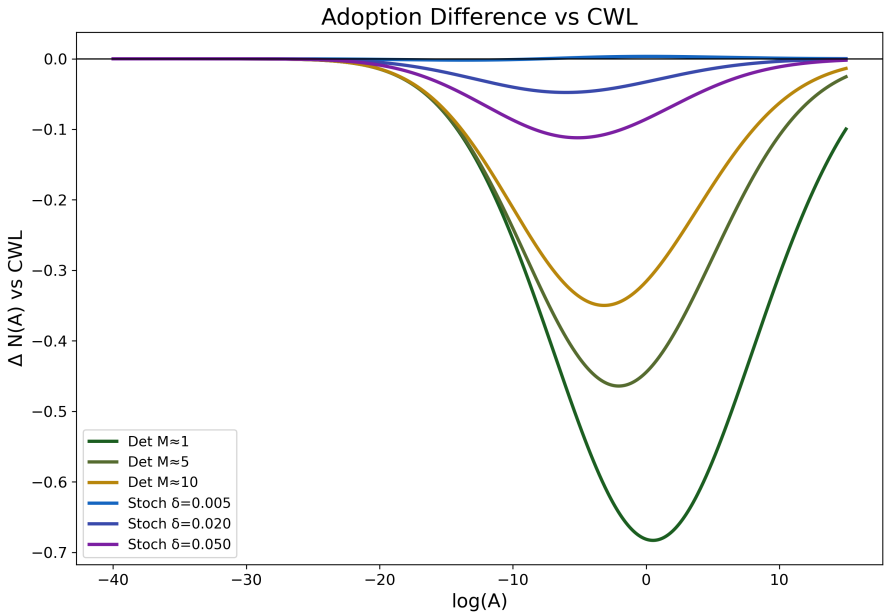


Figure 1.5: Difference in adoption relative to the CWL benchmark. Deterministic maturity generates large negative deviations, especially in the intermediate activity region where adoption is most responsive to incentives. Stochastic termination produces only mild deviations, remaining close to the benchmark across the state space.

results show that this conclusion would be too strong. What fails under deterministic maturity is not tokenization per se, but the possibility of capitalizing future platform growth once the horizon is known to end with certainty.

1.7.6 Pricing Summary at Intermediate and High Productivity Levels

The previous figures illustrate global patterns across the full state space. A more local perspective is provided by Table 1.2, which summarizes token values and expected appreciation at two representative productivity checkpoints.

At $\log A = 0$, corresponding to an intermediate region of platform activity, the CWL benchmark delivers a high valuation together with a positive expected return. The low-hazard stochastic specification remains very close to this outcome, while higher hazard rates reduce prices but still preserve positive expected appreciation. The deterministic specification, instead, displays much lower prices and, for short and intermediate maturities, clearly negative expected returns. This confirms that the distortion created by deterministic expiration is already substantial well before the platform reaches the high-productivity region.

At $\log A = 10$, corresponding to a mature platform environment, the ranking

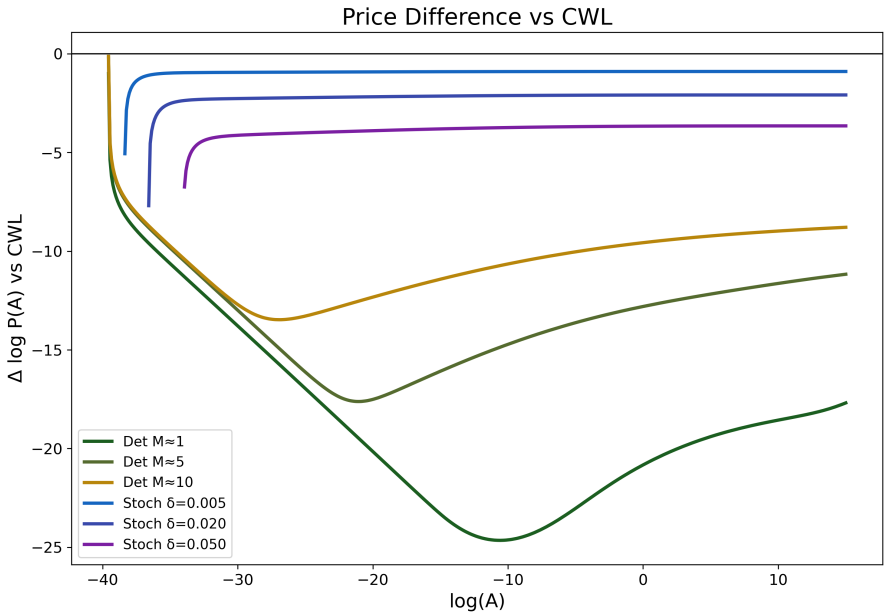


Figure 1.6: Difference in log token prices relative to the CWL benchmark. Deterministic maturity leads to substantial valuation losses, particularly for short horizons. Stochastic termination instead produces moderate and stable price discounts, consistent with a survival-based reduction in continuation value rather than a hard truncation.

of models remains unchanged. Stochastic prices continue to track the perpetual benchmark relatively closely, whereas deterministic prices remain much lower. Expected returns converge across the stochastic and perpetual economies, but remain negative in the deterministic models. The implication is clear: even when platform fundamentals become strong, deterministic truncation continues to weigh heavily on valuation and does not disappear asymptotically over the relevant productivity range.

These local statistics complement the global figures by showing that the deterministic specification is not problematic only in a narrow transitional region. Rather, the pricing drag associated with known expiration is persistent and economically meaningful across the state space. Stochastic duration, by contrast, behaves as a disciplined but robust extension of the perpetual benchmark.

1.7.7 Implied adoption volatility and Platform Stability

A final dimension of interest concerns stability. The CWL benchmark predicts that tokenization not only supports adoption, but also smooths user-base dynamics through the effect of expected appreciation on participation incentives. The question is whether this stabilizing logic survives under finite access duration. In the numerical

Model	Param.	$\log P(0)$	$\mu^P(0)$	$\log P(10)$	$\mu^P(10)$
CWL	–	10.904	0.022	20.909	0.020
Deterministic	$m \approx 1.06$	-9.924	-7.002	2.349	-6.184
Deterministic	$m \approx 4.99$	-1.900	-0.854	9.272	-0.793
Deterministic	$m \approx 9.76$	1.332	-0.327	11.926	-0.336
Stochastic	$\delta = 0.005$	10.003	0.022	20.008	0.020
Stochastic	$\delta = 0.020$	8.810	0.024	18.819	0.020
Stochastic	$\delta = 0.050$	7.233	0.028	17.252	0.021

Table 1.2: Pricing summary across models at two productivity checkpoints. The first checkpoint captures an intermediate region, while the second captures a high-productivity region.

implementation, the reported volatility measure should be interpreted as an implied local volatility of adoption, reflecting the sensitivity of the user base to underlying productivity shocks rather than a full stochastic simulation of user dynamics.

The figure shows that the volatility profile remains hump-shaped across all specifications, with volatility low at very low and very high adoption levels and peaking in the intermediate region. This is consistent with the general intuition that adoption is most sensitive to shocks when the platform is neither almost empty nor almost fully diffused.

The quantitative differences across models are again informative. Deterministic maturity increases implied adoption volatility relative to CWL, with the effect strongest for short maturities. This is economically coherent: once the horizon is short and expected appreciation is weak or negative, user participation becomes more fragile and more sensitive to changes in productivity. Stochastic termination, instead, leaves the volatility profile almost unchanged relative to the perpetual benchmark. Even when hazard rates are higher, the stochastic curves remain very close to CWL.

This finding reinforces the broader interpretation of the numerical results. Deterministic expiration not only depresses prices and slows adoption, but also makes the tokenized platform less stable. Stochastic termination preserves both the incentive structure and the stabilizing properties of tokenization much more effectively.

1.7.8 Economic Interpretation

Taken together, the numerical results deliver a coherent and economically sharp message. The perpetual-access benchmark of CWL is not robust to all forms of finite duration, but neither is it invalidated by realistic access constraints in general. The critical distinction lies in the temporal structure of expiration.

When access expires deterministically, the token loses much of its ability to capital-

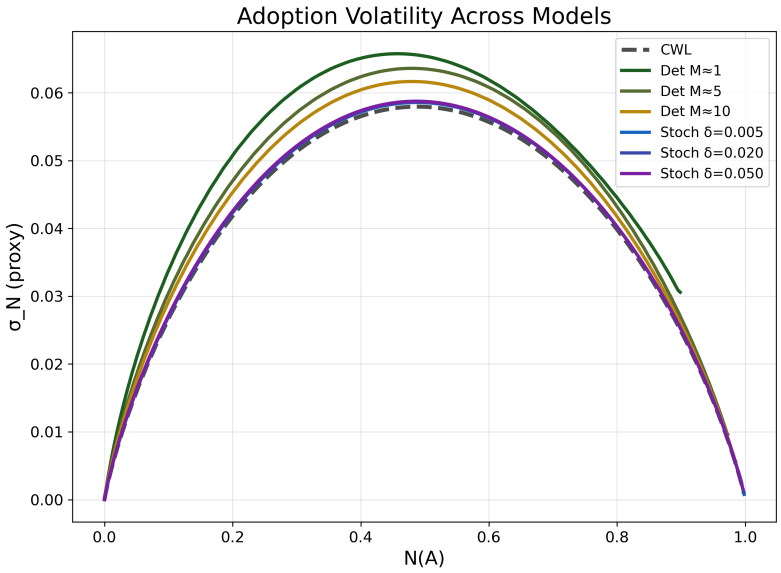


Figure 1.7: Implied adoption volatility as a function of platform activity. Volatility is hump-shaped across all specifications, peaking at intermediate adoption levels. Deterministic maturity increases volatility, reflecting more fragile participation incentives, while stochastic termination preserves a profile close to the perpetual benchmark.

ize future platform growth. The pricing equation incorporates a direct maturity-decay term, expected appreciation weakens sharply and may turn negative, adoption is delayed, and volatility rises. In this regime, tokenization becomes much less attractive as a participation technology and may cease to provide a strong participation advantage relative to non-tokenized access.

When access is instead subject to stochastic termination, the outcome is fundamentally different. Termination risk reduces valuation by shortening the effective horizon, but does so through probabilistic discounting rather than hard truncation. Continuation value survives, expected appreciation remains positive, adoption stays close to the perpetual benchmark, and platform stability is largely preserved. Hence, the token mechanism is not destroyed by finite duration as such. It is destroyed only when finite duration is modeled as a deterministic and perfectly anticipated endpoint.

This conclusion is important for the interpretation of tokenized access rights in real-world digital platforms. Many economically relevant environments are not well approximated by a common deterministic expiration date. Access relationships are often uncertain, renewable, cancellable, or subject to idiosyncratic termination. In such contexts, the stochastic specification provides a more realistic and more economically favorable representation of tokenized participation. Conversely, the de-

terministic model should be understood as a stringent stress test of the capitalization mechanism rather than as its most plausible empirical benchmark.

From the perspective of the thesis as a whole, these results provide the necessary bridge to the next chapter. They show that finite maturity can either severely weaken or largely preserve token attractiveness depending on how duration risk is modeled. This naturally motivates the subsequent comparison between tokenized access and an external asset, where the deterministic framework can be used as the harshest benchmark against which token dominance is tested.

Chapter 2

Token Valuation Beyond Isolation

The analysis developed in Chapter 1 establishes that the core capitalization mechanism underlying token valuation remains operative even when access rights are no longer perpetual. By introducing finite maturity, both in deterministic and stochastic form, the model shows that while expiration reduces the effective valuation horizon, it does not eliminate the incentive channel through which expected platform growth is capitalized into token prices.

However, this analysis is conducted in isolation. The token is evaluated within the platform economy, without considering alternative investment opportunities available to agents. In practice, this represents a strong abstraction. Economic agents do not operate in closed systems: participation in a tokenized platform competes with a broad set of outside opportunities, ranging from traditional financial assets to other forms of capital allocation (Campbell and Viceira [4], Munk [13]).

This is the new assumption introduced in the chapter relative to the CWL benchmark and to Chapter 1. CWL shows how token prices can support adoption inside the platform, and Chapter 1 asks whether this mechanism survives finite access duration. Here the question is stricter: even if the token has positive value inside the platform, is that value high enough to dominate an external asset?

This raises a natural and economically relevant question: *under which conditions does the token remain competitive when compared to an external investment opportunity?*

To address this question, this chapter embeds the finite-horizon token developed in Chapter 1 into a competitive environment in which agents face a direct choice between holding the platform token and investing in an external asset (Munk and Rubtsov [14]). The latter is modeled as a standard financial asset, evolving independently of platform fundamentals and priced under the same risk-neutral measure. The analysis is conducted under both deterministic and stochastic specifications of access

termination, allowing us to compare a conservative benchmark with a more realistic environment in which expiration occurs probabilistically.

The comparison between the two assets is formalized through a dominance framework. In particular, we introduce a dominance function that captures the difference between the equilibrium token price and the value of the external asset. This formulation allows us to characterize the conditions under which the token delivers a higher valuation, and to identify a critical threshold in platform activity separating the regions of dominance (Dixit and Pindyck [8]).

The core contribution of the chapter is the analytical characterization of this dominance threshold. We show that, under the structural properties established in Chapter 1, the threshold exists, is unique, and depends systematically on the relevant state variables of the model, including the remaining maturity in the deterministic specification and the parameters governing termination risk in the stochastic case. This provides a clear mapping between platform fundamentals, time constraints, and external competitive pressure.

We then study the comparative statics of the threshold, highlighting how changes in external conditions and in the effective horizon affect the competitiveness of the token. The analysis reveals strong nonlinearities, particularly in the short-maturity region of the deterministic case, where small changes in maturity or external value can lead to large shifts in the required level of platform activity.

Overall, the chapter extends the valuation framework beyond a closed platform setting, showing that token competitiveness is jointly determined by internal dynamics and external market conditions. While the capitalization mechanism remains central, its effectiveness is shaped by the interaction between platform growth, finite horizons, and the availability of alternative investment opportunities. The resulting dominance threshold is therefore the main innovation of the chapter: it translates the internal token valuation model into a relative valuation problem against an outside opportunity.

2.1 Competitive Economic Environment

State Variables and Timing

In this chapter we move from the token in isolation setting of Chapter 1 to a competitive environment, where the token is evaluated side by side with an external investment opportunity. The objective of this subsection is simply to make precise what information is relevant for valuation at each point in time, and how the

deterministic finite maturity enters the state space.

Relative to CWL, the platform is no longer treated as the only relevant economic environment. Agents can compare platform participation with an outside asset, so token valuation must be strong not only in absolute terms, but also relative to the competing use of capital. This is why the state description includes both the platform variables that determine token value and the external benchmark used for comparison.

We work in continuous time on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$. As in Chapter 1, we adopt a risk-neutral pricing perspective: unless stated otherwise, expectations and present values are taken under the measure $\mathbb{Q}$, and the risk-free rate r is constant. This choice is not a modelling detail: it allows us to compare the token and the external asset on the same pricing footing, abstracting from preference-driven asset allocation motives.

A distinctive feature of the token studied here is its deterministic finite maturity T. Rather than indexing everything by calendar time alone, it is convenient to keep track of the remaining time to maturity, $m_t := T - t, t \in [0, T]$, that measures how much of the future access value is still alive and can be capitalized into the token price. Since maturity is deterministic, m_t evolves mechanically and introduces no additional uncertainty; it simply shrinks at unit speed as time passes.

In the stochastic specification, instead, access termination occurs at a random time τ_i^*, defined as the first jump time of an idiosyncratic Poisson process with intensity δ, as introduced in Chapter 1. In this case, there is no deterministic remaining-maturity variable. The effective horizon is governed by termination risk, and valuation must therefore incorporate the possibility that access rights may expire unexpectedly over time.

We summarize the aggregate state of the platform by an activity variable A_t, which captures the scale of economic interactions on the platform (adoption, usage, and the associated flow value). We specify the dynamics of A_t directly under the risk-neutral measure as

$$\frac{dA_t}{A_t} = \mu_A \, dt + \sigma_A \, dZ_t,$$

where Z_t is a standard Brownian motion and the functions μ_A and σ_A satisfy the usual regularity conditions ensuring a well-defined diffusion.

Taken together, the competitive environment is characterized by platform activity and by the relevant specification of access termination.

The Finite-Maturity Token

The token considered in this chapter coincides with the finite-maturity contract analyzed in Chapter 1. Rather than re-deriving its equilibrium characterization, we recall here the pricing structure that will be relevant for the competitive comparison developed below.

Deterministic Termination At each time t, the token grants access to the platform over the remaining horizon $m_t = T - t$. Its equilibrium price reflects the discounted capitalization of future platform-generated benefits, as determined by the current level of activity A_t and by the remaining lifetime m_t. As shown in Chapter 1 (see (1.13) for further details), the token price admits the closed-form representation

$$P^{det}(A_t, m_t) = \frac{N_t S_t A_t}{M} \left(\frac{1-\alpha}{r - \mu^P(A_t, m_t)} \right)^{\frac{1}{\alpha}}, \tag{2.1}$$

where $\mu^P(A_t, m_t)$ denotes the risk-neutral growth component of the token price and captures the interaction between platform dynamics and maturity effects.

This expression makes explicit the two forces shaping token valuation. First, the level of platform activity A_t directly scales the capitalization of platform-generated surplus. Second, the effective carry term $r - \mu^P(A_t, m_t)$ governs the discounting of future benefits and embeds the impact of finite maturity: as m_t decreases, the horizon over which growth can be capitalized contracts, affecting the growth-adjusted discount rate.

For the purposes of the competitive analysis, we treat $P^{det}(A_t, m_t)$ as a well-defined equilibrium price function satisfying the structural properties established in Chapter 1. In particular, for any fixed m, the function $P^{det}(A, m)$ is continuous in A, strictly increasing in A (see Remark 1.7), and satisfies $P^{det}(A, m) > 0$ for all $A > 0$. These properties ensure that improvements in platform fundamentals translate into higher equilibrium token valuations and will be central in characterizing well-defined dominance regions relative to an external asset.

Stochastic Termination Access rights expire at a random time τ_i^* governed by a Poisson process with intensity δ, as introduced in Chapter 1. In this setting, the token no longer features a deterministic remaining-maturity variable. Instead, the effective horizon is determined by the arrival of the termination event, which occurs unpredictably over time.

The equilibrium price reflects the discounted capitalization of future platform-generated benefits, adjusted for the possibility of termination. As shown in Chapter 1

(see (1.29)), the token price admits a static representation of the form

$$P^{stoch}(A_t) = \frac{N_t S_t A_t}{M} \left(\frac{1-\alpha}{r+\delta-\mu^P(A_t)} \right)^{\frac{1}{\alpha}}, \tag{2.2}$$

where $\mu^P(A_t)$ denotes the risk-neutral growth component of the token price in the stochastic environment.

Relative to the deterministic case, the key difference lies in the effective discounting of future benefits. The hazard rate δ enters the pricing equation as an additional discounting component, reflecting the risk that access rights may terminate before future gains are realized. As a result, stochastic termination induces a uniform erosion of token value, rather than the time-dependent contraction associated with deterministic maturity.

We treat $P^{stoch}(A_t)$ as a well-defined equilibrium price function satisfying the structural properties established in Chapter 1. In particular, the function is continuous and strictly increasing in A, and satisfies $P^{stoch}(A) > 0$ for all $A > 0$. These properties ensure that improvements in platform fundamentals continue to translate into higher equilibrium valuations, and allow for a direct comparison with the deterministic benchmark within the dominance framework developed below.

The External Asset

In order to assess the competitive viability of the token, we introduce an alternative investment opportunity that is external to the platform economy. Economically, this asset represents the opportunity cost of allocating capital to the token. A user or investor who buys the platform token gives up the possibility of investing in this outside asset. The comparison is therefore not simply whether the token has value, but whether its value is large enough to justify choosing it over a competing financial opportunity. In the spirit of Merton [12], the problem can be interpreted as a continuous-time allocation decision between alternative investment opportunities. Our setting is closely related to the dynamic asset allocation framework developed in Munk [13], where investors optimally allocate wealth across assets in the presence of stochastic investment opportunities. We abstract from richer features such as stochastic interest rates and ambiguity, as considered for instance in Munk and Rubtsov [14], to focus on a tractable comparison between two alternatives.

We denote by X_t the price process of the external asset. Under the risk-neutral measure, and consistently with the pricing framework adopted throughout this

chapter, X_t follows a geometric diffusion of the form

$$\frac{dX_t}{X_t} = r\,dt + \sigma_X\,dW_t,$$

where r is the constant risk-free rate, $\sigma_X > 0$ is the asset volatility, and W_t is a standard Brownian motion.

The asset is assumed to be frictionless and perfectly tradable, with no additional carry costs beyond the risk-free financing rate. In the absence of intermediate cash flows, its value at time t coincides with its current market price X_t, which already reflects the discounted value of future payoffs under $\mathbb{Q}$.

Formally, under the risk-neutral measure, the time-t value of holding the external asset can be written as

$$V^X(X_t) = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-r(s-t)}X_s\right], \tag{2.3}$$

for any horizon $s \geq t$. To evaluate this expression, consider the discounted price process $\tilde{X}_t := e^{-rt}X_t$. Under $\mathbb{Q}$, since X_t follows a geometric diffusion with drift r, the process $(\tilde{X}_t)_{t\geq 0}$ is a martingale. It follows that

$$\mathbb{E}_t^{\mathbb{Q}}[\tilde{X}_s] = \tilde{X}_t.$$

Using the identity

$$e^{-r(s-t)}X_s = e^{rt}\tilde{X}_s,$$

we obtain

$$\mathbb{E}_t^{\mathbb{Q}}\left[e^{-r(s-t)}X_s\right] = e^{rt}\mathbb{E}_t^{\mathbb{Q}}[\tilde{X}_s] = e^{rt}\tilde{X}_t = X_t,$$

which implies

$$V^X(X_t) = X_t. \tag{2.4}$$

Importantly, the valuation of X_t is independent of platform fundamentals and of the specification of token termination. In contrast to the token price, which embeds the capitalization of platform-generated surplus over a finite deterministic or stochastic horizon, the external asset provides exposure to a separate source of financial risk. This structural distinction will be central in the relative valuation analysis that follows.

Dependence Structure

Up to this point, the dynamics of platform activity A_t and of the external asset price X_t have been introduced separately. For the competitive analysis developed in the

subsequent sections, it is useful to clarify how the underlying sources of uncertainty may be related.

Let Z_t and W_t denote the Brownian motions driving A_t and X_t, respectively. We allow for a non-trivial instantaneous correlation between these shocks, specified by

$$d\langle Z, W\rangle_t = \rho\, dt, \qquad \rho \in [-1, 1].$$

The parameter ρ captures the degree of exposure of the external asset to the same aggregate risk factors affecting platform fundamentals. When $\rho = 0$, the token and the external asset are driven by independent sources of uncertainty. For $\rho \neq 0$, shocks to platform activity are partially transmitted to the external asset, generating a direct linkage between the two valuation processes.

It is important to emphasize that the structural definitions of the token price, both in the deterministic (P^{det}) and in the stochastic specification (P^{stoch}), and V^X do not require any specific value of ρ. The correlation parameter is retained as part of the broader dynamic environment, but it does not affect the pointwise dominance threshold studied below.

The competitive problem can now be formulated as a pointwise comparison between the token price and the external asset value. The next section formalizes this comparison and characterizes the regions of the state space in which each asset dominates.

2.2 Relative Valuation

The Dominance Function

At any given time t, the investor faces a direct comparison between two valuation objects defined on the same stochastic environment: the equilibrium token price and the market value of the external asset X_t. Depending on the specification of access termination, the token price is given by $P^{det}(A_t, m_t)$ in the deterministic case or by $P^{stoch}(A_t)$ in the stochastic case. Since all quantities are expressed under the risk-neutral measure and refer to the same information set $\mathcal{F}_t$, their comparison is well-defined and purely economic in nature.

Definition 2.1 (**Dominance Function**). *For any fixed time t, define the dominance function as the difference between the token price and the value of the external asset.*

Deterministic Specification *For a given state* (A_t, m_t, X_t)*,*

$$D^{det} : \mathbb{R}_+ \times [0,T] \times \mathbb{R}_+ \to \mathbb{R}, \qquad D^{det}(A, m; X) := P^{det}(A, m) - X, \tag{2.5}$$

which at time t yields

$$D^{det}(A_t, m_t; X_t) = P^{det}(A_t, m_t) - X_t.$$

Stochastic Specification *For a given state* (A_t, X_t)*,*

$$D^{stoch} : \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}, \qquad D^{stoch}(A; X) := P^{stoch}(A) - X, \tag{2.6}$$

which at time t yields

$$D^{stoch}(A_t; X_t) = P^{stoch}(A_t) - X_t.$$

The sign of the dominance function determines which asset delivers the higher equilibrium value at time t. In particular, for either specification,

$$D(\cdot) \geq 0 \quad \Longleftrightarrow \quad \text{the token weakly dominates the external asset,}$$

whereas $D(\cdot) < 0$ implies that the external asset yields a higher valuation.

The token price is strictly increasing in platform activity, reflecting the fact that higher levels of A_t raise the capitalization of platform-generated surplus into the token valuation. By contrast, the value of the external asset is independent of platform fundamentals and remains equal to X_t. As a result, variations in A_t affect only the token component of the comparison, implying that the relative attractiveness between the two assets is primarily driven by the level of platform activity, conditional on the remaining maturity in the deterministic case and on termination risk in the stochastic specification, as well as on the current level of the external asset price.

The behavior of the dominance function with respect to platform activity is particularly informative. In both specifications, the token price is strictly increasing in A, while the value of the external asset remains equal to X and is independent of platform fundamentals.

As a consequence we have

$$\lim_{A \downarrow 0} D(\cdot) = -X < 0,$$

for any fixed $X > 0$, where $D(\cdot)$ denotes either $D^{det}(A, m; X)$ or $D^{stoch}(A; X)$. Hence,

for sufficiently low levels of platform activity, the external asset strictly dominates the token.

Moreover, provided that the growth-adjusted carry term remains positive so that the token price grows proportionally with A, we obtain

$$\lim_{A \to +\infty} D(\cdot) = +\infty.$$

Thus, for sufficiently high platform activity, the token eventually dominates the external asset in both specifications.

These limit properties, combined with continuity and monotonicity, will underpin the threshold characterization developed in the next section. In the deterministic case, the threshold depends on the remaining maturity m as a state variable, while in the stochastic specification it depends on the intensity of termination δ, which governs the effective horizon through a constant hazard rate.

Characterization of Dominance Regions

The dominance function defined above induces a natural partition of the state space into regions where either the token or the external asset delivers the higher valuation.

In the deterministic specification, for any fixed (m, X), define the *token-dominance region* as

$$\mathcal{D}^{+,det}(m, X) := \{A \in \mathbb{R}_+ : D^{det}(A, m; X) \geq 0\},$$

and the corresponding *external-asset dominance region* as

$$\mathcal{D}^{-,det}(m, X) := \{A \in \mathbb{R}_+ : D^{det}(A, m; X) < 0\}.$$

In the stochastic specification, for any fixed X, define analogously

$$\mathcal{D}^{+,stoch}(X) := \{A \in \mathbb{R}_+ : D^{stoch}(A; X) \geq 0\},$$

and

$$\mathcal{D}^{-,stoch}(X) := \{A \in \mathbb{R}_+ : D^{stoch}(A; X) < 0\}.$$

By construction, these regions form a partition of $\mathbb{R}_+$ for any given (m, X) in the deterministic case and for any given X in the stochastic case. In both specifications, the structural properties established in the previous subsection imply that the mapping

$$A \longmapsto D(\cdot)$$

is continuous and strictly increasing in A. Combined with the boundary behavior

$$\lim_{A\downarrow 0} D(\cdot) = -X < 0, \qquad \lim_{A\to+\infty} D(\cdot) = +\infty,$$

this ensures that the token-dominance region is non-empty for sufficiently large values of A, while the external-asset dominance region is non-empty for sufficiently small values of A.

Economically, this characterization highlights a fundamental trade-off. When platform activity is low, the capitalization of future platform surplus is limited and the external asset—whose value is independent of A—provides a higher equilibrium payoff. As platform activity increases, the token price rises monotonically, eventually exceeding the value of the external asset.

The relative attractiveness of the two assets is therefore governed by the interaction between platform fundamentals, the effective horizon of the token, and the current level of the external asset price.

The formal identification of a boundary separating these regions, and the conditions under which such a boundary is unique, will be established in the next section.

2.3 Existence and Uniqueness of the Dominance Threshold

The previous section introduced the dominance function, which compares the equilibrium value of the token to the time-t value of the external asset. Depending on the specification of access termination, the token price is given by $P^{det}(A_t, m_t)$ in the deterministic case and by $P^{stoch}(A_t)$ in the stochastic specification, leading to the corresponding dominance functions D^{det} and D^{stoch}.

For a given external asset price X, and conditional on the relevant variables governing the token horizon, dominance regions were defined as the subsets of the activity space where the dominance function is nonnegative (token dominance) or negative (external-asset dominance).

The purpose of this subsection is to formalize the boundary between these regions by establishing that, under mild regularity conditions inherited from Chapter 1, a unique activity level separates them. This boundary will be referred to as the *dominance threshold*.

For expositional clarity, we first derive the result under the deterministic termination specification. The stochastic case follows by analogous arguments and is

discussed in the subsequent remark.

Theorem 2.2 (**Existence and uniqueness of the dominance threshold**). *Fix $m \in (0, T]$ and $X > 0$. Suppose that the token price function $P^{det}(\cdot, m)$ satisfies the following properties:*

*(A1) (*Continuity*) $P^{det}(\cdot, m)$ is continuous on $\mathbb{R}_+$.*

*(A2) (*Strict monotonicity*) $P^{det}(\cdot, m)$ is strictly increasing on $\mathbb{R}_+$.*

*(A3) (*Boundary behavior*) $\lim_{A \downarrow 0} P^{det}(A, m) = 0$ and $\lim_{A \to +\infty} P^{det}(A, m) = +\infty$.*

Define $D^{det}(\cdot, m; X) : \mathbb{R}_+ \to \mathbb{R}$ by $D^{det}(A, m; X) = P^{det}(A, m) - X$. Then there exists a unique $A^*(m, X) \in \mathbb{R}_+$ such that

$$D^{det}(A^*(m, X), m; X) = 0, \qquad \textit{equivalently} \qquad P^{det}(A^*(m, X), m) = X. \tag{2.7}$$

Moreover, the dominance regions admit the representation

$$\mathcal{D}^{-,det}(m, X) = \{A \in \mathbb{R}_+ : A < A^*(m, X)\}, \qquad \mathcal{D}^{+,det}(m, X) = \{A \in \mathbb{R}_+ : A \geq A^*(m, X)\}.$$

Proof. Fix $m \in (0, T]$ and $X > 0$. By (A1), the mapping $A \mapsto D^{det}(A, m; X) = P^{det}(A, m) - X$ is continuous on $\mathbb{R}_+$. Using (A3), we have

$$\lim_{A \downarrow 0} D^{det}(A, m; X) = \lim_{A \downarrow 0} P^{det}(A, m) - X = -X < 0,$$
$$\lim_{A \to +\infty} D^{det}(A, m; X) = \lim_{A \to +\infty} P^{det}(A, m) - X = +\infty.$$

Hence there exists $A_0 > 0$ such that $D^{det}(A_0, m; X) > 0$. Since $D^{det}(\cdot, m; X)$ is continuous and takes values of opposite sign at $A \downarrow 0$ and at A_0, the Intermediate Value Theorem implies the existence of at least one $A^* \in (0, A_0)$ such that $D^{det}(A^*, m; X) = 0$.

To prove uniqueness, suppose that there exist $A_1 < A_2$ such that $D^{det}(A_1, m; X) = D^{det}(A_2, m; X) = 0$. Then $P^{det}(A_1, m) = X = P^{det}(A_2, m)$, contradicting (A2) since $P^{det}(\cdot, m)$ is strictly increasing. Therefore the solution is unique and we denote it by $A^*(m, X)$, establishing (2.7).

Finally, strict monotonicity of $D^{det}(\cdot, m; X)$ in A follows from (A2).

Hence $D^{det}(A, m; X) < 0$ for all $A < A^*(m, X)$ and $D^{det}(A, m; X) \geq 0$ for all $A \geq A^*(m, X)$, which yields the stated characterization of $\mathcal{D}^{-,det}(m, X)$ and $\mathcal{D}^{+,det}(m, X)$.

□

Remark 2.3 (**Stochastic specification**). An analogous result holds in the stochastic specification.

Let $D^{stoch}(A; X) = P^{stoch}(A) - X$. Under the same regularity conditions as in Theorem 2.2 (continuity, strict monotonicity, and boundary behavior), there exists a unique threshold $A^*(\delta, X) \in \mathbb{R}_+$ such that

$$D^{stoch}(A^*(\delta, X); X) = 0, \qquad \text{equivalently} \qquad P^{stoch}(A^*(\delta, X)) = X.$$

The proof follows from the same intermediate value and monotonicity arguments, noting that in this case the dominance function depends only on A, while the effect of termination enters through the parameter δ embedded in the price function.

Theorem 2.2 shows that, for any given external asset price X, the competitive comparison between the token and the external asset is summarized by a single cutoff in platform activity.

In the deterministic case, this cutoff is given by $A^*(m, X)$ and depends on the remaining maturity. When platform activity is below this level, the capitalization of platform-generated surplus into the finite-horizon token is insufficient to match the value of the external investment opportunity, so the external asset dominates. When activity exceeds the cutoff, the token price overtakes X and the token dominates.

In the stochastic specification, the cutoff takes the form $A^*(\delta, X)$ and depends on the intensity of termination. While the structure of the result is unchanged, the economic interpretation differs: rather than being driven by a deterministic contraction of the horizon, the dominance threshold reflects a uniform discounting effect induced by termination risk.

The dependence of the cutoff on (m, X) in the deterministic case and on (δ, X) in the stochastic specification has a clear economic meaning. The effective horizon of the token governs how much future platform growth can be capitalized into current prices, while the external asset price X captures the prevailing benchmark the token must exceed. In the next section, we study these comparative-static implications formally.

2.4 Comparative Statics of the Threshold

Dependence on the External Asset Value X

The goal of this subsection is to study how this threshold reacts to changes in the value of the external asset. In other words, we analyze how the activity level

required for the token to match the competing investment opportunity varies when the benchmark value X shifts.

Proposition 2.4 (Dependence of the dominance threshold on the external asset value)**.** *Suppose the token price function $P^{det}(A, m)$ is continuously differentiable and strictly increasing in the activity level A. Then the dominance threshold $A^*(m, X)$ is strictly increasing in the value of the external asset X, and satisfies*

$$\frac{\partial A^*(m, X)}{\partial X} = \frac{1}{P_A^{det}(A^*(m, X), m)} > 0. \tag{2.8}$$

Proof. Recall that the dominance threshold $A^*(m, X)$ is defined implicitly by the condition

$$P^{det}(A^*(m, X), m) = X, \tag{2.9}$$

which corresponds to the boundary between the dominance regions identified by the dominance function

$$D^{det}(A, m; X) = P^{det}(A, m) - X.$$

Since the token price function $P^{det}(A, m)$ ((2.1)) is continuously differentiable and strictly increasing in A, the partial derivative $P_A^{det}(A, m)$ is strictly positive. These regularity conditions ensure that the Implicit Function Theorem applies to the equation above.

Differentiating the identity

$$P^{det}(A^*(m, X), m) = X$$

with respect to X while keeping m fixed yields

$$P_A^{det}(A^*(m, X), m) \frac{\partial A^*(m, X)}{\partial X} = 1.$$

Solving for the derivative gives

$$\frac{\partial A^*(m, X)}{\partial X} = \frac{1}{P_A^{det}(A^*(m, X), m)}.$$

Because $P_A^{det}(A, m) > 0$ for all admissible (A, m), it follows immediately that

$$\frac{\partial A^*(m, X)}{\partial X} > 0,$$

which proves that the dominance threshold is strictly increasing in the external asset value X. □

Remark 2.5 (**Stochastic specification**). An analogous result holds in the stochastic specification.

Let the dominance threshold $A^*(\delta, X)$ be defined by

$$P^{stoch}(A^*(\delta, X)) = X.$$

If $P^{stoch}(A)$ is continuously differentiable and strictly increasing in A, then the threshold is strictly increasing in X and satisfies

$$\frac{\partial A^*(\delta, X)}{\partial X} = \frac{1}{P_A^{stoch}(A^*(\delta, X))} > 0.$$

The result follows directly from the Implicit Function Theorem, noting that in this case the dependence on δ is embedded in the price function rather than appearing as a state variable.

The external asset value X represents the market valuation of the competing investment opportunity. When X increases, the benchmark that the token must match in order to dominate becomes higher. As a consequence, a larger level of platform activity is required for the token price to reach that benchmark.

In the deterministic case, the adjustment occurs through the threshold $A^*(m, X)$, conditional on the remaining maturity. In the stochastic specification, the same effect is captured by the threshold $A^*(\delta, X)$, where the intensity of termination shapes the effective horizon of the token.

Equivalently, improvements in outside investment opportunities tighten the conditions under which the token is able to dominate the alternative asset. The platform must generate greater activity in order to compensate for the higher opportunity cost faced by investors. Conversely, when X declines, the token becomes relatively more attractive and the activity threshold required for token dominance shifts downward.

Dependence on Remaining Maturity m

The dominance threshold also depends on the remaining maturity of the token. We now study how the level of platform activity required for the token to match the external asset varies when the remaining maturity m changes.

Proposition 2.6 (Dependence of the dominance threshold on remaining maturity)**.** *Suppose the token price function $P^{det}(A, m)$ is continuously differentiable in both arguments and strictly increasing in A. Then the dominance threshold $A^*(m, X)$ is*

differentiable with respect to m and satisfies

$$\frac{\partial A^*(m,X)}{\partial m} = -\frac{P_m^{det}(A^*(m,X),m)}{P_A^{det}(A^*(m,X),m)}. \tag{2.10}$$

Proof. Recall that the dominance threshold is defined implicitly by the equation

$$P^{det}(A^*(m,X),m) = X.$$

Since $P^{det}(A,m)$ is continuously differentiable and $P_A^{det}(A,m) > 0$, the Implicit Function Theorem applies and guarantees that $A^*(m,X)$ is differentiable with respect to m.

Differentiating the identity

$$P^{det}(A^*(m,X),m) = X$$

with respect to m while keeping X fixed yields

$$P_A^{det}(A^*(m,X),m)\frac{\partial A^*(m,X)}{\partial m} + P_m^{det}(A^*(m,X),m) = 0.$$

Solving for the derivative gives

$$\frac{\partial A^*(m,X)}{\partial m} = -\frac{P_m^{det}(A^*(m,X),m)}{P_A^{det}(A^*(m,X),m)},$$

which establishes the result. □

Corollary 2.7 (**Sign under a natural maturity-capitalization condition**)**.** *Assume in addition that the deterministic token value is weakly increasing in remaining maturity, i.e. $P_m^{det}(A,m) \geq 0$ for all admissible (A,m). Then the dominance threshold is weakly decreasing in m:*

$$\frac{\partial A^*(m,X)}{\partial m} \leq 0, \tag{2.11}$$

with strict inequality whenever $P_m^{det}(A^*(m,X),m) > 0$.

Proof. Assume that the deterministic token value is weakly increasing in remaining maturity, i.e.

$$P_m^{det}(A,m) \geq 0$$

for all admissible (A,m). This condition captures the natural economic restriction that, holding platform activity fixed, a longer remaining access horizon cannot reduce

the value of the token.

Using Proposition 2.6, the derivative of the dominance threshold with respect to remaining maturity is

$$\frac{\partial A^*(m,X)}{\partial m} = -\frac{P_m^{det}(A^*(m,X),m)}{P_A^{det}(A^*(m,X),m)}.$$

Since $P_A^{det}(A,m) > 0$ by strict monotonicity in platform activity, and $P_m^{det}(A,m) \geq 0$ by assumption, it follows that

$$\frac{\partial A^*(m,X)}{\partial m} \leq 0.$$

The inequality is strict whenever $P_m^{det}(A^*(m,X),m) > 0$. □

Remaining maturity governs the horizon over which future platform activity can be capitalized into the token price. When a longer remaining maturity increases (or at least does not reduce) the risk-neutral appreciation component $\mu^P(A,m)$, it effectively relaxes the growth-adjusted discounting term $r-\mu^P(A,m)$ and raises token valuation for any fixed activity level. As a result, the platform requires *less* current activity to make the token competitive: the dominance boundary shifts toward lower A as m increases. Conversely, as maturity shortens, the capitalization window contracts mechanically, making it harder for the token to dominate unless platform activity is sufficiently high. These comparative statics results can be interpreted as a decision under uncertainty, where the agent selects between competing investment opportunities based on evolving state variables. In this context, the dominance threshold $A^*(m,X)$ acts as a state-dependent trigger, determining when the token becomes preferable to the external asset. This interpretation is closely related to the real options framework of Dixit and Pindyck [8], as well as to the literature on strategic asset allocation for long-term investors, where portfolio choices depend on the evolution of state variables (see, Campbell and Viceira [4]).

Dependence on the Termination Intensity δ

We now turn to the stochastic specification and study how the dominance threshold reacts to changes in the termination intensity δ. Before proceeding, a notational clarification is needed. Throughout the previous analysis, the stochastic token price has been written as $P^{stoch}(A)$, so as to emphasize that, unlike in the deterministic case, the stochastic model does not feature an additional time-to-maturity state variable. This notation, however, should not be interpreted as implying that the

price is independent of δ.

Rather, the dependence on δ has been left implicit for notational convenience. In the stochastic specification, the hazard rate δ is not a state variable, since it does not evolve over time and does not constitute part of the time-varying state vector of the economy. It is instead a structural parameter governing the random termination of access rights. Nevertheless, because δ enters the pricing equation through the killing term and affects the effective discounting of future token payoffs, the equilibrium price must in general be understood as $P^{stoch}(A;\delta)$, even when the dependence on δ is suppressed in the notation.

This distinction becomes essential in the present subsection. While it was sufficient in the previous sections to write $P^{stoch}(A)$ when studying monotonicity and threshold existence with respect to platform activity, the analysis of comparative statics with respect to δ requires the parameter dependence to be made explicit. The stochastic dominance threshold must therefore be written as $A^*(\delta, X)$, reflecting the fact that changes in the termination intensity alter the token valuation and, in turn, the activity level required for the token to match the external asset.

Proposition 2.8 (Dependence of the dominance threshold on the termination intensity)**.** *Suppose the stochastic token price function $P^{stoch}(A;\delta)$ is continuously differentiable in both arguments and strictly increasing in A. Then the dominance threshold $A^*(\delta, X)$ is differentiable with respect to δ and satisfies*

$$\frac{\partial A^*(\delta, X)}{\partial \delta} = -\frac{P^{stoch}_{\delta}(A^*(\delta, X), \delta)}{P^{stoch}_{A}(A^*(\delta, X), \delta)}. \tag{2.12}$$

Proof. Recall that the dominance threshold is defined implicitly by the equation

$$P^{stoch}(A^*(\delta, X); \delta) = X.$$

Since $P^{stoch}(A;\delta)$ is continuously differentiable and $P^{stoch}_{A}(A;\delta) > 0$, the Implicit Function Theorem applies and guarantees that $A^*(\delta, X)$ is differentiable with respect to δ.

Differentiating the identity

$$P^{stoch}(A^*(\delta, X); \delta) = X$$

with respect to δ while keeping X fixed yields

$$P^{stoch}_{A}(A^*(\delta, X), \delta)\frac{\partial A^*(\delta, X)}{\partial \delta} + P^{stoch}_{\delta}(A^*(\delta, X), \delta) = 0.$$

Solving for the derivative gives

$$\frac{\partial A^*(\delta, X)}{\partial \delta} = -\frac{P_\delta^{stoch}(A^*(\delta, X), \delta)}{P_A^{stoch}(A^*(\delta, X), \delta)},$$

which establishes the result. □

Corollary 2.9 (**Sign under a natural termination-risk condition**). *Assume in addition that an increase in the termination intensity reduces the token value, i.e.* $P_\delta^{stoch}(A, \delta) \leq 0$ *for all admissible* (A, δ)*. Then the dominance threshold is weakly increasing in* δ*:*

$$\frac{\partial A^*(\delta, X)}{\partial \delta} \geq 0, \tag{2.13}$$

with strict inequality whenever $P_\delta^{stoch}(A^*(\delta, X), \delta) < 0$*.*

Proof. From Proposition 2.8, we have

$$\frac{\partial A^*(\delta, X)}{\partial \delta} = -\frac{P_\delta^{stoch}(A^*(\delta, X), \delta)}{P_A^{stoch}(A^*(\delta, X), \delta)}.$$

Since $P_A^{stoch} > 0$ by strict monotonicity in A, the sign of the derivative is determined by $-P_\delta^{stoch}$. Under the assumption $P_\delta^{stoch} \leq 0$, it follows that

$$\frac{\partial A^*(\delta, X)}{\partial \delta} \geq 0,$$

with strict inequality whenever $P_\delta^{stoch} < 0$. □

Termination intensity governs the effective horizon of the token in the stochastic specification. A higher δ increases the likelihood of early termination, reducing the expected duration over which future platform activity can be capitalized into the token price. As a result, for any fixed activity level, the token valuation declines as δ increases.

Consequently, a higher level of platform activity is required for the token to match the external asset, and the dominance threshold shifts toward higher values of A as δ increases. Conversely, when the termination intensity is lower, the effective horizon of the token expands, making it easier for the token to compete with the external asset and reducing the activity threshold required for dominance.

These comparative statics results parallel those obtained in the deterministic case, but arise from a fundamentally different mechanism. While remaining maturity directly determines the length of the capitalization window in the deterministic specification, in the stochastic case the effective horizon is shaped by termination

risk. The dominance threshold therefore reflects not a mechanical contraction of time, but a probabilistic erosion of future value.

As in the deterministic setting, the dominance threshold $A^*(\delta, X)$ can be interpreted as a state-dependent trigger in a dynamic decision problem, determining when the token becomes preferable to the external asset. This interpretation connects naturally to the real options framework of Dixit and Pindyck [8] and to the literature on dynamic asset allocation under uncertainty.

2.5 Economic Interpretation and Limiting Cases

The Dominance Threshold as a Decision Rule The dominance threshold provides a state-dependent boundary that translates the pricing problem into an economically meaningful decision rule. Rather than interpreting token valuation purely in terms of prices, the threshold identifies the minimum level of platform activity required for tokenization to become competitive relative to the external asset.

From this perspective, the comparison between the token and the external asset can be understood as an investment decision driven by platform fundamentals. When platform activity lies below the threshold, the capitalization of future platform growth is insufficient to offset the opportunity cost represented by the external asset, and the external investment dominates. When activity exceeds the threshold, the token price reflects sufficiently strong growth expectations, making the token the preferred asset.

The threshold therefore converts a continuous valuation object into a binary economic outcome: a regime in which tokenization is not viable, and a regime in which it becomes self-sustaining.

Aggregation of Economic Forces A key feature of the dominance threshold is that it aggregates the main forces of the model into a single observable condition. In particular, its location reflects the interaction between three elements: the expected capitalization of future platform growth, the effective horizon of the token, and the external benchmark value.

The threshold thus identifies the point at which endogenous growth expectations are sufficiently strong to compensate for the external return. This interpretation highlights that token competitiveness is not determined by valuation in isolation, but by valuation relative to an alternative investment opportunity.

Deterministic vs Stochastic Termination While the structure of the threshold result is identical across specifications, the economic mechanisms underlying it differ substantially.

In the deterministic specification, the effective horizon is governed by the remaining maturity m. As maturity decreases, the window over which future platform growth can be capitalized contracts mechanically. This directly reduces token valuation and shifts the dominance threshold toward higher levels of platform activity. As a result, tokenization requires increasingly strong fundamentals to remain competitive.

In the stochastic specification, the effective horizon is shaped by the termination intensity δ. Rather than imposing a fixed cutoff, stochastic termination introduces a probabilistic discounting of future value. This generates a smoother and more gradual erosion of valuation, preserving the qualitative structure of the perpetual benchmark. Consequently, the dominance threshold remains closer to the benchmark case and is less sensitive to moderate levels of termination risk.

This distinction confirms that it is not finite duration per se that affects token viability, but the way in which duration enters the valuation problem.

Limiting Cases The limiting behavior of the model further clarifies the role of the effective horizon.

In the deterministic case, as $m \to 0$, the token loses its ability to capitalize future platform growth, and its value collapses toward zero. In this limit, the dominance threshold diverges, and the external asset dominates for all relevant levels of activity.

Conversely, as $m \to \infty$, the model converges to the perpetual benchmark, and the dominance threshold approaches the corresponding cutoff in the CWL economy.

In the stochastic specification, as $\delta \to 0$, termination risk vanishes and the model again converges to the perpetual benchmark. As δ increases, the effective horizon shortens, and the threshold shifts upward. However, unlike the deterministic case, this shift is gradual and does not generate a sharp collapse in valuation.

Economic Implications Taken together, these results highlight a central insight of the chapter: tokenization is not universally beneficial, but becomes viable only beyond a critical scale of platform activity.

The dominance threshold captures this idea explicitly. It identifies the level of fundamentals required for the endogenous growth mechanism embedded in token prices to overcome the opportunity cost of alternative investments. As such, it provides a natural metric for assessing when token-based architectures are economically justified.

This perspective also suggests that policy or design choices affecting the effective horizon of the token—such as maturity structure or termination risk—play a crucial role in determining whether tokenization can emerge as a competitive mechanism in practice.

2.6 Numerical Characterization of Token-Asset Dominance

This subsection provides the numerical counterpart of the dominance framework developed in Sections 2.2–2.5. The objective is not merely graphical. The figures are used to translate the analytical threshold characterization into economically interpretable objects: dominance frontiers in the (A, X) space, maps of relative valuation, and comparative-statics profiles with respect to maturity and termination risk.

The numerical analysis should be read as the quantitative expression of the chapter's departure from the closed-platform CWL environment. The token price functions are inherited from Chapter 1; what is new here is the external benchmark X. The results therefore show how the same token economy changes once its value is judged relative to an alternative investment opportunity.

Throughout, the token price is evaluated using the equilibrium objects already computed in Chapter 1. In the deterministic specification, we work with the finite-maturity price function $P^{\text{det}}(A, m)$ for three representative maturity levels, corresponding to short, intermediate, and long remaining horizons. In the stochastic specification, we use the price function $P^{\text{stoch}}(A; \delta)$ for three representative values of the termination intensity, ranging from low to high hazard. This preserves full consistency with the previous chapter: the present numerical exercise does not recalibrate the token economy, but places the same equilibrium price schedules in direct competition with an outside asset of value X.

The comparison is conducted under the same risk-neutral valuation logic adopted throughout the chapter. For each pair (m, X) in the deterministic case, and for each pair (δ, X) in the stochastic case, the dominance threshold A^* is defined implicitly by the equation

$$P^{\text{det}}(A^*, m) = X \qquad \text{or} \qquad P^{\text{stoch}}(A^*; \delta) = X.$$

Whenever the token price exceeds X, the token dominates the external asset; when it falls short of X, the outside opportunity is preferred.

A final numerical clarification is important. In the theory, the threshold exists and is unique under the monotonicity and boundary conditions established earlier in the chapter. Accordingly, a missing or non-displayed threshold in the numerical output should not be interpreted as theoretical non-existence. It only means that, for that parameter combination, the crossing point is located outside the finite numerical activity domain used in the Python implementation. In other words, the theorem guarantees existence; the numerics may simply fail to bracket the crossing on the plotted grid if the required activity level is extremely small or extremely large.

Table 2.1 reports the main threshold values used in the numerical discussion. Bringing these values into the chapter is useful because the economic interpretation depends not only on the shape of the figures, but also on the scale of activity required for token dominance. The table shows that short deterministic maturity is the most restrictive case: when $m = 1.06$, the token fails to dominate very large external opportunities within the numerical activity domain. Longer maturity sharply improves competitiveness, reducing the activity threshold for any fixed X. In the stochastic specification, thresholds remain available over a wide range of external benchmarks, but they rise with the termination intensity δ, confirming that stronger termination risk weakens the token's ability to compete with outside assets.

Table 2.1: Main numerical results for Chapter 2. For each specification, parameter value, and external-asset benchmark X, the table reports whether the dominance threshold exists, the corresponding activity threshold $A^*(X)$ when it exists, and the scale ratio $X/P_{\max}$. Missing thresholds indicate that no crossing is bracketed on the numerical activity domain.

Specification	Parameter	X	Threshold exists	$A^*(X)$	$X/P_{\max}$
Deterministic finite-maturity specification					
Deterministic	$m = 1.06$	1×10^3	Yes	1.14×10^6	0.000
		1×10^4	Yes	6.99×10^6	0.003
		1×10^5	Yes	3.81×10^7	0.030
		1×10^6	Yes	2.02×10^8	0.305
		1×10^7	No	–	3.045
		1×10^8	No	–	30.453
Deterministic	$m = 4.99$	1×10^3	Yes	2.59×10^3	0.000
		1×10^4	Yes	2.08×10^4	0.000
		1×10^5	Yes	1.69×10^5	0.000
		1×10^6	Yes	1.40×10^6	0.002

Continued on next page

Specification	Parameter	X	Threshold exists	$A^*(X)$	$X/P_{\max}$
		1×10^7	Yes	1.17×10^7	0.019
		1×10^8	Yes	1.00×10^8	0.186
Deterministic	$m = 9.76$	1×10^3	Yes	1.85×10^2	0.000
		1×10^4	Yes	1.64×10^3	0.000
		1×10^5	Yes	1.48×10^4	0.000
		1×10^6	Yes	1.35×10^5	0.000
		1×10^7	Yes	1.25×10^6	0.002
		1×10^8	Yes	1.16×10^7	0.021
Stochastic termination specification					
Stochastic	$\delta = 0.005$	1×10^8	Yes	4.50×10^3	0.000
		1×10^9	Yes	4.50×10^4	0.000
		1×10^{10}	Yes	4.50×10^5	0.001
		1×10^{11}	Yes	4.50×10^6	0.009
		1×10^{12}	Yes	4.50×10^7	0.093
Stochastic	$\delta = 0.020$	1×10^8	Yes	1.48×10^4	0.000
		1×10^9	Yes	1.48×10^5	0.000
		1×10^{10}	Yes	1.48×10^6	0.003
		1×10^{11}	Yes	1.48×10^7	0.030
		1×10^{12}	Yes	1.48×10^8	0.305
Stochastic	$\delta = 0.050$	1×10^8	Yes	7.09×10^4	0.000
		1×10^9	Yes	7.08×10^5	0.001
		1×10^{10}	Yes	7.08×10^6	0.015
		1×10^{11}	Yes	7.08×10^7	0.146
		1×10^{12}	No	–	1.460

The rows in which the threshold is not reported should be interpreted as cases of numerical non-attainment within the simulated activity range, not as a failure of the theoretical threshold logic. In those cases, the external asset is so valuable relative to the token price generated on the numerical grid that the equation $P(A) = X$ is never reached. This is visible in the column $X/P_{\max}$: whenever this ratio exceeds one, the external benchmark X is larger than the maximum token value computed in the simulation. Economically, these entries describe parameter configurations in which the token would need platform activity beyond the simulated range in order to become competitive. They therefore reinforce the main message of the chapter: short deterministic maturity and high stochastic termination risk make token dominance much harder to obtain when the outside opportunity is sufficiently attractive.

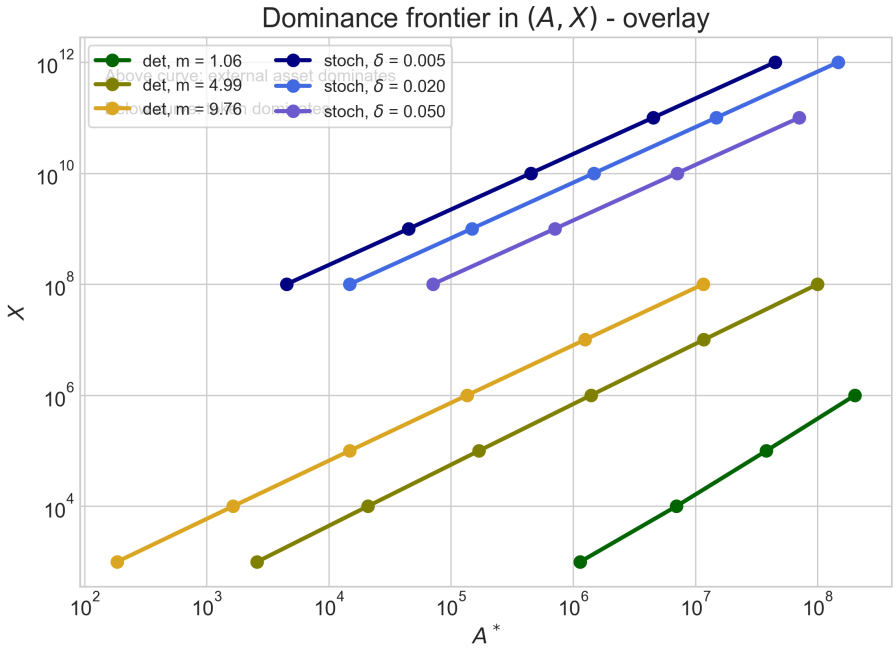


Figure 2.1: Dominance frontiers in the (A^*, X) plane for representative deterministic maturities and stochastic termination intensities. Each curve identifies the activity threshold at which the token value equals the external asset value. Frontiers located further to the left indicate stronger token competitiveness, since a lower level of platform activity is sufficient for the token to dominate the outside option.

2.6.1 Dominance Threshold: Geometry and Interpretation

The numerical analysis begins by visualizing the dominance threshold derived in the theoretical part of the chapter. The purpose of this subsection is twofold. First, it shows how the threshold varies with the value of the external asset across deterministic and stochastic specifications. Second, it illustrates how the threshold is generated by the intersection between token price schedules and fixed external-asset benchmarks.

Dominance Frontier Figure 2.1 reports the dominance frontiers in the (A^*, X) plane for representative deterministic maturities and stochastic termination intensities. Each curve collects the combinations of platform activity and external-asset value for which the investor is exactly indifferent between the two assets, that is,

$$P(A^*) = X.$$

Points to the right of a given frontier correspond to token dominance, while points to the left correspond to external-asset dominance.

Several conclusions emerge from the figure. First, all frontiers are upward sloping.

A higher external-asset value requires a higher level of platform activity for the token to remain competitive. This is the numerical counterpart of the analytical result that the dominance threshold is increasing in X: as the outside option becomes more valuable, the platform must generate stronger fundamentals in order for the token price to match it.

Second, the deterministic and stochastic specifications are clearly separated. For the parameter values considered, stochastic frontiers lie systematically to the left of deterministic ones. Economically, this means that stochastic termination preserves token competitiveness more effectively than deterministic expiry. For a given value of the external asset, the token requires less platform activity to dominate when the access horizon is eroded probabilistically rather than truncated mechanically.

Third, the within-model rankings are consistent with the comparative statics developed above. In the deterministic specification, longer remaining maturity shifts the frontier to the left, because the token can capitalize platform benefits over a longer horizon. In the stochastic specification, a higher termination intensity shifts the frontier to the right, because a higher hazard rate reduces expected continuation value and weakens token competitiveness.

The frontiers are also numerically regular. The activity level required for dominance increases smoothly with the value of the external asset, indicating a stable mapping between platform fundamentals and competitive viability. The model therefore produces not only isolated threshold values, but a coherent family of dominance boundaries across a wide range of external conditions.

Price-Based Representation Figure 2.2 provides a complementary representation of the same threshold mechanism. Instead of plotting the threshold directly, it displays token price schedules together with horizontal external-asset benchmarks. The intersections between the two determine the activity levels at which dominance switches from the external asset to the token.

This representation makes the construction of the dominance threshold explicit. For any fixed value of X, the intersection between the corresponding horizontal line and a token price curve determines the level of platform activity at which the investor is indifferent between the two assets. Since all token price schedules are increasing in A, higher platform activity favors the token, while a higher external benchmark shifts the crossing point to the right.

The ranking across models is visible in level terms. Stochastic price schedules lie above deterministic ones for the parameter values considered, implying that, at the same level of platform activity, stochastic termination delivers a higher token

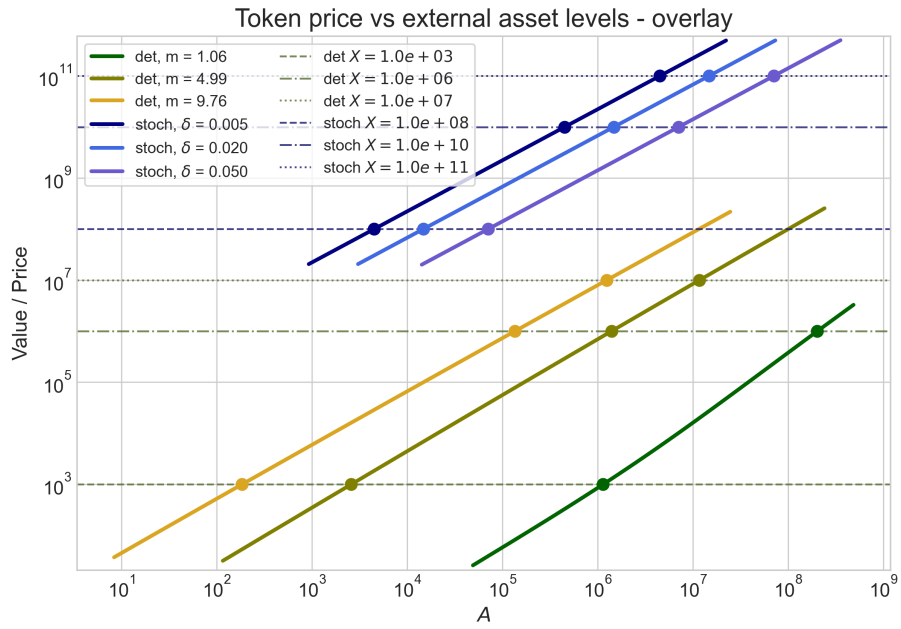


Figure 2.2: Token price schedules and external-asset benchmarks. Upward-sloping curves represent equilibrium token prices under deterministic and stochastic specifications, while horizontal lines represent fixed external-asset values X. Intersections identify the dominance thresholds A^*: to the left of each intersection the external asset dominates, while to the right the token dominates.

valuation and reaches the dominance threshold earlier. Conversely, short deterministic maturity produces the lowest price schedule and therefore the latest crossing with any given external benchmark.

This is the relative-valuation counterpart of the result established in Chapter 1. Deterministic expiry depresses token valuation more severely than stochastic termination, and this effect translates directly into a higher activity requirement for dominance. The external asset therefore plays a disciplinary role: it does not change the token pricing function itself, but determines whether that price is sufficient once compared with an alternative use of capital.

2.6.2 Comparative Statistics in Maturity m

Figure 2.3 isolates the dependence of the deterministic threshold on remaining maturity for a fixed external benchmark.

The downward slope is exactly what Proposition 2.6 predicts under the natural condition $P_m^{\text{det}} \geq 0$. Longer maturity expands the capitalization window, increases the value of the token for any fixed activity level, and therefore reduces the activity threshold required for the token to dominate the outside asset.

What is particularly striking numerically is the nonlinearity of this effect. The

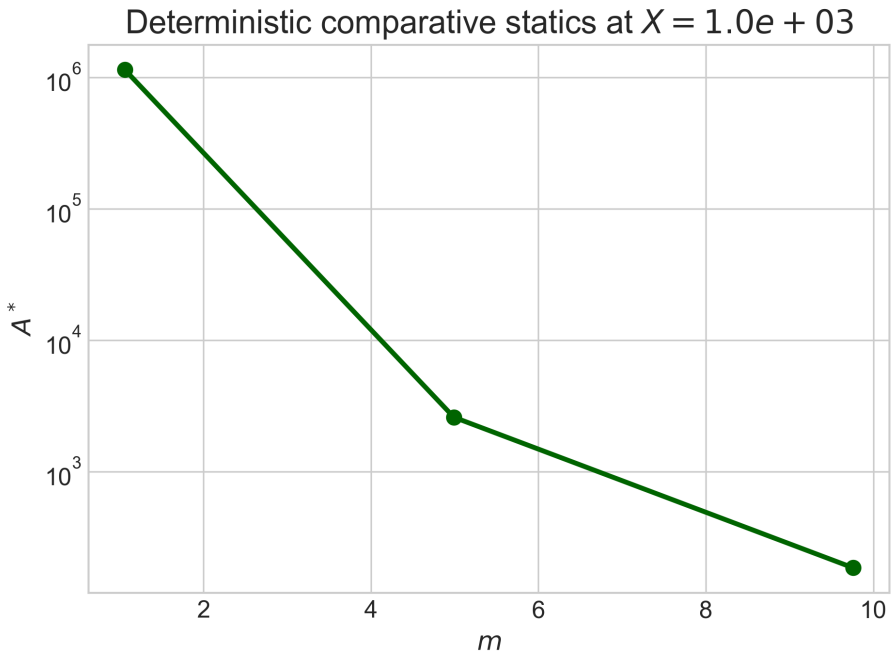


Figure 2.3: Deterministic comparative statics of the dominance threshold at fixed X. The threshold A^* decreases as remaining maturity increases, confirming that longer access horizons enhance the token's ability to capitalize future platform growth and therefore improve competitiveness relative to the external asset.

reduction in A^* is especially pronounced when moving away from very short maturity levels. This means that maturity extensions are economically most valuable when the token is close to expiration. Put differently, adding a few extra units of horizon matters far more in a compressed-horizon regime than in an already long-lived one. That is a useful applied message: when access duration is short, even modest contractual changes can materially alter competitiveness.

This figure also helps clarify why deterministic maturity is the harshest environment for token valuation. Since the horizon contracts mechanically and predictably, the token loses future optionality at every instant. The threshold therefore reacts strongly to changes in m, especially near the short-horizon region where continuation value is already fragile.

2.6.3 Comparative Statistics in δ

Figure 2.4 reports the stochastic analogue by plotting the threshold against the termination intensity.

The relationship is positive, as predicted by Proposition 2.8 and Corollary 2.9. A higher hazard rate increases the risk that access rights expire before future platform benefits are realized, so the token price falls for any given level of activity and the

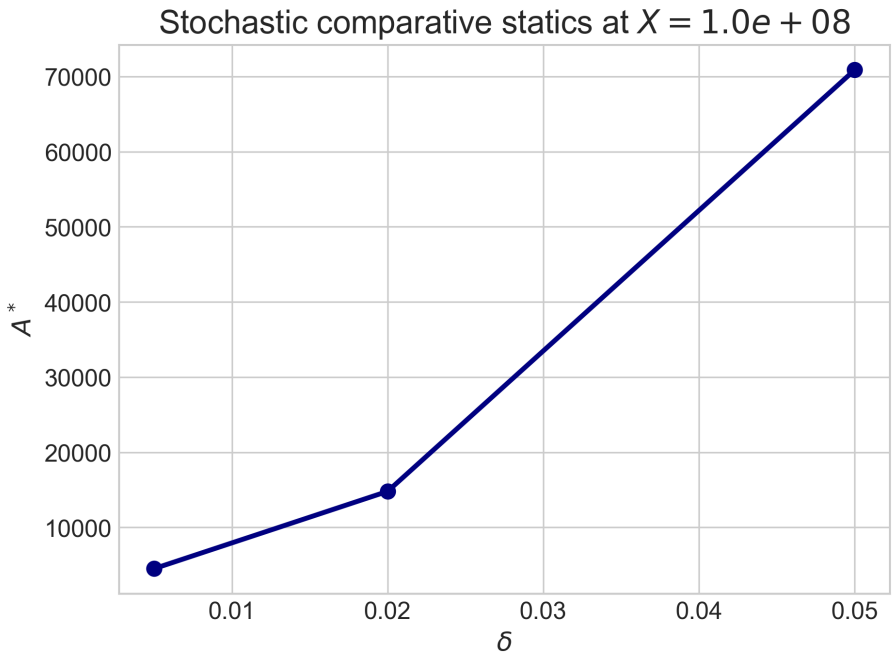


Figure 2.4: Stochastic comparative statics of the dominance threshold at fixed X. Higher termination intensity shifts A^* upward, showing that stronger exit risk weakens continuation value and requires more platform activity for the token to remain competitive.

threshold moves upward.

At the same time, the economic mechanism differs from the deterministic case. Here the horizon is not shortened mechanically; rather, its expected duration is reduced through a survival probability effect. This distinction matters because the erosion of competitiveness is smoother. Even when the threshold rises materially as δ increases, the token retains a broader region of viability than under deterministic expiry. The stochastic specification is therefore less fragile: worsening the horizon hurts, but does not destroy the capitalization channel as abruptly as a fixed maturity does.

Numerically, the steep increase between the higher hazard values also suggests that termination risk can become highly relevant once it exceeds moderate levels. In practical terms, a platform may tolerate some uncertainty about access duration without losing competitiveness, but beyond a certain point the required activity level begins to escalate quickly.

2.6.4 Deterministic vs Stochastic: Key Differences

The numerical evidence collected across the figures points to a sharp and systematic asymmetry between the two horizon structures.

In the deterministic specification, maturity acts as a hard cap on capitalization. Future platform growth can be valued only up to a known terminal date, and this produces three cumulative effects: lower token prices, steeper competitiveness requirements, and stronger sensitivity of the threshold to horizon compression. As the remaining maturity becomes short, the token behaves increasingly like a wasting asset whose value must be justified almost entirely by current, rather than prospective, platform activity.

In the stochastic specification, by contrast, the horizon remains uncertain but not mechanically truncated. Termination risk enters valuation as an additional discounting force, yet continuation value is preserved in expectation. This generates a much smoother deterioration in competitiveness. Thresholds are lower, the dominance region is broader, and the token can remain attractive even against substantially larger outside opportunities.

From a modeling perspective, the deterministic case can therefore be interpreted as a conservative lower-bound scenario for token competitiveness, while the stochastic case provides a more realistic benchmark for environments in which access rights are finite but not contractually extinguished at a fixed date. The numerical results are especially valuable because they show that this distinction is not cosmetic. It materially changes the region of the state space in which tokenization remains economically viable.

2.6.5 Economic Interpretation

The numerical analysis translates the analytical threshold results into a clear economic characterization of token competitiveness. The dominance threshold A^* summarizes the interaction between platform fundamentals, the effective horizon of tokenized access, and the external investment benchmark.

Stronger platform activity improves token competitiveness, but internal strength alone is not sufficient. What matters is the extent to which future platform value can be capitalized, which depends critically on the horizon structure. A longer or weakly constrained horizon allows the token to embed future growth and compete even at moderate activity levels, while a short or heavily penalized horizon requires substantially stronger current performance.

This extends the insight of Chapter 1. Finite duration does not necessarily eliminate the capitalization mechanism, but once an external benchmark is introduced, the way duration is modeled becomes decisive. Deterministic expiry significantly restricts the range of states in which the token is competitive, whereas stochastic termination preserves viability over a broader region of the state space.

More generally, token design cannot be evaluated in isolation. A token may sustain participation internally and still fail as a competitive asset if its contractual horizon is too restrictive. Conversely, sufficiently persistent access rights allow the token to remain viable even in the presence of meaningful external alternatives.

The central implication is therefore not simply whether a token has value, but under which conditions that value is strong enough to compete with alternative uses of capital. The dominance framework makes this trade-off explicit and identifies horizon design as one of its key determinants.

Chapter 3

Reserve-Based Token Pricing with Optimal Issuance

Chapters 1 and 2 analyzed tokenized asset systems within the Cong–Wang–Li (CWL) framework and its subsequent extensions. The central focus was the equilibrium interaction between token adoption, market liquidity, and asset valuation. While the introduction of finite and stochastic maturities enriched the original setting, the fundamental structure remained unchanged: token supply was treated as exogenous and valuation emerged from decentralized market equilibrium.

Many contemporary tokenized ecosystems, however, operate under a different paradigm. In reserve-backed protocols, token issuance is not fixed *ex ante* but constitutes an explicit policy decision.

Reserve-backed token systems have attracted growing attention in the broader literature on asset-backed digital currencies and stable-value crypto-assets, where reserve management plays a central role in sustaining market confidence and valuation (see Lyons and Viswanath-Natraj [11]).

Protocol operators actively determine how many new tokens are released into circulation, balancing growth objectives against the potential dilution of token value. More generally, the design of issuance and monetary rules has emerged as a central theme in the economics of digital currencies and tokenized systems (see Benigno et al. [2]). As a consequence, the analysis naturally shifts from an equilibrium problem to a control problem.

The motivation for the present chapter originates from the study of *RIV Technologies*, a fintech company developing a reserve-backed token ecosystem (Rivera et al. [16]) designed to connect token issuance to the growth of productive underlying assets. In the proposed architecture, newly issued tokens are supported by a reserve whose value evolves over time, while the protocol actively controls the issuance process.

This setting raises a natural quantitative question: how should a reserve-backed protocol determine its issuance policy in order to promote growth without excessively diluting token value?

Motivated by this observation, the present chapter develops a reserve-based token pricing framework in which the issuance rate is chosen endogenously by the protocol. Unlike the CWL environment, where prices emerge from trading frictions and adoption decisions, the value of the token is linked to the evolution of an underlying reserve backing the system. Although inspired by the economic architecture proposed by RIV Technologies, the model is formulated in a general setting in order to isolate the fundamental mechanisms governing reserve growth, token dilution, and optimal issuance decisions.

The key question is therefore no longer whether agents adopt the token, but rather how a protocol should optimally manage token issuance in order to maximize long-run value creation.

3.1 Model Environment

The framework describes a reserve-backed token ecosystem managed by a protocol that controls token issuance. The purpose is not to reproduce the institutional details of a specific implementation, but to isolate the core mechanism suggested by the RIV architecture (Rivera et al. [16]): token value depends jointly on reserve accumulation, platform activity, and the dilutionary effect of issuance.

Let $t \geq 0$ denote continuous time. The protocol issues a token whose circulating supply is denoted by k_t. The control variable of the protocol is the relative issuance rate

$$x_t \geq 0, \tag{3.1}$$

so that higher values of x_t correspond to more aggressive token creation. The supply process satisfies

$$dk_t = q_t \, dt, \tag{3.2}$$

where q_t denotes the instantaneous quantity of newly issued tokens. Equivalently, the relative issuance rate is

$$x_t = \frac{q_t}{k_t}. \tag{3.3}$$

Thus, x_t measures issuance intensity relative to the existing token base, rather than issuance in absolute units.

The reserve value of the protocol is denoted by V_t. It represents the aggregate value of the assets, cash flows, or productive activities backing the token ecosystem.

The reserve evolves according to

$$dV_t = \mu_V V_t \, dt + P_t q_t \, dt + \sigma_V V_t \, dZ_t, \tag{3.4}$$

where μ_V is the organic growth rate of reserves, σ_V is reserve volatility, and $(Z_t)_{t\geq 0}$ is a standard Brownian motion. The term $P_t q_t \, dt$ captures the contribution of newly issued tokens to the reserve base: when the protocol issues tokens at price P_t, the corresponding proceeds increase reserves. This specification reflects the central economic feedback of the model: issuance expands the system but may also dilute token value.

The reserve-backed intrinsic value per token is defined as

$$I_t = \frac{V_t}{k_t}. \tag{3.5}$$

This quantity represents the reserve value attributable to one token. It increases when reserves grow faster than token supply and decreases when issuance expands the token base faster than reserves. In this sense, I_t provides the fundamental valuation anchor of the system. The use of reserve backing as a valuation anchor is consistent with the broader literature on collateralized digital assets and stable-value token systems, where reserve quality and reserve adequacy play a central role in sustaining confidence and market value (see Gorton and Zhang [10]).

Market valuation may exceed intrinsic value when the token is associated with platform utility, adoption, or network effects. Let $A_t \geq 0$ denote the activity level of the ecosystem. The activity-driven premium is modeled by

$$1 + \phi A_t^{\beta}, \tag{3.6}$$

where $\phi > 0$ measures the sensitivity of market valuation to activity and $\beta > 0$ governs the shape of network effects. If $\beta < 1$, activity has diminishing marginal impact on valuation; if $\beta = 1$, the premium is linear in activity; if $\beta > 1$, activity generates convex amplification effects.

Issuance also affects market valuation through dilution and absorption pressure. Large emissions may reduce the market premium by increasing circulating supply faster than demand can absorb it. This effect is captured by the compression factor

$$e^{-\gamma x_t}, \qquad \gamma > 0, \tag{3.7}$$

where γ measures the sensitivity of market valuation to issuance intensity. The higher x_t is, the stronger the compression of the market premium.

The parameters ϕ, β, and γ summarize market-specific characteristics governing activity premia and issuance-induced valuation effects. While treated as exogenous constants in the theoretical analysis, they can be estimated or calibrated from observed market data in practical implementations. Combining intrinsic value, activity premium, and issuance compression, the token price is specified as

$$P_t = I_t \left(1 + \phi A_t^{\beta}\right) e^{-\gamma x_t}. \tag{3.8}$$

This pricing equation decomposes the token price into three components: a reserve-backed intrinsic value I_t, an activity-driven valuation premium, and an endogenous discount induced by issuance. Activity increases valuation, while issuance reduces the excess premium above intrinsic value. Unlike the equilibrium pricing equations derived in Chapters 1 and 2, where token prices emerge endogenously from agents' optimal decisions and market-clearing conditions, the specification in (3.8) is introduced as a reduced-form pricing relationship. It summarizes, in a tractable and economically interpretable manner, the joint effects of reserve backing, platform activity, and issuance pressure on market valuation. Reserve-backed token architectures of this type have been studied in recent literature on digital asset stability and institutional token design (see Lyons and Viswanath-Natraj [11], Gorton and Zhang [10]).

It is useful to define the premium ratio

$$\Pi_t = \frac{P_t}{I_t}. \tag{3.9}$$

Using (3.8), this ratio is given by

$$\Pi_t = \left(1 + \phi A_t^{\beta}\right) e^{-\gamma x_t}. \tag{3.10}$$

The premium ratio measures the market price of the token relative to its reserve-backed intrinsic value. Values $\Pi_t > 1$ indicate that the token trades above intrinsic value, while values close to one correspond to a market price aligned with the reserve backing.

The protocol therefore faces a trade-off. Increasing issuance can raise reserves through the proceeds of token sales, but excessive issuance compresses the market premium and may reduce the value attributed to each token. The optimization problem introduced below formalizes this trade-off by allowing the protocol to choose x_t so as to balance reserve accumulation, intrinsic value growth, premium stabilization, and the costs associated with aggressive issuance.

3.2 Reserve and Intrinsic Value Dynamics

The intrinsic value process is the object that connects the reserve side of the protocol with the token supply side. Its dynamics (from (3.5)) depend on two forces moving in opposite directions. Reserve accumulation increases the amount of value backing each token, while issuance increases the number of claims written on the reserve. The role of this section is to make this interaction explicit and to derive the law of motion for I_t implied by the reserve and supply dynamics introduced above.

Proposition 3.1 (Intrinsic value dynamics)**.** *Assume that the reserve process and token supply satisfy* (3.4) *and* (3.2)*, with* $k_t > 0$*. Then the reserve-backed intrinsic value* (3.5) *evolves according to*

$$dI_t = \left[\mu_V I_t + \frac{q_t}{k_t}(P_t - I_t)\right] dt + \sigma_V I_t \, dZ_t. \tag{3.11}$$

Equivalently, using the relative issuance rate (3.3)*, one obtains*

$$dI_t = \left[\mu_V I_t + x_t(P_t - I_t)\right] dt + \sigma_V I_t \, dZ_t. \tag{3.12}$$

Proof. Let

$$I_t = f(V_t, k_t) = \frac{V_t}{k_t}. \tag{3.13}$$

The relevant partial derivatives are

$$\frac{\partial f}{\partial V_t} = \frac{1}{k_t}, \qquad \frac{\partial f}{\partial k_t} = -\frac{V_t}{k_t^2}, \qquad \frac{\partial^2 f}{\partial V_t^2} = 0. \tag{3.14}$$

Since $dk_t = q_t dt$ has finite variation, $(dk_t)^2 = 0$ and $dV_t \, dk_t = 0$. Moreover, the only non-zero quadratic variation term generated by V_t is multiplied by $\partial^2 f / \partial V_t^2 = 0$. Hence, Itô's formula gives

$$dI_t = \frac{1}{k_t} \, dV_t - \frac{V_t}{k_t^2} \, dk_t. \tag{3.15}$$

Substituting (3.4) and (3.2) yields

$$dI_t = \frac{1}{k_t} \left(\mu_V V_t \, dt + P_t q_t \, dt + \sigma_V V_t \, dZ_t\right) - \frac{V_t}{k_t^2} q_t \, dt \tag{3.16}$$

$$= \mu_V \frac{V_t}{k_t} \, dt + \frac{q_t}{k_t} P_t \, dt - \frac{q_t}{k_t} \frac{V_t}{k_t} \, dt + \sigma_V \frac{V_t}{k_t} \, dZ_t. \tag{3.17}$$

Using (3.5), this becomes

$$dI_t = \left[\mu_V I_t + \frac{q_t}{k_t}(P_t - I_t)\right] dt + \sigma_V I_t \, dZ_t. \tag{3.18}$$

Finally, since $x_t = q_t/k_t$, equation (3.12) follows immediately. □

The final expression separates the evolution of intrinsic value into three economically distinct components. The first term,

$$\mu_V I_t, \tag{3.19}$$

reflects the organic growth of the reserve base. Holding issuance fixed, a higher reserve growth rate increases the value backing each token proportionally to current intrinsic value.

The second term,

$$x_t(P_t - I_t), \tag{3.20}$$

is the net contribution of issuance to intrinsic value. This term is positive whenever newly issued tokens are sold above their intrinsic value, $P_t > I_t$. In that case, each new token brings into the reserve more value than the claim it creates, so issuance increases the reserve-backed value per token. If $P_t = I_t$, issuance is neutral for intrinsic value: the reserve receives exactly the amount required to back the additional token. If $P_t < I_t$, issuance is value destroying, since new tokens are effectively issued below their reserve-backed value.

The last term,

$$\sigma_V I_t \, dZ_t, \tag{3.21}$$

transfers reserve uncertainty directly to the intrinsic value process. Since I_t is proportional to V_t, shocks to the reserve affect intrinsic value proportionally.

The issuance component (3.20) is the key mechanism used in the protocol optimization problem. It captures the central trade-off faced by a reserve-backed token system: issuance can increase intrinsic value when the market price exceeds the reserve-backed value, but excessive issuance also affects the market price through the issuance-compression channel introduced in the pricing specification. The protocol therefore cannot simply choose a high issuance rate. It must choose x_t by balancing the positive reserve accumulation effect generated by issuance against the negative impact that issuance has on the market premium.

3.3 The Protocol Optimization Problem

The intrinsic value dynamics derived in Proposition 3.1 show that token issuance can either create or destroy value depending on the relationship between market price and intrinsic value. The protocol therefore faces the problem of determining an

issuance policy that exploits favorable market conditions while preserving long-run ecosystem stability. To formalize this trade-off, we introduce an objective function that combines value creation, reserve accumulation, premium stabilization, and issuance costs.

The protocol pursues four main objectives. First, it aims to increase intrinsic value through value-accretive issuance. Second, it seeks to accumulate reserves and support ecosystem growth. Third, it attempts to maintain a stable relationship between market price and intrinsic value. Finally, it must avoid excessively aggressive issuance policies, since large emissions may generate liquidity stress, market absorption frictions, and reputational costs.

To capture these objectives, the protocol evaluates the instantaneous payoff

$$J(x_t) = \theta x_t(P_t - I_t) + \omega x_t k_t(P_t - I_t) - \chi(\Pi_t - 1)^2 - C(x_t), \tag{3.22}$$

with Π_t defined in (3.9) and

$$C(x_t) = \frac{\lambda}{2} x_t^2, \qquad \lambda > 0, \tag{3.23}$$

represents the cost of token issuance. The parameters $\theta, \omega, \chi, \lambda > 0$ determine the relative weight assigned to intrinsic value creation, reserve accumulation, premium stabilization, and issuance costs, respectively.

The first term in (3.22),

$$\theta x_t(P_t - I_t), \tag{3.24}$$

captures the contribution of issuance to intrinsic value growth. As shown in Proposition 3.1, the issuance component of the intrinsic value dynamics is proportional to (3.20). Issuance is therefore value-accretive whenever newly issued tokens are sold above intrinsic value, namely when $P_t > I_t$. Conversely, if the market price falls below intrinsic value, additional issuance dilutes existing token holders and reduces the reserve-backed value per token. The coefficient θ measures how strongly the protocol values this direct intrinsic-value channel.

The second term,

$$\omega x_t k_t(P_t - I_t), \tag{3.25}$$

reflects the reserve accumulation channel. Since $x_t = q_t/k_t$, the product $x_t k_t$ corresponds to the absolute flow of newly issued tokens. When these tokens are sold at a premium over intrinsic value, issuance contributes positively to the reserve base. The parameter ω captures the importance assigned by the protocol to aggregate reserve growth and ecosystem expansion.

The third component,

$$-\chi(\Pi_t - 1)^2, \tag{3.26}$$

penalizes deviations of the market price from intrinsic value. The premium ratio (3.9) measures the degree to which the token trades above or below its reserve-backed value. A value of $\Pi_t = 1$ corresponds to exact alignment between market price and intrinsic value. Large deviations from this benchmark may indicate either excessive overvaluation or loss of market confidence. The quadratic penalty therefore promotes long-run consistency between market valuation and fundamentals.

Finally, the issuance cost (3.23) introduces convex costs of aggressive issuance. This term captures liquidity stress, market impact, governance frictions, reputational costs, and other inefficiencies that become more severe as the relative issuance rate increases. The convex specification implies that moderate issuance may be sustainable, while very large issuance rates become increasingly inefficient.

Taken together, the four components of (3.22) reflect a fundamental tension faced by reserve-backed token protocols. Aggressive issuance may accelerate reserve growth and ecosystem expansion, but it may also compress market premia and increase issuance costs. The protocol must therefore balance short-term expansion against the preservation of long-term token value.

The protocol chooses the relative issuance rate according to

$$x_t^* = \arg\max_{x_t \geq 0} J(x_t). \tag{3.27}$$

This static optimization problem characterizes the issuance decision at a given point in time, taking the current state variables as given.

The static formulation is adopted for analytical tractability and transparency. A fully dynamic treatment would require the protocol to solve an intertemporal stochastic control problem over the joint state space of reserve value, token supply, and platform activity, leading to a Hamilton–Jacobi–Bellman equation whose solution would generally not admit a simple analytical characterization (see Dixit and Pindyck [8]). The present approach preserves the core economic trade-offs identified by the model while allowing for explicit and economically interpretable results.

3.4 Existence and Uniqueness of the Optimal Issuance Rate

The purpose of this section is to show that, under economically meaningful restrictions on admissible issuance, the protocol's problem admits a unique solution. This

ensures that the issuance rule is well defined before deriving an analytically tractable approximation in the next section.

Throughout the section, the state variables I_t, k_t, and A_t are taken as given at time t. Define

$$B_t := 1 + \phi A_t^{\beta}, \qquad H_t := \theta + \omega k_t, \tag{3.28}$$

so that the pricing specification can be written as

$$P_t(x_t) = I_t B_t e^{-\gamma x_t}, \qquad \Pi_t(x_t) = B_t e^{-\gamma x_t}.$$

The protocol payoff is therefore

$$J(x_t) = H_t x_t \Big(P_t(x_t) - I_t\Big) - \chi\Big(\Pi_t(x_t) - 1\Big)^2 - \frac{\lambda}{2} x_t^2.$$

The admissible set is assumed to be a compact interval

$$\overline{X} = [0, \bar{x}], \qquad \bar{x} > 0, \tag{3.29}$$

where $\bar{x}$ represents the maximum relative issuance rate allowed by the protocol.

Theorem 3.2 (Existence and uniqueness of the optimal issuance rate)**.** *Assume that*

$$I_t > 0, \qquad k_t > 0, \qquad A_t > 0, \qquad \theta, \omega, \chi, \lambda, \phi, \beta, \gamma > 0.$$

Suppose further that the admissible issuance interval satisfies

$$\bar{x} \leq \frac{2}{\gamma} \qquad \textit{and} \qquad \Pi_t(x_t) > \frac{1}{2} \quad \textit{for all } x_t \in [0, \bar{x}]. \tag{3.30}$$

Then the optimization problem

$$x_t^* \in \arg\max_{x_t \in [0, \bar{x}]} J(x_t)$$

admits a unique solution. If, in addition,

$$F(\bar{x}) < 0,$$

where $F(x_t) := J'(x_t)$, then the solution is interior and is the unique root of

$$F(x_t) = 0. \tag{3.31}$$

Proof. Using (3.4), differentiate J with respect to x_t. Since

$$\frac{\partial P_t}{\partial x_t} = -\gamma P_t, \qquad \frac{\partial \Pi_t}{\partial x_t} = -\gamma \Pi_t,$$

the first derivative is

$$\begin{aligned} F(x_t) = J'(x_t) &= H_t\Big(P_t - I_t\Big) + H_t x_t \frac{\partial P_t}{\partial x_t} - 2\chi(\Pi_t - 1)\frac{\partial \Pi_t}{\partial x_t} - \lambda x_t \\ &= H_t\Big(P_t - I_t\Big) - \gamma H_t x_t P_t + 2\chi\gamma\Pi_t(\Pi_t - 1) - \lambda x_t. \end{aligned}$$

Equivalently,

$$F(x_t) = H_t\Big[P_t(1 - \gamma x_t) - I_t\Big] + 2\chi\gamma\Pi_t(\Pi_t - 1) - \lambda x_t. \tag{3.32}$$

The second derivative is obtained by differentiating (3.32). The price-related component gives

$$\frac{\partial}{\partial x_t}\left[H_t(P_t - I_t) - \gamma H_t x_t P_t\right] = H_t P_t(\gamma^2 x_t - 2\gamma),$$

while the premium-stabilization component gives

$$\frac{\partial}{\partial x_t}\left[2\chi\gamma\Pi_t(\Pi_t - 1)\right] = -2\chi\gamma^2\Pi_t(2\Pi_t - 1).$$

Therefore

$$F'(x_t) = J''(x_t) = H_t P_t(\gamma^2 x_t - 2\gamma) - 2\chi\gamma^2\Pi_t(2\Pi_t - 1) - \lambda. \tag{3.33}$$

Under (3.30), one has $\gamma^2 x_t - 2\gamma \le 0$ and $2\Pi_t - 1 > 0$ for all $x_t \in [0, \bar{x}]$. Since $H_t > 0$, $P_t > 0$, $\chi > 0$, $\gamma > 0$, and $\lambda > 0$, each term in (3.33) is non-positive and the final term is strictly negative. Hence

$$J''(x_t) = F'(x_t) < 0 \qquad \text{for all } x_t \in [0, \bar{x}]. \tag{3.34}$$

Thus J is strictly concave on the admissible set.

Since J is continuous on the compact interval $[0, \bar{x}]$, the Weierstrass theorem implies the existence of at least one maximizer. Strict concavity implies that this maximizer is unique.

It remains to characterize the interior case. At $x_t = 0$, the first derivative is

$$\begin{aligned} F(0) &= H_t\Big(P_t(0) - I_t\Big) + 2\chi\gamma\Pi_t(0)\Big(\Pi_t(0) - 1\Big) \\ &= H_t I_t \phi A_t^{\beta} + 2\chi\gamma(1 + \phi A_t^{\beta})\phi A_t^{\beta} > 0, \end{aligned} \tag{3.35}$$

where the strict inequality follows from $A_t > 0$ and the positivity of the parameters. Since F is strictly decreasing by (3.34), if $F(\bar{x}) < 0$, the intermediate value theorem implies that there exists a unique $x_t^* \in (0, \bar{x})$ such that $F(x_t^*) = 0$. By strict concavity, this root is the unique maximizer of J on $[0, \bar{x}]$. If $F(\bar{x}) \geq 0$, strict concavity implies that the unique maximizer is attained at the upper boundary $x_t^* = \bar{x}$. This completes the proof. □

The theorem establishes that the protocol's static issuance decision is well posed. The admissibility condition $\bar{x} \leq 2/\gamma$ restricts attention to issuance rates that do not make the price-compression channel excessively nonlinear. Economically, it excludes unrealistically aggressive issuance policies for which the marginal effect of emissions on market valuation may become dominated by extreme compression effects. The condition $\Pi_t(x_t) > 1/2$ requires the token not to trade too far below intrinsic value over the relevant issuance range. This is also economically natural: the protocol optimization problem is meaningful when the market price remains sufficiently connected to the reserve-backed value of the token.

The first-order condition (3.32) also clarifies the forces determining optimal issuance. The term

$$H_t\Big(P_t - I_t\Big)$$

encourages issuance when the market price exceeds intrinsic value. The term

$$-\gamma H_t x_t P_t$$

reflects the fact that issuance itself compresses the price through the factor $e^{-\gamma x_t}$. The premium-stabilization term

$$2\chi\gamma\Pi_t(\Pi_t - 1)$$

accounts for the effect of issuance on the distance between market price and intrinsic value, while the term

$$-\lambda x_t$$

represents the marginal cost of issuance. The optimal issuance rate is therefore determined by the balance between the value created by selling tokens above intrinsic

value and the marginal costs generated by price compression, premium stabilization, and aggressive emissions. From an economic perspective, Theorem 3.2 guarantees that the protocol does not face multiple competing issuance regimes. Under the maintained assumptions, the interaction between reserve accumulation, premium compression, and issuance costs generates a single optimal issuance policy. This eliminates potential coordination ambiguities and provides a well-defined benchmark for protocol design.

Although Theorem 3.2 guarantees that the optimal issuance rate is uniquely determined, the exact first-order condition does not yield a useful elementary closed-form solution. Substituting $P_t(x_t) = I_t B_t e^{-\gamma x_t}$ and $\Pi_t(x_t) = B_t e^{-\gamma x_t}$ into (3.32) produces a nonlinear equation involving both $x_t e^{-\gamma x_t}$ and $e^{-2\gamma x_t}$, together with the linear cost term $-\lambda x_t$. This mixture of polynomial and exponential terms makes the exact solution analytically opaque and economically difficult to interpret.

For this reason, the next section develops an approximate closed-form solution based on a first-order expansion of the issuance-compression factor. The approximation preserves the central economic mechanisms of the model while converting the first-order condition into a tractable quadratic equation for the optimal issuance rate.

3.5 Approximate Closed-Form Solution

This section derives an approximate closed-form solution by linearizing the issuance-compression term. The objective is not to replace the exact characterization, but to obtain an analytically tractable formula that clarifies how the optimal issuance rate depends on market activity, intrinsic value, premium stabilization, and issuance costs.

Recall the pricing specification P_t (3.8) and the definition of B_t and H_t (3.28).

To obtain economically transparent comparative statics, we approximate the nonlinear first-order condition around the relevant operating region of the model. This approach follows the general philosophy of analytical approximation methods commonly employed in nonlinear dynamic economic models (see Dixit [7]).

$$e^{-\gamma x_t} \simeq 1 - \gamma x_t, \tag{3.36}$$

which is accurate for small or moderate values of γx_t. This approximation corresponds economically to a regime in which issuance affects market valuation, but the compression effect is not so extreme as to dominate the whole pricing relation.

Under (3.36),

$$P_t \simeq I_t B_t(1-\gamma x_t), \qquad \Pi_t \simeq B_t(1-\gamma x_t). \tag{3.37}$$

Substituting (3.37) into the exact first-order condition gives

$$\begin{aligned} 0 \simeq\ & H_t I_t \left[B_t(1-\gamma x_t) - 1\right] - H_t \gamma x_t I_t B_t (1-\gamma x_t) \\ & + 2\chi\gamma B_t(1-\gamma x_t)\left[B_t(1-\gamma x_t) - 1\right] - \lambda x_t. \end{aligned}$$

Expanding and collecting powers of x_t yields the quadratic equation

$$a_t x_t^2 + b_t x_t + c_t = 0, \tag{3.38}$$

where

$$a_t = \gamma^2 B_t H_t I_t + 2\chi\gamma^3 B_t^2, \tag{3.39}$$

$$b_t = -2\gamma H_t I_t B_t - 2\chi\gamma^2 B_t(2B_t - 1) - \lambda, \tag{3.40}$$

$$c_t = (H_t I_t + 2\chi\gamma B_t)(B_t - 1). \tag{3.41}$$

These coefficients have clear signs under the maintained assumptions. Since $\gamma, B_t, H_t, I_t, \chi > 0$, one has $a_t > 0$. Moreover, because each term in (3.40) is negative, $b_t < 0$. Finally, $c_t \geq 0$ because $B_t \geq 1$.

The discriminant associated with (3.38) is

$$\Delta_t = b_t^2 - 4a_t c_t. \tag{3.42}$$

Whenever $\Delta_t \geq 0$, the approximate first-order condition admits the two roots

$$x_{t,\pm} = \frac{-b_t \pm \sqrt{\Delta_t}}{2a_t}. \tag{3.43}$$

Since $a_t > 0$ and $b_t < 0$, the quadratic function in (3.38) is convex and initially nonnegative when $c_t \geq 0$. The economically relevant optimizer is the smaller root,

$$\boxed{\widehat{x}_t^* = \frac{-b_t - \sqrt{\Delta_t}}{2a_t}} \tag{3.44}$$

provided that this value lies in the admissible region $\widehat{x}_t^* \geq 0$. The larger root corresponds to the point at which the approximate first-order condition becomes positive again and is therefore not associated with the local maximum generated by the initial value-creating region of the payoff.

The approximate solution highlights a simple economic principle. The protocol issues more aggressively when reserve backing and activity-related premia create additional value from token sales, and more conservatively when issuance has a stronger compressive effect on market valuation or when issuance costs become more significant.

Remark 3.3 (Non-negativity of the Approximate Solution). The approximate optimal issuance rate is given by (3.44) with Δ_t defined as in (3.42)

Under the assumptions of Theorem 3.2, we have $a_t > 0$, $c_t \geq 0$, and $b_t < 0$. Since $c_t \geq 0$,

$$\Delta_t = b_t^2 - 4a_t c_t \leq b_t^2,$$

which implies

$$\sqrt{\Delta_t} \leq |b_t| = -b_t.$$

Therefore,

$$-b_t - \sqrt{\Delta_t} \geq 0.$$

Because $2a_t > 0$, it follows that

$$\hat{x}_t^* = \frac{-b_t - \sqrt{\Delta_t}}{2a_t} \geq 0.$$

Hence, the approximate closed-form solution is economically admissible and satisfies the non-negativity constraint on the issuance rate.

Remark 3.4 (Reality of the Approximate Solution). The approximate closed-form solution is meaningful only in the region where the discriminant (3.42) is non-negative. The sign conditions $a_t > 0$, $b_t < 0$, and $c_t \geq 0$ do not, by themselves, imply $\Delta_t \geq 0$. When $c_t = 0$, which corresponds to $B_t = 1$ and therefore $A_t = 0$, one has

$$\Delta_t = b_t^2 > 0.$$

For $c_t > 0$, a sufficient condition for the approximate solution to be real is

$$b_t^2 \geq 4a_t c_t.$$

Equivalently,

$$c_t \leq \frac{b_t^2}{4a_t}.$$

Hence, the approximation is economically and mathematically admissible in the region

$$0 \leq c_t \leq \frac{b_t^2}{4a_t}.$$

In terms of the model, this condition requires the constant component of the quadratic approximation to be sufficiently small relative to the curvature and slope terms. Economically, the activity-driven premium and issuance-compression effects must be such that the local quadratic approximation of the first-order condition admits real roots. Outside this region, the quadratic approximation does not generate a real closed-form issuance rate, and the exact first-order condition should be solved numerically.

The closed-form approximation in (3.44) makes explicit the main economic forces behind optimal issuance. A larger intrinsic value I_t or a stronger activity premium B_t increases the potential benefit of issuing tokens above reserve-backed value. At the same time, higher issuance sensitivity γ, stronger premium stabilization χ, and larger issuance cost λ reduce the optimal issuance intensity by increasing the marginal cost of aggressive emissions.

The approximation also clarifies why the exact solution is not analytically convenient. In the full model, the exponential term $e^{-\gamma x_t}$ appears both in the price level and in the premium ratio, generating a nonlinear first-order condition that does not reduce to a polynomial expression. The Taylor approximation transforms this implicit condition into a quadratic equation, allowing the optimizer to be represented explicitly while preserving the key economic trade-off between reserve growth, market premium, dilution pressure, and issuance costs.

It is important to emphasize that (3.44) is an approximate analytical formula, not an exact replacement for the optimizer characterized in the previous Section. Its role is interpretive: it provides a tractable expression that highlights how the protocol's optimal issuance decision reacts to the economic state of the system and to the structural parameters governing market valuation.

Remark 3.5 (Degenerate Linear Case). The quadratic approximation also admits a useful limiting interpretation when the curvature coefficient becomes negligible. If $a_t \to 0$, the approximate first-order condition

$$a_t x_t^2 + b_t x_t + c_t = 0$$

degenerates into the linear equation

$$b_t x_t + c_t = 0.$$

Since $b_t < 0$, the corresponding approximate issuance rate is

$$\hat{x}_t^* \approx -\frac{c_t}{b_t}.$$

This expression provides a simple economic benchmark. When the curvature of the approximate first-order condition is small, the optimal issuance rate is governed mainly by the ratio between the constant incentive to issue, represented by c_t, and the marginal dampening effect, represented by $-b_t$. Hence, stronger value-creation incentives increase the optimal issuance rate, while stronger marginal compression or cost effects reduce it.

This chapter proposed a reserve-based framework for token valuation with endogenous issuance. By linking reserve dynamics, intrinsic value, and issuance decisions, the model provides a tractable benchmark for the analysis of reserve-backed digital assets. Extensions incorporating dynamic issuance policies and time-varying market conditions are left for future research.

3.6 Next Step: Dynamic Token Pricing

The analysis developed in this chapter provides a complete static characterization of the issuance problem faced by a reserve-backed token protocol. The objective was deliberately focused: to isolate the trade-off between reserve-backed value creation and dilution, and to characterize the optimal issuance rate at a given state of the system. This static formulation is appropriate for the purpose of the chapter because it identifies the local economic forces that determine whether issuing additional tokens increases or reduces value.

Within this framework, the protocol observes the current level of reserve backing, intrinsic value, market activity, and issuance costs, and chooses the issuance rate that optimally balances growth incentives against valuation compression. The resulting model therefore offers a self-contained benchmark for understanding reserve-based token pricing with endogenous issuance. Its contribution is not to simulate the full time path of a token economy, but to clarify the economic mechanism that governs the issuance decision at each state.

A natural extension of this benchmark is a dynamic token-pricing model in which reserves, activity, token supply, intrinsic value, and market premia evolve jointly over time. In such a setting, the issuance rule would affect not only current token

value, but also the future trajectory of the system through feedback effects. Higher issuance today could expand the reserve base and support future growth, while also influencing dilution pressure, market expectations, and future protocol incentives.

This dynamic perspective would provide a useful framework for future numerical simulations. Once explicit laws of motion are imposed on reserve growth, activity, and token demand, one could compare alternative issuance rules in terms of price stability, reserve accumulation, dilution pressure, and long-run token value. The dynamic extension should therefore be viewed as a natural continuation of the present analysis rather than as a missing component of it: the static model developed here supplies the analytical foundation on which a richer intertemporal version can be built.

Conclusions

A token, by itself, has no intrinsic economic significance.

Its value depends on the rules that govern access, participation, competition, and issuance. Change those rules, and the same token may become highly valuable, economically irrelevant, or even value-destructive. This simple observation lies at the core of the analysis developed throughout this thesis.

Rather than treating token valuation as a purely pricing problem, the results suggest a broader perspective. The economic properties of a token emerge from the interaction between valuation mechanisms and institutional design. Horizon structure determines how future growth is capitalized. Competitive pressure determines whether tokenized participation remains attractive relative to alternative opportunities. Issuance policy determines how value is distributed between existing and future participants.

The three chapters translate this perspective into concrete analytical results. Chapter 1 established that finite access duration does not eliminate the adoption-enhancing mechanism identified by Cong, Li, and Wang, but that the temporal structure of expiration is decisive: deterministic maturity depresses token values and adoption incentives substantially, while stochastic termination largely preserves the qualitative structure of the perpetual benchmark through probabilistic survival. Chapter 2 showed that token competitiveness against an external asset is governed by a unique activity threshold, with comparative statics revealing strong nonlinearities in the short-maturity region and confirming that it is not finite duration per se that threatens token viability, but the way in which duration enters the valuation problem. Chapter 3, motivated by direct engagement with the reserve-backed token architecture developed by RIV Technologies, established conditions under which the optimal issuance rate of a protocol exists and is unique, and derived an approximate closed-form solution characterizing how the optimal rate responds to reserve backing, platform activity, market premium, and issuance costs. The chapter contributes a formal optimization framework for a class of token design problems that existing literature addresses only qualitatively.

The numerical implementations of Chapters 1 and 2 were carried out using proxy parameters, in the absence of real platform data. The calibration choices were guided by internal consistency and comparability with the CWL benchmark [6] rather than by empirical estimation, and the quantitative results should therefore be interpreted as illustrative rather than predictive. Replacing these proxies with data from actual tokenized platforms would be a direct and meaningful empirical extension of the theoretical framework.

A natural direction for future work concerns the dynamic extension of the issuance model developed in Chapter 3. The static formulation adopted in the chapter was a deliberate methodological choice, made in order to isolate the core trade-offs in their most transparent and analytically tractable form. A fully dynamic treatment, in which the protocol optimizes intertemporally over the joint evolution of reserve value, token supply, and market price, represents a well-defined and economically meaningful extension that goes beyond the scope of the present thesis and is left for future research. The practical implementation of the model — including the calibration of reserve dynamics, activity parameters, and issuance cost structure — is left to RIV Technologies, should they choose to adopt the framework as a quantitative basis for their treasury and issuance policy. The theoretical structure is in place; its empirical application depends on proprietary data that lie outside the scope of this thesis.

The challenge faced by reserve-backed token ecosystems is not merely to determine the value of a token at a given moment, but to understand how the token system should be managed over time in order to preserve and create value. Answering this question rigorously requires exactly the kind of quantitative framework developed here: one that connects reserve dynamics, market valuation, and supply policy within a coherent formal model.

If there is one conclusion that emerges clearly from the analysis, it is that token valuation and token design are ultimately inseparable problems. A token is not simply an asset to be priced. It is the product of an institutional architecture, and its value reflects the quality of that architecture as much as the fundamentals of the platform it represents.

Appendix A

Numerical Implementations: Chapter 1

Numerical Implementation of the CWL Benchmark

This appendix documents in detail the numerical implementation used to recover the perpetual benchmark of Cong, Li and Wang (CWL) and to generate the benchmark curves employed in Chapter 1. The implementation is entirely carried out in Python and is organized into six modules: `cwl_params.py`, `type_distribution.py`, `adoption_block.py`, `flow_block.py`, `ode_solver.py`, and `plot_cwl.py`. Their roles are complementary: the first file fixes the economic and numerical parameters; the second implements the truncated-normal type distribution analytically; the third solves the adoption block; the fourth computes the flow term entering the pricing equation; the fifth solves the equilibrium boundary-value problem; and the sixth produces the benchmark plots and diagnostic outputs. The source code is not included as part of the thesis submission. For this reason, the appendix reports the numerical architecture, equations, parameter values, grid choices, boundary conditions, solver routines, and diagnostic checks in sufficient detail to allow an independent reconstruction of the computational pipeline. The purpose of the implementation is to recover the tokenized perpetual economy described by the CWL pricing equation,

$$rP(A) = \mu_A A P_A(A) + \frac{1}{2}\sigma_A^2 A^2 P_{AA}(A) + (1-\alpha)\left(\frac{N(A)S(A)A}{P(A)M}\right)^{\alpha} P(A), \quad (3.45)$$

subject to the lower and upper boundary conditions

$$P(0) = 0, \qquad P(\bar{A}) = \frac{S\bar{A}}{M}\left(\frac{1-\alpha}{r-\mu_A}\right)^{1/\alpha}, \quad (3.46)$$

where $\bar{A}$ is a sufficiently large upper activity bound and S denotes the full-adoption value of the aggregate transaction-needs term. The central numerical difficulty is that the flow term in (3.45) depends on the endogenous objects $N(A)$ and $S(A)$,

which themselves depend on equilibrium pricing through the carry cost. The code therefore does not solve the ODE in isolation, but repeatedly solves a coupled local equilibrium block along the activity grid.

Calibration and Numerical Controls All model and solver parameters are stored in the dataclass `CWLParams`. The benchmark calibration used in the code is

$$\begin{aligned} &r = 0.05, \quad \sigma_A = 2.0, \quad \eta\rho = 1.0, \quad \hat{\mu}_A = 2.02, \quad \mu_A^Q = 0.02, \\ &\alpha = 0.3, \quad \phi = 1, \quad M = 10^{10}, \quad \mathrm{Var}(u) = 50. \end{aligned} \tag{3.47}$$

The code stores the risk-neutral drift directly as `mu_A_q = 0.02`; this is the quantity entering the pricing equation. The physical-measure drift is stored as `mu_A_hat = 2.02`, consistently with the paper calibration in which the change of measure implies a large upward adjustment under the physical measure.

The computational activity domain is specified in logarithmic coordinates:

$$\log A \in [\log A_{\min}, \log A_{\max}] = [-50, 20], \tag{3.48}$$

with `n_A = 2000` grid points for the final diagnostic and plotting grid. The boundary-value problem itself is initialized on a coarser grid of `bvp_n_init = 100` points and then refined internally by `scipy.integrate.solve_bvp`. The continuation path for the volatility parameter is set to

$$\sigma_A \in \{0.35,\ 1.0,\ 2.0\}, \tag{3.49}$$

so that the nonlinear problem is solved first in an easier regime and then gradually deformed to the final target volatility.

In addition, the code introduces numerical safeguards, namely a positivity floor $eps = 10^{-12}$, a cutoff tolerance $cutoff_tol = 10^{-12}$, a minimum tail probability $min_tail_prob = 10^{-12}$, and trimming rules for residual diagnostics.

Analytic Construction of the Type Distribution The module `type_distribution.py` implements the cross-sectional distribution of user types. Rather than approximating the type distribution numerically, the code uses an *analytic truncated normal* on the compact support

$$[\underline{u}, \bar{u}] = [-6\theta, 6\theta], \qquad \theta = \sqrt{\mathrm{Var}(u)} = \sqrt{50}. \tag{3.50}$$

Hence,

$$\underline{u} = -6\sqrt{50}, \qquad \bar{u} = 6\sqrt{50}. \tag{3.51}$$

This choice is consistent with the finite-support formulation of CWL and avoids interpolation noise when computing tail masses and truncated exponential moments.

Let $\Phi(\cdot)$ denote the standard normal distribution function. The normalization constant associated with truncation is

$$Z = \Phi\left(\frac{\bar{u}}{\theta}\right) - \Phi\left(\frac{\underline{u}}{\theta}\right). \tag{3.52}$$

For any cutoff u^* in the support, the code computes the tail mass analytically as

$$N(u^*) = \int_{u^*}^{\bar{u}} dG(u) = \frac{\Phi\left(\frac{\bar{u}}{\theta}\right) - \Phi\left(\frac{u^*}{\theta}\right)}{Z}, \tag{3.53}$$

and the aggregate transaction-needs term as

$$S(u^*) = \int_{u^*}^{\bar{u}} e^u \, dG(u) = \frac{e^{\theta^2/2}}{Z}\left[\Phi\left(\frac{\bar{u} - \theta^2}{\theta}\right) - \Phi\left(\frac{u^* - \theta^2}{\theta}\right)\right]. \tag{3.54}$$

The full-adoption value S used in the Gordon boundary condition is simply obtained by evaluating (3.54) at the lower bound of the support, which is numerically equivalent to setting the cutoff to $-\infty$ in the code interface.

The type density itself is also evaluated analytically:

$$g(u) = \frac{1}{\theta\sqrt{2\pi}Z}\exp\left(-\frac{u^2}{2\theta^2}\right), \qquad u \in (\underline{u}, \bar{u}), \tag{3.55}$$

with density set to zero outside the support. This density is used when the cutoff is solved by Newton iterations, because the derivative of the tail mass satisfies $N_u(u^*) = -g(u^*)$.

Adoption Block and Cutoff Computation The module `adoption_block.py` solves the static adoption block for a given activity level A and carry cost. In the CWL benchmark, the participation cutoff is characterized by

$$u^*(A) = -\log N(A) + \log\left(\frac{\phi}{A^\alpha}\right) - \frac{1-\alpha}{\alpha}\log\left(\frac{1-\alpha}{r - \mu^P(A)}\right). \tag{3.56}$$

Equivalently, for a given carry cost

$$c(A) := r - \mu^P(A) > 0, \tag{3.57}$$

the code solves the residual equation

$$F(u; A, c) = u + \log N(u) - \log\left(\frac{\phi}{A^{\alpha}}\right) + \frac{1-\alpha}{\alpha}\log\left(\frac{1-\alpha}{c}\right) = 0, \tag{3.58}$$

where $N(u)$ is the tail mass in (3.53). The algorithm proceeds in two stages.

First, the code attempts a Newton step initialized at the previous cutoff value, which is efficient because the solution varies smoothly across nearby grid points. The derivative used in the Newton update is

$$F_u(u; A, c) = 1 - \frac{g(u)}{N(u)}. \tag{3.59}$$

If the Newton step fails to converge, exits the support, or produces a nearly singular derivative, the code switches to a robust bracketing method using `scipy.optimize.brentq`.

Second, if no interior root exists in the support, the code classifies the regime explicitly. If the residual is positive at both endpoints, the implied root lies to the left of the support and the model is assigned to the full-adoption regime. If the residual is negative at both endpoints, the implied root lies to the right of the support and the model is assigned to the zero-adoption regime. This regime classification is encoded in the function `classify_cutoff_regime`.

Once the cutoff is determined, the adoption block returns the triple

$$\Big(u^*(A),\, N(A),\, S(A)\Big), \tag{3.60}$$

with

$$N(A) = N\Big(u^*(A)\Big), \qquad S(A) = S\Big(u^*(A)\Big). \tag{3.61}$$

This block is used twice in the full pipeline: first inside the tokenized equilibrium solver, where the carry cost is endogenous, and second in the tokenless comparison branch, where the code simply fixes `mu_p = 0`, so that the carry cost reduces to r.

Flow Term entering the Pricing Equation The file `flow_block.py` translates the output of the adoption block into the flow term entering the pricing equation. Given an activity level A, a candidate expected appreciation μ^P, and a candidate token price P, the code computes the ratio

$$\text{ratio}(A) = \frac{N(A)S(A)A}{P(A)M}, \tag{3.62}$$

and then evaluates the dividend-like flow contribution as

$$D(A) = (1 - \alpha)\,\mathrm{ratio}(A)^{\alpha} P(A). \tag{3.63}$$

This is exactly the CWL flow term. Numerically, the code enforces positivity of P through a small floor, so as to avoid meaningless divisions in the very-low-price region near the left boundary.

Although `flow_block.py` is a separate module, the same logic is also embedded directly inside the ODE solver through the function `_solve_local_state`, because the boundary-value problem requires repeated local evaluation of the carry cost, the cutoff, and the flow term at each collocation node.

Transformation of the Pricing Equation into Log-Activity Coordinates The core numerical solver is implemented in `ode_solver.py`. The pricing equation is not solved directly in the activity variable A, but in logarithmic coordinates

$$x = \log A. \tag{3.64}$$

This transformation is numerically convenient because the equilibrium must be resolved over several orders of magnitude in activity, from extremely low values to a large upper-bound region where the Gordon asymptotic condition becomes accurate.

Write the price as a function of x, still denoted by $P(x) = P(e^x)$. Then

$$P_A = \frac{P_x}{A}, \qquad P_{AA} = \frac{P_{xx} - P_x}{A^2}. \tag{3.65}$$

Substituting (3.65) into (3.45) yields the ODE solved numerically by the code:

$$\frac{1}{2}\sigma_A^2 P_{xx}(x) + \left(\mu_A^Q - \frac{1}{2}\sigma_A^2\right) P_x(x) - rP(x) + D(x) = 0, \tag{3.66}$$

where $D(x)$ is the endogenous flow term computed from the local state block.

In the code, the ODE is written as a first-order system for the vector

$$y(x) = \begin{pmatrix} P(x) \\ P_x(x) \end{pmatrix}, \tag{3.67}$$

so that

$$y_1' = y_2, \qquad y_2' = P_{xx}(x) = \frac{2}{\sigma_A^2}\left[rP(x) - \left(\mu_A^Q - \frac{1}{2}\sigma_A^2\right) P_x(x) - D(x)\right]. \tag{3.68}$$

This is exactly the expression implemented in the internal function `_ode_rhs`.

Local Equilibrium State at each Collocation Node A key feature of the implementation is that the solver does not pre-compute adoption externally and then solve the pricing ODE once. Instead, at each evaluation of the ODE right-hand side it recomputes the local equilibrium objects implied by the current candidate price profile. This is done by the function `_solve_local_state`.

For each pair (A, P), the code solves the tokenized local equilibrium from the static market-clearing condition. Starting from

$$P(A) = \frac{N(A)S(A)A}{M}\left(\frac{1-\alpha}{r-\mu^P(A)}\right)^{1/\alpha}, \tag{3.69}$$

it follows that the carry cost can be written as

$$r - \mu^P(A) = (1-\alpha)\left(\frac{N(A)S(A)A}{P(A)M}\right)^{\alpha}. \tag{3.70}$$

The code therefore uses $P(A)$ to infer the carry cost directly from (3.70), rather than introducing $\mu^P(A)$ as an additional unknown in the boundary-value problem. Given that carry, it solves for the cutoff, computes $N(A)$ and $S(A)$, and then reconstructs the flow term.

Operationally, the local solver returns the quintuple

$$\Big(\text{carry}(A),\, u^*(A),\, N(A),\, S(A),\, D(A)\Big). \tag{3.71}$$

This local block is evaluated sequentially over the grid, using the previous cutoff as the initial guess for the next one, which stabilizes and accelerates convergence.

Boundary Conditions and Gordon Boundary at High Activity The left boundary condition is imposed exactly as

$$P(x_{\min}) = 0, \tag{3.72}$$

which corresponds to $P(0) = 0$ in the limit of vanishing activity.

The right boundary condition is constructed from the Gordon full-adoption formula. The code first computes the full-adoption aggregate transaction-needs term,

$$S = \int_{\underline{u}}^{\bar{u}} e^u \, dG(u), \tag{3.73}$$

and then sets

$$P(\bar{A}) = \frac{S\bar{A}}{M}\left(\frac{1-\alpha}{r-\mu_A^Q}\right)^{1/\alpha}, \qquad \bar{A} = e^{\log A_{\max}}. \tag{3.74}$$

This is implemented in the function `upper_boundary_value`. The logic is identical to CWL: once activity is sufficiently high and adoption is effectively complete, the economy behaves like its asymptotic full-adoption counterpart and the price becomes linear in A.

Initial Guess and Continuation Strategy Because the ODE is nonlinear through the flow term, the solver is initialized carefully. The initial guess is not an arbitrary polynomial, but the Gordon linear-in-A profile rescaled to the current right boundary value:

$$P_0(x) = P(\bar{A})\frac{A}{\bar{A}}. \tag{3.75}$$

Its derivative with respect to x is then obtained numerically through a centered finite difference. This seed is implemented by the function `initial_guess`.

The code then solves the nonlinear boundary-value problem by continuation in volatility. Specifically, it first solves the ODE for a low volatility $\sigma_A = 0.35$, then uses that solution as the initial guess for $\sigma_A = 1.0$, and finally uses the latter as the initial guess for the target benchmark value $\sigma_A = 2.0$. This continuation strategy is implemented in `_continuation_solve` and significantly improves robustness relative to attempting the full benchmark solve from a cold start.

The actual collocation problem is solved with *scipy.integrate.solve_bvp*, using the tolerance $bvp_tol = 10^{-4}$ and allowing refinement up to $bvp_max_nodes = 8000$. At each step, the boundary conditions are reimposed explicitly through the helper function *_bc*.

Recovery of Derivatives, Expected Return, and Volatility Once the boundary-value problem has converged, the solution is evaluated on the final plotting grid of 2000 evenly spaced points in log-activity. The solver returns $P(x)$ and $P_x(x)$. The code then reconstructs the second derivative from the ODE itself,

$$P_{xx}(x) = \frac{2}{\sigma_A^2}\left[rP(x) - \left(\mu_A^Q - \frac{1}{2}\sigma_A^2\right)P_x(x) - D(x)\right], \tag{3.76}$$

thereby avoiding a noisy second finite-difference differentiation.

Using the derivative transformation (3.65), the code also recovers the derivatives

with respect to the original activity variable:

$$P_A(A) = \frac{P_x(x)}{A}, \qquad P_{AA}(A) = \frac{P_{xx}(x) - P_x(x)}{A^2}. \tag{3.77}$$

The expected token return under the risk-neutral measure is then reconstructed from the carry cost as

$$\mu^P(A) = r - \mathrm{carry}(A), \tag{3.78}$$

while the token return volatility is recovered from Itô's lemma as

$$\sigma^P(A) = \sigma_A \frac{P_x(x)}{P(x)}. \tag{3.79}$$

These quantities are stored in the `SolverResult` object together with the cutoff, user base, aggregate transaction-needs term, flow term, and residual diagnostics.

Residual Diagnostics and Numerical Validation To verify the numerical quality of the computed solution, the code evaluates the residual of the log-activity ODE,

$$\mathcal{R}(x) = \frac{1}{2}\sigma_A^2 P_{xx}(x) + \left(\mu_A^Q - \frac{1}{2}\sigma_A^2\right) P_x(x) - rP(x) + D(x). \tag{3.80}$$

It also computes a scaled residual,

$$\mathcal{R}_{\mathrm{rel}}(x) = \frac{|\mathcal{R}(x)|}{1 + |rP(x)| + |D(x)| + |P_x(x)| + |P_{xx}(x)|}, \tag{3.81}$$

which makes it possible to assess the residual in a scale-free way across the entire activity domain.

The left and right boundary residuals are also imposed explicitly by setting

$$\mathcal{R}(x_{\min}) = P(x_{\min}), \qquad \mathcal{R}(x_{\max}) = P(x_{\max}) - P_{\mathrm{Gordon}}(x_{\max}). \tag{3.82}$$

For summary diagnostics, the code trims a small number of points near the boundaries and reports the median and maximum relative residual on the interior grid. These diagnostics are printed by `print_diagnostics` and are also saved in the compressed output file.

Construction of the Comparison Branchs and Figures The file `plot_cwl.py` is responsible for generating the benchmark figures. After solving the tokenized CWL equilibrium, the script constructs two additional comparison objects.

First, it computes the tokenless equilibrium branch by evaluating the adoption

block at the same activity grid with `mu_p = 0`. This corresponds exactly to setting the expected token appreciation equal to zero, so that the carry cost collapses to the risk-free rate r.

Second, it computes the planner threshold

$$A^{PL} = \frac{\phi}{\alpha \left(\frac{1-\alpha}{r}\right)^{\frac{1-\alpha}{\alpha}} S}, \tag{3.83}$$

where S again denotes the full-adoption aggregate transaction-needs term. This threshold is used to draw the vertical line in the benchmark adoption plot.

The plotting script then produces the following outputs:

1. **User base vs. log-activity** ("Figure 2 style"): plots the tokenized user base $N(A)$, the tokenless user base, and the planner threshold line.

2. **User-base volatility** ("Figure 3A style"): computes

$$\sigma_N(A) = \sigma_A \left| \frac{dN}{d \log A} \right|, \tag{3.84}$$

   using a numerical gradient on the log-activity grid, and plots it as a function of the adoption level.

3. **Expected token return** ("Figure 3B style"): plots $\mu^P(A)$ together with the asymptotic benchmark line μ_A^Q.

4. **Log price over adoption stages** ("Figure 4 style"): plots $\log P(A)$ as a function of the adoption level after sorting the curve monotonically in $N(A)$ and removing duplicate values.

5. **Extra diagnostics**: plots the cutoff $u^*(A)$, the aggregate transaction-needs term $S(A)$, the volatility of user-base growth $\sigma_N(A)/N(A)$, and the relative residual.

In addition, the script stores the solution arrays in a compressed `.npz` archive, including the activity grid, log-activity grid, token price, expected token return, user base, cutoff, transaction-needs term, and residual measures. This archive is then used as the benchmark input for subsequent comparative analysis.

Execution Order of the Full CWL Pipeline The entire simulation of the CWL benchmark follows the sequence below:

1. instantiate the benchmark parameter object `CWLParams`;

2. build the truncated-normal type distribution analytically;
3. solve the nonlinear pricing ODE by continuation in volatility using `solve_cwl_ode`;
4. evaluate the solution on the final log-activity grid;
5. recover $u^*(A)$, $N(A)$, $S(A)$, $\mu^P(A)$, and $\sigma^P(A)$ pointwise;
6. compute residual and relative residual diagnostics;
7. solve the tokenless comparison branch on the same activity grid;
8. compute the planner threshold;
9. save the numerical arrays to a compressed output file;
10. generate the benchmark figures and diagnostic plots.

This sequence is executed by running `plot_cwl.py`, which acts as the top-level script. The resulting benchmark is therefore not a stylized or manually constructed reference object, but the numerical solution of the perpetual CWL equilibrium recovered from the same computational architecture later used for the finite-maturity deterministic and stochastic extensions.

Summary The perpetual CWL benchmark is simulated as a nonlinear boundary-value problem in log-activity space, closed pointwise by an endogenous adoption block and solved through continuation in volatility. The implementation is deliberately structured to avoid interpolation artefacts in the type distribution, to ensure consistent recovery of the local equilibrium objects entering the pricing equation, and to provide explicit residual diagnostics for validation. As a result, the CWL benchmark used throughout Chapter 1 is numerically transparent, internally consistent, and documented in sufficient detail to be independently reconstructed.

Numerical Implementation of the Deterministic Finite-Maturity Model

This appendix documents in detail the numerical implementation used to obtain the deterministic finite-maturity results reported in Chapter 1. The goal of the implementation is to solve the equilibrium pricing problem

$$\begin{aligned} rP(A,m) =& \mu_A A P_A(A,m) + \frac{1}{2}\sigma_A^2 A^2 P_{AA}(A,m) - P_m(A,m) + \\ &+ (1-\alpha)\left(\frac{N(A,m)\,S(A,m)\,A}{P(A,m)\,M}\right)^{\alpha} P(A,m), \end{aligned}$$

subject to the boundary conditions

$$P(0, m) = 0, \qquad P(A, 0) = 0,$$

and to recover, from the resulting price surface, the endogenous objects used in the plots, namely $N(A, m)$, $u(A, m)$, $S(A, m)$, $\mu^P(A, m)$, $\sigma^P(A, m)$, the pricing residual, and the selected maturity slices used in the figures.

The implementation is modular. Some files are specific to the deterministic extension, while others are reused from the CWL benchmark without modification. This was done deliberately. Wherever the deterministic model does not alter the economic structure of the perpetual benchmark, the same code was retained in order to preserve comparability and to ensure that any difference in the numerical output is generated by the finite-maturity mechanism itself, rather than by an unnecessary change in auxiliary routines.

File Structure and Reused Components The deterministic implementation relies on the following files:

- `det_fm_solver.py`: main solver for the deterministic finite-maturity PDE;
- `det_fm_params.py`: deterministic-model plotting and verification parameters;
- `plot_det_fm.py`: export of representative maturity slices and internal verification plots;
- `plot_det_fm1.py`: explicit verification of the very-high-maturity limit against the CWL benchmark;
- `type_distribution.py`: analytic truncated-normal distribution of user types;
- `adoption_block.py`: CWL adoption block, retained for consistency of the participation mapping;
- `cwl_params.py`: benchmark economic calibration used as common baseline;
- `ode_solver.py`: CWL ODE solver, reused only for benchmark comparison and high-maturity verification.

Three files were reused from the benchmark structure without conceptual modification:

1. `type_distribution.py`. The cross-sectional distribution of user types is unchanged relative to CWL. The deterministic extension modifies the pricing equation through the maturity state variable m, but it does not modify the distributional assumption on heterogeneity. For this reason, the same analytic truncated-normal representation was retained. This choice guarantees that the mapping from the cutoff u to the user base N and to the exponential moment S is identical to the benchmark case.

2. `adoption_block.py`. The deterministic model does not alter the underlying participation logic: users still enter through a cutoff rule obtained from the optimality condition and the type distribution. What changes is the equilibrium price $P(A, m)$, and hence the effective carry cost entering the equilibrium mapping. Since the functional structure of the adoption block remains the same, the benchmark implementation was reused. This avoids introducing artificial differences in the adoption routine when comparing the deterministic model to CWL.

3. `cwl_params.py` and `ode_solver.py`. These files are not used to solve the deterministic PDE itself. They are retained for two specific purposes:

1. to keep the deterministic calibration anchored to the same economic benchmark;
2. to compare the deterministic solution with the perpetual CWL benchmark, especially in the verification stage where the very-high-maturity drift is checked against the CWL expected return.

No change was necessary because the benchmark comparison must be performed against the original CWL numerical environment, not against a reimplemented or approximated version.

Common Economic Structure Retained from CWL The deterministic finite-maturity model preserves the benchmark user-side structure. Let u denote the individual type, N the mass of participating users, and S the aggregate transaction-needs shifter

$$S(u) = \int_u^{\bar{U}} e^z \, dG(z).$$

For a given candidate price $P(A, m)$, the local state is recovered by solving the same nonlinear cutoff equation used in the tokenized benchmark. In the code, this is encoded through the residual

$$\log\left(\frac{\alpha}{\phi}\right) + u + \alpha \log N + \alpha \log A + (1-\alpha)\log(MP) - (1-\alpha)\log S = 0.$$

Once the cutoff is found, the code computes

$$N(A,m) = 1 - G(u(A,m)), \qquad S(A,m) = S(u(A,m)),$$

and reconstructs the implied carry cost via

$$r - \mu^P(A,m) = (1-\alpha)\left(\frac{N(A,m)S(A,m)A}{P(A,m)M}\right)^{\alpha}.$$

This is precisely the benchmark tokenized mapping. It was intentionally preserved, because the deterministic extension is meant to isolate the effect of the maturity term $-P_m$, not to change the user-side equilibrium logic.

Deterministic-Specific Solver: `det_fm_solver.py` The core numerical work is carried out in `det_fm_solver.py`. In contrast to the CWL benchmark, where the equilibrium pricing problem reduces to an ordinary differential equation in A, the deterministic finite-maturity case requires solving a two-dimensional problem in (A, m).

State Grids

The solver constructs two grids:

Activity grid. The activity state is represented on a uniform grid in $\log A$:

$$\log A_j \in [\log A_{\min}, \log A_{\max}], \qquad A_j = e^{\log A_j}.$$

The use of a log-activity grid is the same as in CWL and is retained for numerical stability and comparability of plots across models.

Maturity grid. The maturity variable is discretized over $[0, T]$ using a nonuniform transformation

$$m_k = T\, s_k^p, \qquad s_k \in [0,1],$$

where $p =$ `m_grid_power`. This implies that the maturity grid is denser near $m = 0$, where the solution is most curved because the terminal condition $P(A, 0) = 0$ creates a strong short-horizon effect. The nonuniformity is therefore not cosmetic: it is introduced to stabilize the numerical treatment of the maturity-decay region.

Upper Boundary at Large Activity

At the right boundary in activity, the solver does not impose an arbitrary numerical cap. Instead, it uses the closed-form full-adoption expression implied by the deterministic finite-horizon model:

$$P(A_{\max}, m) = \frac{S_{\text{full}} A_{\max}}{M} \left[\frac{1-\alpha}{r-\mu_A} \left(1 - e^{-\alpha(r-\mu_A)m} \right) \right]^{1/\alpha},$$

where S_{full} is computed as the full-support exponential moment of the type distribution. This boundary condition is implemented in the function `upper_boundary_value(·)`.

This is the natural deterministic analogue of the Gordon-type boundary used in CWL. Keeping the right boundary model-consistent is essential, because the expected return and price-level comparisons in the figures become unreliable if the large-A region is contaminated by an ad hoc numerical closure.

Terminal Condition at Zero Maturity

The exact terminal boundary condition is imposed at $m = 0$:

$$P(A, 0) = 0.$$

This is encoded by setting the first maturity slice identically equal to zero. Economically, this reflects the fact that an access right with no remaining life has no continuation value.

Bootstrap for the First Positive Maturity Slice

The slice immediately above $m = 0$ is not obtained by a crude linear guess. Instead, the solver constructs a dedicated short-maturity approximation through the routine `_small_m_profile`. The logic is:

1. start from a zero or boundary-compatible profile;
2. compute the local flow term from the current tentative price;
3. update the price using a short-horizon approximation

$$P(A, m_1) \approx m_1 \times \text{flow}(A, m_1);$$

4. damp the update and iterate until convergence.

This bootstrap is necessary because the maturity derivative is most singular near the terminal boundary. A naive initialization would propagate instability into all subsequent maturity slices.

Semi-Implicit Marching Scheme in Maturity

After initializing the terminal and near-terminal slices, the PDE is solved by a semi-implicit finite-difference march forward in maturity.

At each maturity level m_k, the solver:

1. takes the current price slice $P(\cdot, m_k)$;
2. recovers the local equilibrium objects $u_k(A), N_k(A), S_k(A)$, carry, and flow;
3. constructs the linear spatial operator in $\log A$;
4. solves a tridiagonal linear system for the next slice $P(\cdot, m_{k+1})$.

In matrix form, the interior nodes satisfy

$$(I - \Delta m\, L)\, P^{k+1} = P^k + \Delta m\, \text{flow}^k,$$

where L is the discretized activity-space operator incorporating drift, diffusion, and discounting. The tridiagonal banded system is assembled in `_build_banded_matrix(·)` and solved with `solve_banded` from `scipy.linalg`.

The method is semi-implicit because the spatial pricing operator is treated implicitly, while the nonlinear flow term is evaluated from the current slice. This is a deliberate compromise: it is far more stable than a fully explicit scheme, but remains materially lighter than a fully nonlinear implicit solve at every maturity step.

Recovery of Derivatives and Economic Objects Once the full price surface $P(A, m)$ has been computed, the solver reconstructs the objects used in the figures and in the numerical diagnostics.

Spatial derivatives

For each maturity slice, the code computes first and second derivatives with respect to $\log A$:

$$P_x = \frac{\partial P}{\partial \log A}, \qquad P_{xx} = \frac{\partial^2 P}{\partial (\log A)^2},$$

using local finite-difference formulas implemented in `_spatial_derivatives(·)`. These are then converted into derivatives with respect to A through

$$P_A = \frac{P_x}{A}, \qquad P_{AA} = \frac{P_{xx} - P_x}{A^2}.$$

This conversion is stored in `convert_log_derivatives_to_A(·)`.

Maturity derivative

The derivative P_m is reconstructed ex post using finite differences across the maturity grid. Forward, central, and backward formulas are used at the lower boundary, interior nodes, and upper boundary respectively. This quantity is essential because the deterministic pricing equation contains the explicit maturity-decay term $-P_m$.

Expected return and volatility

The code computes two versions of the expected return:

Static return. Using the equilibrium carry relation,

$$\mu^P_{\text{static}}(A, m) = r - \text{carry}(A, m).$$

Itô-implied return. Using the computed derivatives,

$$\mu^P_{\text{Ito}}(A, m) = \frac{(\mu_A - \frac{1}{2}\sigma_A^2)P_x + \frac{1}{2}\sigma_A^2 P_{xx} - P_m}{P}.$$

The reporting version $\mu^P(A, m)$ is not taken blindly from one formula. The code compares the two expressions and only retains values in regions where:

- the price is not too close to zero;
- the point is not too close to the artificial activity boundary;
- the static and Itô reconstructions are numerically consistent within specified tolerances.

This is implemented through the boolean mask `mu_valid_mask`. The purpose is to avoid displaying numerically fragile expected returns in regions where μ^P is theoretically defined but computationally unreliable because P is extremely small or the slice is dominated by boundary effects.

Token return volatility is reconstructed as

$$\sigma^P(A,m) = \sigma_A \frac{P_x}{P}.$$

Residual diagnostics

To verify numerical consistency, the solver evaluates the PDE residual

$$\mathcal{R}(A,m) = (\mu_A - \tfrac{1}{2}\sigma_A^2)P_x + \tfrac{1}{2}\sigma_A^2 P_{xx} - P_m - rP + \text{flow},$$

and the relative residual obtained by dividing by a scale factor involving the main terms of the equation. These diagnostics are stored in `residual` and `rel_residual`. The summary routine `print_diagnostics(·)` reports, among other things, the median and maximum relative residual in the core of the grid.

Role of `det_fm_params.py` The file `det_fm_params.py` controls the deterministic plotting and verification configuration. Among the uploaded parameters, the following are explicit:

- the maturity values used in the main plots,

$$m \in \{1.0,\ 3.0,\ 5.0,\ 10.0\};$$

- the three representative maturity slices exported as standalone `NPZ` files,

$$m \in \{1.0,\ 5.0,\ 10.0\};$$

- a very-high-maturity diagnostic value

$$m = 20{,}000,$$

  used exclusively for verification against the CWL limit;

- the vertical zoom window used in the verification drift plots.

These values are not arbitrary. The exported maturities are chosen to represent, respectively, a short but still interpretable horizon, an intermediate horizon, and a long finite horizon. The very-high-maturity level is used to verify numerically that the deterministic model converges, in the long-horizon limit, to the perpetual CWL benchmark.

Generation of Deterministic Outputs: `plot_det_fm.py` The file `plot_det_fm.py` serves two purposes.

1. Export of selected maturity slices

After loading the deterministic solution stored in `det_fm_solution.npz`, the script extracts the maturity slices closest to the target values $m = 1, 5, 10$. For each target maturity, it saves a standalone `NPZ` file containing the one-dimensional objects required for plotting:

$$\log A, \quad P, \quad \mu^P, \quad N, \quad \sigma^P, \quad \text{etc.}$$

The file name records both the target maturity and the actual grid maturity selected. This is important because the maturity grid is nonuniform, so the exact grid point need not coincide perfectly with the target nominal value.

2. Internal verification against CWL

The same script also constructs a verification plot comparing the deterministic expected return at very high maturity with the benchmark CWL expected return. To do this, it solves the CWL benchmark through `ode_solver.py`, interpolates the CWL drift onto the deterministic $\log A$-grid, and plots:

1. the two drift curves directly;
2. their difference.

This provides a transparent numerical check that the deterministic model converges to CWL when the maturity horizon becomes arbitrarily large.

Explicit Benchmark Verification: `plot_det_fm1.py` The file `plot_det_fm1.py` performs the same high-maturity verification in a more path-explicit way. The reason for this duplication is technical rather than conceptual.

Both the deterministic and CWL folders contain modules with similar names (`cwl_params.py`, `type_distribution.py`, `ode_solver.py`, etc.). To avoid namespace collisions, `plot_det_fm1.py` loads the CWL benchmark modules explicitly from a user-specified directory using `importlib.util.spec_from_file_location`. This ensures that the benchmark comparison is carried out against the exact original CWL codebase, rather than against whatever similarly named module happens to be visible in the local Python path.

This script therefore has a purely verification purpose: it makes the deterministic-to-CWL comparison robust to file-structure ambiguity.

How the Main Deterministic Figures Are Obtained The deterministic figures reported in Chapter 1 are obtained by selecting representative maturity slices from the computed price surface and plotting, for each selected m, the relevant object against either $\log A$ or the endogenous adoption level $N(A, m)$.

Concretely, the procedure is:

1. solve the two-dimensional deterministic PDE once on the full (A, m) grid;
2. compute the associated equilibrium objects on the entire grid;
3. extract the selected maturity slices $m \approx 1$, $m \approx 5$, $m \approx 10$ (and, where useful, $m \approx 3$);
4. compare those slices either:
    - across maturities within the deterministic model, or
    - against the CWL benchmark solved independently by the one-dimensional ODE solver.

The deterministic figures therefore do not come from separate independent simulations for each maturity. They come from a single coherent numerical solution of the entire finite-maturity pricing surface, from which one-dimensional slices are subsequently extracted. This matters because it preserves internal consistency across maturities and avoids re-solving the equilibrium independently at each horizon.

Why Reusing Benchmark Components Was Numerically Appropriate The reuse of benchmark files was not merely convenient. It was methodologically necessary.

- Reusing `type_distribution.py` guarantees that heterogeneity is identical in CWL and deterministic finite maturity.
- Reusing the CWL adoption logic guarantees that any difference in $N(A)$ is generated by finite maturity through prices, not by a rewritten participation routine.
- Reusing the benchmark calibration and solver stack for comparison guarantees that the deterministic-to-CWL gap is measured against the actual perpetual benchmark, not against a numerically altered proxy.

In other words, the deterministic implementation changes only what the theory changes: the pricing problem gains the maturity state variable and the decay term $-P_m$. Everything else is intentionally kept as close as possible to the benchmark environment.

Summary The deterministic finite-maturity implementation is built as a controlled extension of the CWL numerical architecture. The benchmark user heterogeneity, participation mapping, and calibration are preserved. The genuinely new element is the two-dimensional pricing problem in (A, m), solved by a semi-implicit finite-difference scheme with:

- exact terminal boundary at $m = 0$,
- model-consistent large-A boundary,
- nonlinear bootstrap for the first positive maturity slice,
- ex post recovery of P_A, P_{AA}, P_m, μ^P, σ^P, and residuals,
- explicit verification of the high-maturity convergence toward CWL.

This design ensures that the deterministic figures in Chapter 1 are generated in a way that is numerically stable, theoretically aligned with the model, and directly comparable to the perpetual benchmark.

Numerical Implementation of the Stochastic-Termination Model

This appendix documents in detail the numerical implementation used to obtain the stochastic-termination results reported in Chapter 1. The objective is to solve the equilibrium pricing problem

$$rP(A) = \mu_A A P_A(A) + \frac{1}{2}\sigma_A^2 A^2 P_{AA}(A) - \delta P(A) + (1-\alpha)\left(\frac{N(A)\,S(A)\,A}{P(A)\,M}\right)^{\alpha} P(A),$$

subject to the lower boundary condition

$$P(0) = 0,$$

and to recover, from the resulting solution, the endogenous objects used in the figures, namely $N(A)$, $u(A)$, $S(A)$, the effective carry, the expected token return $\mu^P(A)$, the

token return volatility $\sigma^P(A)$, user-base volatility, and the direct comparison objects used against the perpetual CWL benchmark.

The implementation is modular. Several files are specific to the stochastic extension, while others are reused from the benchmark environment without conceptual modification. This was done intentionally. The stochastic model changes the pricing equation by introducing the killing term $-\delta P$ and by modifying the user-side effective carry to $r+\delta-\mu^P$. It does not change the type distribution, the benchmark economic calibration, or the general logic that maps a carry cost into a participation cutoff. Reusing those parts of the CWL architecture therefore preserves comparability and avoids introducing numerical discrepancies unrelated to the stochastic termination mechanism itself.

File Structure and Reused Components The stochastic implementation relies on the following files:

- `stochastic_params.py`: economic and numerical parameters specific to the stochastic model;
- `stochastic_ode_solver.py`: one-dimensional boundary-value solver for the pricing ODE with killing;
- `stochastic_adoption_block.py`: stochastic version of the local adoption block, consistent with the carry $r + \delta - \mu^P$;
- `plot_stochastic.py`: generation of all stochastic outputs, benchmark comparisons, selected-δ files, and summary tables;
- `type_distribution.py`: analytic truncated-normal distribution of user types;
- `cwl_params.py`: baseline benchmark calibration;
- `flow_block.py`: CWL flow routine, reused only as conceptual benchmark and not as the main stochastic evaluator;
- `adoption_block.py`: benchmark tokenless adoption block, reused only to construct the tokenless comparison branch in the plotting stage.

The reused files serve precise purposes.

1. `type_distribution.py`. The stochastic extension leaves user heterogeneity unchanged. The benchmark truncated-normal representation of types was therefore reused exactly. This guarantees that the mapping from cutoff u to adoption N and to the exponential moment S remains identical across CWL, deterministic finite maturity, and stochastic termination.

2. `cwl_params.py`. The benchmark calibration is reused as the economic anchor of the entire exercise. The stochastic model introduces δ, but the remaining primitives are kept aligned with CWL so that any deviation in prices, adoption, or expected returns is attributable to stochastic termination rather than to a reparametrization of the baseline economy.

3. `adoption_block.py`. This file is not used to solve the stochastic tokenized equilibrium itself. It is reused in the plotting stage only to compute the *tokenless* benchmark branch under the same activity grid. This is useful when constructing planner or tokenless comparison objects. No modification was needed because the tokenless participation condition is exactly the CWL one, with no stochastic token appreciation term.

4. `flow_block.py`. This module encodes the CWL flow term. It is not the main computational engine of the stochastic model, because the stochastic solver reconstructs its own local flow internally from the modified carry $r + \delta - \mu^P$. It is nevertheless retained as part of the shared project architecture, since the stochastic pricing term remains the same object economically, up to the changed carry structure.

Economic Structure Specific to the Stochastic Model The stochastic extension differs from CWL in two places.

Pricing side. The asset-pricing equation includes the killing term $-\delta P(A)$, reflecting the expected loss of continuation value due to stochastic termination.

User-side carry. To keep valuation and user optimization internally consistent, the local effective carry is not $r - \mu^P$, but

$$r + \delta - \mu^P.$$

This is explicitly stated in the docstring of `stochastic_params.py` and is implemented throughout the solver and the stochastic adoption block.

Because of this adjustment, the stochastic model is not obtained by simply subtracting δP from the CWL equation while leaving the participation block unchanged. The user-side carry must be modified as well; otherwise, the numerical implementation would not correspond to the valuation logic developed in the theory section.

Parameters: `stochastic_params.py` The file `stochastic_params.py` defines the class `StochasticParams`. It contains both the economic calibration and the numerical controls.

Economic parameters

The benchmark values are

$$r = 0.05, \qquad \sigma_A = 2.0, \qquad \eta\rho = 1.0, \qquad \widehat{\mu}_A = 2.02, \qquad \mu_A^Q = 0.02,$$

together with

$$\alpha = 0.3, \qquad \phi = 1, \qquad M = 10^{10}, \qquad \mathrm{Var}(u) = 50.$$

The stochastic-specific parameter is the hazard rate

$$\delta,$$

whose baseline value in the parameter file is $\delta = 0.02$.

Activity grid and BVP controls

The solution domain is defined over

$$\log A \in [-50, 20],$$

with

$$n_A = 2000$$

grid points for diagnostics and plots. The boundary-value problem is first initialized on a coarser grid and then interpolated on the final dense activity grid.

The BVP continuation controls are explicitly listed:

- $bvp_n_init = 120$,
- $bvp_max_nodes = 10000$,

- $bvp_tol = 10^{-5}$,
- $continuation_deltas = (0.0, 0.005, 0.01, 0.02)$,
- $continuation_sigmas = (0.35, 1.0, 2.0)$.

This means that the stochastic solver does not jump directly to the full nonlinear problem. Instead, it gradually deforms the solution from an easier benchmark configuration toward the target one.

Type Distribution and Moments The file `type_distribution.py`, reused from the benchmark environment, provides an analytic truncated-normal distribution over the compact support

$$[\underline{U}, \overline{U}] = [-6\theta,\, 6\theta], \qquad \theta = \sqrt{\mathrm{Var}(u)}.$$

The file returns:

- the density $g(u)$,
- the tail mass $1 - G(u)$,
- the truncated exponential moment

$$S(u) = \int_u^{\overline{U}} e^z \, dG(z),$$

- the full-support exponential moment S_{full}.

These routines are reused without modification because the stochastic model does not alter the cross-sectional distribution of types. Any change here would have polluted the comparison with CWL and deterministic maturity.

Stochastic Adoption Block: `stochastic_adoption_block.py` The local participation block is implemented in `stochastic_adoption_block.py`. Its purpose is to recover, for each candidate (A, P), the local equilibrium objects

$$u(A), \qquad N(A), \qquad S(A),$$

consistent with the stochastic carry

$$\text{carry} = r + \delta - \mu^P.$$

Cutoff residual

The code solves the cutoff equation through the residual

$$u - \left[-\log N(u) + \log\left(\frac{\phi}{A^{\alpha}}\right) - \frac{1-\alpha}{\alpha}\log\left(\frac{1-\alpha}{\text{carry}}\right)\right] = 0,$$

where

$$N(u) = 1 - G(u).$$

Relative to the CWL adoption block, the structure is identical except for the fact that the carry entering the logarithm is the stochastic carry $r + \delta - \mu^P$.

Solution method

The code first attempts a Newton step initialized at the previous cutoff value. If Newton fails because the derivative is too small, leaves the support, or becomes numerically unstable, the code falls back to a Brent root finder on the full support. If the residual does not cross zero over the support, the block classifies the state as either full adoption or zero adoption.

This logic is intentionally conservative. Since the stochastic pricing routine repeatedly solves the local equilibrium block across the entire activity grid, cutoff failures would propagate immediately into distorted prices. The two-step Newton/Brent structure therefore acts as a safeguard rather than as an unnecessary complication.

Recovered objects

For a given carry, the block returns

$$u(A), \qquad N(A) = 1 - G(u(A)), \qquad S(A) = S(u(A)).$$

A wrapper then maps these objects from a given μ^P by setting

$$\text{carry} = r + \delta - \mu^P.$$

Core Solver: `stochastic_ode_solver.py` The main numerical problem is solved in `stochastic_ode_solver.py`. Unlike the deterministic finite-maturity model, the stochastic case remains one-dimensional: the state variable is only A, because stochastic termination enters the pricing equation through the killing term rather than through an explicit maturity state.

Activity grid

The solver works on a uniform grid in $\log A$:

$$x = \log A, \qquad A = e^x.$$

As in CWL, this transformation is used to stabilize the numerical treatment of a wide activity domain and to make plots directly comparable across models.

Upper boundary

At the right boundary, the solver uses a closed-form full-adoption expression implemented in `upper_boundary_value(·)`:

$$P(A_{\max}) = \frac{S_{\text{full}} A_{\max}}{M} \left(\frac{1-\alpha}{r + \delta - \mu_A^Q} \right)^{1/\alpha}.$$

This is the stochastic analogue of the CWL Gordon-type asymptotic boundary. It is not an arbitrary cap: it is the model-consistent full-adoption price under constant hazard δ.

Initial guess

The solver starts from a simple linear initial guess

$$P_0(x) = \text{right_bc} \cdot \frac{x - x_{\min}}{x_{\max} - x_{\min}},$$

with derivative obtained by finite differences. This is encoded in `initial_guess(·)`. The initialization is intentionally simple because the nonlinear continuation routine, not the initial guess, is what carries the main stabilization burden.

Local state reconstruction

For each candidate (A, P), the solver reconstructs the local equilibrium objects through the internal routine `_solve_local_state`. The logic is:

1. given A and a candidate price P, solve the local cutoff equation in u;
2. recover N and S;
3. compute the ratio
$$\frac{NSA}{PM};$$

4. recover the carry

$$\text{carry} = (1-\alpha)\left(\frac{NSA}{PM}\right)^{\alpha};$$

5. recover the local flow

$$\text{flow} = \text{carry} \cdot P;$$

6. recover the implied expected return

$$\mu^P = r + \delta - \text{carry}.$$

This is the central local equilibrium mapping of the stochastic solver. The code computes it at each node of the activity grid, using the previous cutoff as the next initial guess. This makes the grid sweep both faster and more stable.

ODE in log-activity space

In the variable $x = \log A$, the stochastic pricing equation is rewritten as a second-order ODE for $P(x)$. Denoting $P_x = \partial P/\partial x$ and $P_{xx} = \partial^2 P/\partial x^2$, the solver uses

$$P_{xx} = \frac{2}{\sigma_A^2}\left[(r+\delta)P - \left(\mu_A^Q - \frac{1}{2}\sigma_A^2\right)P_x - \text{flow}\right].$$

The right-hand side is assembled in `_ode_rhs(·)`.

This is a key difference from the deterministic implementation. There is no maturity derivative P_m. Instead, stochastic termination appears entirely through the modified discount term $(r+\delta)P$.

Boundary-value solver

The ODE is solved with `solve_bvp` from `scipy.integrate`. For each pair (δ, σ_A) in the continuation schedule, the solver imposes the boundary conditions

$$P(x_{\min}) = 0, \qquad P(x_{\max}) = \text{right_bc}.$$

The auxiliary function `_solve_one_bvp` handles one BVP stage, while `_continuation_solve` chains all stages.

Continuation in δ and σ_A

The solution is not obtained in one shot. Instead, the solver follows a continuation path:

- first through the sequence of hazard rates in `continuation_deltas`,
- and, for each δ, through the sequence of volatilities in `continuation_sigmas`.

This means that the final target problem is reached by gradually deforming the BVP from a smoother, easier case toward the full calibration.

This continuation structure plays the same conceptual role as in CWL: it prevents the BVP solver from failing when started directly on the most nonlinear or most volatile version of the problem.

Recovery of Derivatives and Reported Objects Once the final BVP solution has been obtained, the solver evaluates it on the dense $\log A$-grid and reconstructs all objects used in the figures.

Price derivatives

The solution object returns

$$P(x), \qquad P_x(x).$$

The second derivative P_{xx} is reconstructed from the stochastic ODE itself:

$$P_{xx} = \frac{2}{\sigma_A^2}\left[(r+\delta)P - \left(\mu_A^Q - \frac{1}{2}\sigma_A^2\right)P_x - \text{flow}\right].$$

These log-derivatives are then transformed into derivatives with respect to A via

$$P_A = \frac{P_x}{A}, \qquad P_{AA} = \frac{P_{xx} - P_x}{A^2}.$$

Expected return

The expected return used in the output is

$$\mu^P(A) = r + \delta - \text{carry}(A).$$

Since the local carry is already reconstructed from the pricing fixed point, this is the economically consistent drift corresponding to the stochastic model.

Token return volatility

The code computes

$$\sigma^P(A) = \sigma_A \frac{P_x}{P},$$

whenever P is sufficiently far from zero.

Residual diagnostics

The solver evaluates the pricing residual

$$\mathcal{R}(A) = \frac{1}{2}\sigma_A^2 P_{xx} + \left(\mu_A^Q - \frac{1}{2}\sigma_A^2\right) P_x - (r+\delta)P + \text{flow},$$

and the relative residual obtained by dividing by a scale term involving the absolute magnitudes of the main ODE components. These diagnostics are stored in `residual` and `rel_residual`, and printed by `print_diagnostics(·)`.

Selected Hazard Rates and Parameter Construction The plotting file works with a selected set of representative hazard rates:

$$\delta \in \{0.005,\ 0.020,\ 0.050\}.$$

This set is declared in `SELECTED_DELTAS` inside `plot_stochastic.py`.

For each selected δ, the function `params_for_delta` creates a dedicated `StochasticParams` instance. The economic parameters are held fixed, while the continuation path is adapted to the target hazard level through the sequence

$$(0,\ 0.25\delta,\ 0.5\delta,\ \delta),$$

up to numerical lower bounds. This avoids reusing one generic continuation schedule for all hazard levels and makes the BVP path smoother at each target δ.

Role of `plot_stochastic.py` The file `plot_stochastic.py` is the post-processing and figure-generation driver. It performs four tasks.

1. Solve the stochastic model for multiple hazard rates

The function `solve_for_deltas` loops over the selected hazard rates, builds the corresponding parameter object, constructs the type distribution, solves the stochastic BVP, and stores the full solver output.

2. Solve the CWL benchmark on the same architecture

The file then reconstructs the perpetual benchmark using the original CWL solver. This is done either by importing the sibling CWL folder, if available, or by falling back to local copies of `cwl_params.py` and `ode_solver.py`. This design ensures that

the stochastic-vs-CWL comparisons are carried out against the actual benchmark codebase rather than against a manual approximation.

3. Save selected outputs in `NPZ` format

For each hazard rate, the script saves a compressed `NPZ` file containing the main one-dimensional equilibrium objects:

$$A, \quad \log A, \quad P, \quad \mu^P, \quad \sigma^P, \quad u, \quad N, \quad S, \quad \text{flow}, \quad \text{carry}, \quad \text{residual}.$$

The file name records the hazard rate using the function `sanitize_delta`.

4. Generate all figures and summary comparisons

The script produces:

- one-model stochastic figures for each selected δ,
- direct baseline-vs-CWL comparisons for a chosen baseline stochastic run,
- multi-δ overlays against CWL,
- difference plots and price ratios,
- summary tables at specified activity checkpoints.

How User-Base Volatility Is Computed The figures involving user-base volatility do not solve an independent stochastic volatility problem. Instead, the script reconstructs the volatility of the user base from the slope of the adoption curve:

$$\sigma_N(A) = \left| \sigma_A \frac{dN}{d\log A} \right|.$$

Numerically, this is computed by finite differences on the dense $\log A$-grid via the function `compute_user_base_volatility`.

This choice is fully consistent with the one-factor structure of the model: adoption volatility is induced by the exposure of the endogenous user base $N(A)$ to the same aggregate activity shock that drives token prices.

How the Main Stochastic Figures Are Obtained The main stochastic figures in Chapter 1 are produced as follows:

1. solve the stochastic ODE independently for each selected δ;

2. solve the CWL benchmark on a comparable activity domain;

3. interpolate the CWL objects onto the stochastic $\log A$-grid whenever direct differences or ratios are needed;

4. plot, for each δ, the objects of interest either against $\log A$ or against the endogenous adoption stage $N(A)$.

The script includes several specific plotting routines:

- `plot_single_run_vs_cwl`: detailed comparison of one stochastic run against CWL;
- `plot_multi_delta_vs_cwl`: overlays all selected δ values against CWL;
- direct log-price, expected-return, adoption, volatility, and difference plots;
- summary panels combining $N^{stoch} - N^{CWL}$, $\mu^{P,stoch} - \mu^{P,CWL}$, P^{stoch}/P^{CWL}, and the effective carry.

The plots are therefore not hand-assembled from partial simulations. They are generated systematically from a set of full stochastic equilibrium solutions, each compared against the benchmark on a common grid.

Summary Tables and Checkpoints The same plotting file also computes summary statistics at selected activity checkpoints, such as $\log A = 0$ and $\log A = 10$. These are extracted by interpolation from the dense activity grid after the full equilibrium solution is available.

Likewise, adoption thresholds such as the values of $\log A$ required to reach $N = 0.50$ or $N = 0.90$ are obtained from the solved adoption curve and then recorded in the summary tables. Since the stochastic model remains one-dimensional in A, these thresholds are read directly from the monotone relation between $\log A$ and $N(A)$.

Why Reusing Benchmark Components Was Numerically Appropriate The reused benchmark files were not retained out of laziness. They were retained because changing them would have been methodologically wrong.

- Reusing `type_distribution.py` ensures that heterogeneity is identical across all model variants.
- Reusing `cwl_params.py` ensures that the stochastic model differs from CWL only through the hazard mechanism.

- Reusing `adoption_block.py` for the tokenless branch allows benchmark comparison objects to be generated under the exact original participation logic.
- Reusing the benchmark CWL solver inside the plotting file ensures that stochastic results are compared against the actual perpetual numerical solution rather than against a stylized proxy.

In short, the stochastic implementation changes only what the theory changes: the pricing equation gains the killing term $-\delta P$, and the user-side carry becomes $r + \delta - \mu^P$. All remaining structures are left unchanged whenever theory does not require an alteration.

Summary The stochastic-termination implementation is a controlled one-dimensional extension of the CWL benchmark. It preserves the benchmark heterogeneity, support, and calibration, while modifying the pricing block through:

- a hazard-adjusted discount term $(r + \delta)P$,
- a user-side effective carry $r + \delta - \mu^P$,
- a local equilibrium mapping consistent with stochastic survival,
- a BVP continuation routine over both δ and σ_A,
- systematic output generation and interpolation-based comparison with CWL.

This architecture ensures that the stochastic figures in Chapter 1 are generated in a way that is numerically stable, theoretically aligned with the model, and directly comparable to both the perpetual benchmark and the deterministic finite-maturity extension.

Appendix B

Numerical Implementations: Chapter 2

This appendix documents the numerical implementation used to obtain the results of Chapter 2. Unlike Chapter 1, the purpose here is not to solve a new equilibrium model from scratch. The equilibrium objects relevant for the tokenized economy have already been computed in the numerical routines of Chapter 1, for the perpetual benchmark, the deterministic finite-maturity specification, and the stochastic-termination specification. The aim of the Chapter 2 pipeline is therefore different: it takes those previously computed equilibrium solutions as inputs, places them in direct competition with an external asset valued under the same risk-neutral measure, identifies the regions in which the token dominates or is dominated, and extracts the threshold objects used throughout the chapter.

As in Appendix A, the purpose of this description is to make the numerical procedure reproducible at the methodological level. The code itself is not submitted with the thesis; therefore, the appendix emphasizes the data flow, equations, numerical checks, threshold rules, and plotting conventions needed to reconstruct the analysis independently.

The full implementation is organized into eight Python modules: `params_2.py`, `io_2.py`, `external_asset.py`, `dominance_functions.py`, `threshold_solver.py`, `comparative_statistics.py`, `plot_2.py`, and `test_chapter2_pipeline.py`. Their roles are complementary. The first file centralizes the numerical configuration and the list of input files. The second loads the numerical outputs generated in Chapter 1 and standardizes them into a common structure without altering their content. The third encodes the external asset under the risk-neutral measure. The fourth constructs the dominance object $D(A; X) = P(A) - X$. The fifth identifies the activity threshold at which token and external asset have equal value. The sixth extracts comparative-statics objects from those thresholds. The seventh produces the figures used in Chapter 2. The eighth performs end-to-end pipeline checks on the real imported datasets.

The guiding numerical principle is the same as in Appendix A: the code is not allowed to manipulate the imported economic objects in order to "make the results work." In particular, it does not smooth prices, does not re-solve the Chapter 1 models under slightly different conventions, does not fabricate thresholds where no numerical bracket exists on the imported domain, and does not force distinct model families onto a common representation before that step is economically justified. The Chapter 2 code is therefore deliberately modular and conservative: it performs comparisons, validations, and threshold extraction, while preserving complete fidelity to the solutions already computed in Chapter 1.

Economic structure implemented in Chapter 2 The object compared to the token is the external asset introduced in Section 2.1. Under the risk-neutral measure Q, its law of motion is

$$\frac{dX_t}{X_t} = r\,dt + \sigma_X\,dW_t, \tag{3.85}$$

so its time-t value satisfies the martingale pricing identity

$$V^X(X_t) = \mathbb{E}_t^Q\Big[e^{-r(s-t)}X_s\Big] = X_t, \tag{3.86}$$

for every $s \geq t$. The Chapter 2 comparison is therefore pointwise in the current state: once the token price P is known from the equilibrium solution inherited from Chapter 1, the relevant dominance object is simply

$$D(A; X) = P(A) - X \tag{3.87}$$

for the perpetual and stochastic cases, and

$$D(A, m; X) = P(A, m) - X \tag{3.88}$$

for a fixed deterministic maturity slice.

The associated activity threshold $A^*(X)$ is defined, whenever it is bracketed on the numerical grid, by the indifference condition

$$P\Big(A^*(X)\Big) = X. \tag{3.89}$$

For deterministic finite maturity, the threshold is computed separately for each selected maturity slice; for stochastic termination, it is computed separately for each selected hazard rate. Once these thresholds are available, the code can recover the regions in which the token dominates, the regions in which the external asset

dominates, and the comparative-statics relations discussed in Chapter 2.

This architecture is intentionally different from the one in Appendix A. Appendix A had to solve coupled nonlinear pricing systems. By contrast, the Chapter 2 implementation is a structured post-processing layer built on top of already solved equilibria. That is exactly the right design choice. Re-solving the Chapter 1 models inside the Chapter 2 files would have been redundant, less transparent, and potentially dangerous, because even tiny implementation differences could have contaminated the comparison between the theory of Chapter 1 and the asset-allocation interpretation of Chapter 2.

Central configuration: `params_2.py` The file `params_2.py` acts as the central configuration block of the entire Chapter 2 pipeline. It introduces three dataclasses:

- `Chapter2FilePaths`, which stores the base directory and the names of the imported solution files;
- `Chapter2NumericalParams`, which stores threshold-solver controls;
- `Chapter2XGrids`, which stores the external-asset grids used for threshold extraction and figure construction.

The default file configuration points to one perpetual benchmark file, three deterministic finite-maturity files, and three stochastic-termination files. This is economically appropriate. Chapter 2 does not need a dense maturity-by-hazard grid. It needs a small number of representative solved economies that are sufficiently distinct to reveal the economic mechanisms of interest. The three deterministic files correspond approximately to short, intermediate, and long remaining maturity, while the three stochastic files correspond to low, moderate, and high termination intensity. This matches the exposition in the chapter and avoids cluttering the comparison with redundant slices.

The external-asset grids are not chosen symmetrically across model families. The deterministic specification uses

$$X \in \{10^3, 10^4, 10^5, 10^6, 10^7, 10^8\}, \tag{3.90}$$

while the stochastic specification uses

$$X \in \{10^8, 10^9, 10^{10}, 10^{11}, 10^{12}\}. \tag{3.91}$$

This asymmetry is deliberate, not accidental. The deterministic and stochastic token prices live on markedly different quantitative scales. Reusing one identical X-grid

for both families would either waste most of the deterministic grid on values that are trivially too large, or compress the stochastic comparison into values that are too small to be informative. The selected grids therefore reflect the magnitudes at which thresholds are actually informative in each family.

The same file also introduces two important numerical safeguards. First, the flag `require_monotone_threshold_solver=True` forces the threshold routine to check that the dominance function is non-decreasing on the imported grid before extracting a crossing. This is economically consistent with the theoretical monotonicity results of Chapter 2. Second, the tolerance `monotone_tol`=10^{-12} is explicitly documented as a machine-noise safeguard only. It does not alter the data and it is intentionally tiny, so that it can absorb floating-point noise without concealing a genuine economic violation.

Finally, `params_2.py` provides explicit validation routines for paths, scalar grids, and numerical controls. This is methodologically useful because the Chapter 2 pipeline is built from imported files rather than from a single monolithic solver. In such a setting, transparent configuration validation is not cosmetic. It is the first line of defense against silent mismatches in file selection, threshold ranges, or numerical conventions.

Importing Chapter 1 outputs without altering them: `io_2.py` The file `io_2.py` is the I/O and standardization layer. Its purpose is not to solve any model but to load the numerical outputs already generated in Chapter 1 and rewrite them into a common dictionary structure that can be consumed uniformly by the Chapter 2 routines.

This standardization is indispensable because the perpetual, deterministic, and stochastic files are not stored under identical field names. For example, the perpetual benchmark uses variables such as `A_grid`, `P`, `mu_p_used`, `N_star`, and `u_star`, while the deterministic slice files store objects such as `P_slice`, `mu_p_slice`, `N_star_slice`, and the selected maturity identifiers `target_m` and `selected_m`. The loader converts these heterogeneous raw archives into a standardized package containing at least

$$\{A, \log A, P, \mu^P, N, u, S, \text{meta}\}, \tag{3.92}$$

plus optional fields such as residuals, carry, volatility, or validity masks.

The key methodological point is that this standardization does not transform the economics. The code is explicit on this issue. It does not fill missing values, does not clip non-finite objects into finite ones, does not smooth noisy series, and does not replace one pricing convention with another. It only relabels and validates.

Several validation choices in `io_2.py` deserve explanation.

Validation of the deterministic expected return In the deterministic maturity files, the imported array for expected return may contain non-finite entries outside the numerically valid region, while the source archive also provides the boolean mask `mu_valid_mask_slice`. The loader therefore validates μ^P only on the model-declared valid region. This is the correct choice. Requiring global finiteness would incorrectly treat a known limitation of the source representation as if it were an economic error. Conversely, replacing invalid entries ex post would create artificial information. The loader does neither. It simply respects the validity region declared by the model that generated the file.

Validation of token return volatility When present, the token-volatility object σ^P is validated only on the region where the imported token price is materially positive. The reason is structural: in the code base inherited from Chapter 1, σ^P is constructed from a ratio proportional to P_A/P. Near the left boundary, where P is zero or numerically indistinguishable from zero, this quantity may be undefined even though the underlying price solution is perfectly correct. The loader therefore constructs a positive-price validation mask based on a relative tolerance and checks finiteness only there. Again, this is not a transformation of the data. It is a mathematically appropriate validation domain.

Grid compatibility is reported, not forced The file also contains a routine `check_same_activity_grid` to verify whether different datasets share the same A-grid and $\log A$-grid. However, exact equality is not imposed at the I/O stage. This is an important design decision. Different archives can legitimately be defined on grids of different lengths, and it would be conceptually wrong for the loader to interpolate one dataset onto another merely to make the arrays look comparable. Grid harmonization, when actually needed, belongs to the later analytical or plotting stage and must be justified case by case. The loader therefore reports compatibility without manufacturing it.

Why `selected_m` is used rather than `target_m` For the deterministic family, the metadata stored by the standardized package include both the requested maturity and the actual selected maturity. The threshold and comparative-statics routines use the actual value, denoted `actual_m` in the standardized dictionary. This is the correct object to report in the thesis because the numerical slices are computed on a pre-existing maturity grid. What matters empirically is the maturity actually

associated with the imported solution, not the ideal target value originally requested by the user.

External asset block: `external_asset.py` The file `external_asset.py` implements the external asset introduced in Section 2.1. It stores the parameters (r, σ_X, ρ) in the dataclass `ExternalAssetParams`, validates their admissibility, and makes the risk-neutral valuation identity explicit through the function `external_asset_value(X_t)`.

This may look almost too simple, which is exactly why it is useful to isolate it in its own module. The whole logic of Chapter 2 rests on the fact that under the risk-neutral measure the relevant present value of the external asset is just its current level X_t. Writing that as an explicit function prevents conceptual drift later in the pipeline. The code thereby enforces the correct comparison object: the Chapter 2 exercise is not a portfolio-optimization problem, not a simulation-based market-timing problem, and not a multi-period rebalancing problem. It is a pointwise state-by-state comparison between an equilibrium token value and an outside asset valued under the same measure.

The same module also includes two auxiliary pieces of functionality: exact one-step risk-neutral simulation of the geometric Brownian motion and a construction of correlated Brownian increments satisfying

$$\operatorname{corr}(dZ_t, dW_t) = \rho. \tag{3.93}$$

These routines are not needed for the threshold analysis itself, but they are still justified. They make the external-asset block self-contained, allow diagnostic checks of the martingale property, and preserve consistency with the theoretical environment of Section 2.1, where the platform shock and the external-asset shock may be correlated. Keeping these tools separate from the dominance routines avoids mixing valuation logic with optional simulation diagnostics.

Dominance mapping: `dominance_functions.py` The file `dominance_functions.py` contains the simplest conceptual step in the pipeline and, for that reason, one of the most important. Given a token-price array P and a scalar or vector of external-asset values X, it constructs the dominance array

$$D = P - X. \tag{3.94}$$

It also provides classification utilities:

- `classify(D)`, which returns $+1$ when the token dominates, -1 when the

external asset dominates, and 0 at indifference;

- `token_mask(D)`, which identifies the weak token-dominance region $D \geq 0$;
- `external_mask(D)`, which identifies the strict external-dominance region $D < 0$.

This separation into a dedicated module may seem excessive, but it is good numerical hygiene. Threshold solvers, comparative statics, figures, and tests all rely on the same primitive object D. Centralizing its definition ensures that the meaning of dominance is identical throughout the pipeline. Without such a module, it would be too easy for one plotting function to use $D = P - X$ while another silently used $D = (P - X)/X$, or for one threshold routine to interpret equality as $D = 0$ and another to work with a ratio. The file prevents this kind of inconsistency.

Threshold extraction on the imported activity grid: `threshold_solver.py`
The central numerical task of Chapter 2 is to extract the threshold at which the token and the external asset are equally attractive. This is handled by `threshold_solver.py`. The input is an imported dataset containing at least (A, P) and a scalar X. The routine computes

$$D(A; X) = P(A) - X, \tag{3.95}$$

and then searches for the first transition from negative dominance to positive dominance along the strictly increasing activity grid.

The routine `find_threshold_on_grid` works as follows.

1. It validates that the activity grid is one-dimensional, finite, non-empty, and strictly increasing.
2. If monotonicity is required, it checks that D is non-decreasing up to the tolerance `monotone_tol`.
3. If an exact grid point satisfies $|D| \leq$ `monotone_tol`, that point is returned as an `exact_grid_zero` threshold.
4. Otherwise, the solver identifies the first sign bracket with $D_i < 0$ and $D_{i+1} > 0$ and linearly interpolates inside that interval.

The interpolation formula is simply

$$A^* = A_i - D_i \frac{A_{i+1} - A_i}{D_{i+1} - D_i}. \tag{3.96}$$

This is intentionally modest. Higher-order interpolation would give the illusion of extra information not present in the imported data, especially in regions where the price curve is steep. Linear interpolation inside a genuine sign bracket is the correct conservative choice.

The same file also provides wrappers that attach model metadata, such as the actual maturity or the hazard rate, to the extracted threshold. This matters because Chapter 2 does not work with one generic threshold but with threshold families indexed by m or δ.

Why monotonicity is checked explicitly The monotonicity check is not merely defensive programming. It is tightly connected to the theory developed in Chapter 2. The threshold interpretation relies on the idea that, for a fixed external-asset value, higher platform activity should not make the token less valuable relative to the outside option. If the imported numerical dominance function were wildly non-monotone, then the notion of a unique threshold would become questionable on the discrete grid. Requiring monotonicity within a tiny tolerance therefore ensures that the numerical threshold has the same economic meaning as the theoretical object.

Why a threshold may be "not bracketed" numerically The solver raises a clear error when D never becomes positive on the supplied grid or is never negative on the supplied grid. This is extremely important for the interpretation of the Chapter 2 tables. A missing entry does not mean that the theoretical threshold fails to exist in the abstract model. It means only that, on the imported numerical domain and for the chosen external-asset level, the crossing is not bracketed by the available activity grid. This distinction is essential and is fully consistent with the discussion in the Chapter 2 results.

Comparative statics from threshold families: `comparative_statistics.py` Once thresholds can be computed for one dataset and one external-asset value, the next step is to study how they move across X, maturity, and termination intensity. This is the purpose of `comparative_statistics.py`.

The function `threshold_curve_vs_X` takes one dataset and an external-asset grid and returns the threshold curve

$$X \mapsto A^*(X), \tag{3.97}$$

together with the numerical method used at each point and the finite-difference

slopes

$$\frac{\Delta A^*}{\Delta X}. \tag{3.98}$$

The file then specializes this logic to the two model families.

For the deterministic case, `deterministic_thresholds_vs_m` orders the datasets by actual maturity and evaluates

$$m \mapsto A^*(m; X) \tag{3.99}$$

for a common external-asset value. The routine reports whether the threshold is non-increasing in m, which is the relevant economic monotonicity in this family: if access lasts longer, the token should require a lower activity level to dominate a given outside option.

For the stochastic case, `stochastic_thresholds_vs_delta` orders the datasets by hazard rate and evaluates

$$\delta \mapsto A^*(\delta; X) \tag{3.100}$$

for a common external-asset value. Here the economically meaningful check is the opposite monotonicity: if termination becomes more likely, the token should require higher activity to dominate the outside option, so the threshold should move upward in δ.

The file also reports finite-difference slopes in both dimensions. These slopes are not treated as structural derivatives in a continuous parametric sense. They are descriptive diagnostics on the discrete family of imported solutions. That distinction matters. The code never pretends that three solved maturities or three solved hazard rates identify a smooth structural derivative. It simply extracts the monotone pattern that Chapter 2 discusses economically.

Figure generation: `plot_2.py` The file `plot_2.py` is the figure-production driver for Chapter 2. Its header states its purpose explicitly: produce a compact set of thesis-grade figures that make the economics visually obvious. This design objective is reflected in the types of figures generated.

The main figures include:

- token price curves together with multiple horizontal external-asset levels for the deterministic and stochastic families;
- threshold curves $A^*(X)$ across maturities and across hazard rates;
- comparative-statics plots of A^* against m and against δ for a common successful external-asset value;

- heat maps of the relative advantage $\log(P/X)$, with the threshold frontier overlaid;
- dominance-region figures in the (A, X) plane.

Several implementation choices in this file are worth justifying.

Why threshold points are shown only where they exist The helper `_safe_threshold_points` attempts threshold extraction for each candidate X and simply skips values for which the solver raises an exception. This is exactly the correct visualization rule. The code refuses to draw a frontier where the numerical threshold is not bracketed. In other words, the figures inherit the same epistemic discipline as the tables: no threshold is plotted unless it has actually been identified from the imported data.

Why the heat map uses $\log(P/X)$ The relative-advantage heat map is based on

$$Z(A, X) = \log\left(\frac{P(A)}{X}\right). \tag{3.101}$$

This is preferable to plotting $P - X$ directly when one wants a two-dimensional picture across several orders of magnitude in both variables. The log ratio makes proportional dominance visually interpretable, and the zero contour still corresponds exactly to the threshold condition $P = X$. The code clips the display range of Z for readability, but this clipping is purely graphical. It affects only the color scale of the figure, not the threshold frontier itself.

Why representative datasets are used in the heat maps The heat maps are built for one representative deterministic slice and one representative stochastic slice, chosen as the middle maturity and the middle hazard rate, respectively. This is a sensible compromise between completeness and readability. A heat map already uses two axes, A and X, so stacking several maturities or several hazard rates inside the same graphic would obscure the message. The middle slice provides a representative picture, while the family-level comparative statics are conveyed separately through threshold curves and line plots.

Why no interpolation is performed across model families The plotting file is explicit on another methodological point: it does not interpolate across different model families. Deterministic and stochastic solutions are plotted as the objects they are, not as if they were observations on one common structural manifold. This is

correct because the two families arise from different pricing equations and different economic mechanisms. Visual comparison should therefore preserve their identity rather than blur it.

End-to-end diagnostics: `test_chapter2_pipeline.py` The file `test_chapter2_pipeline.py` is not an afterthought. It is the module that turns the Chapter 2 implementation from a collection of scripts into a verified pipeline. It performs an extended real-data report based on the actual imported solution files.

The testing sequence is organized in several layers.

1. It validates the global configuration declared in `params_2.py`.
2. It loads all perpetual, deterministic, and stochastic datasets through `io_2.py`.
3. It validates the standardized datasets and reports summary information, including price monotonicity diagnostics.
4. It checks whether the imported datasets live on the same activity grid.
5. It computes threshold-existence and threshold-bracketing diagnostics over the configured external-asset grids.
6. It performs full threshold-curve tests for each dataset.
7. It computes cross-model comparative statics on common successful X values.

A particularly useful feature of this file is that it does not just declare success or failure. It also reports boundary-sign diagnostics, scale diagnostics such as the ratio $X/P_{\max}$, local threshold brackets, and local residual information when residuals are available in the imported archives. This is a very good practice for a threshold-based chapter. When a threshold exists, one wants to know not only that the routine found it, but also where it was bracketed and how large the residuals were around that bracket. When a threshold does not exist numerically, one wants to know whether this happened because the external asset is simply too large relative to the token, because the crossing lies outside the activity grid, or because some monotonicity condition was not satisfied.

The file also distinguishes between raw price monotonicity and monotonicity up to numerical tolerance. This distinction is valuable because imported numerical solutions can contain tiny floating-point oscillations that are economically irrelevant. Reporting both diagnostics makes the testing procedure honest without being mechanically fragile.

Why the Chapter 2 pipeline reuses Chapter 1 outputs One might ask why Chapter 2 was implemented as a post-processing architecture rather than as a self-contained solver. The answer is that this is not merely convenient but methodologically superior.

First, Chapter 2 is conceptually downstream from Chapter 1. The point of the chapter is to reinterpret the already solved token economies in relation to an outside asset, not to change the underlying equilibrium model. Reusing the Chapter 1 outputs therefore preserves exact consistency between the theory, the numerical solution, and the economic comparison.

Second, importing precomputed solutions isolates the new contribution of Chapter 2. The chapter adds the external-asset block, the dominance mapping, the threshold extraction, and the comparative-statics layer. It does not pretend to add a new equilibrium pricing engine where none is theoretically needed.

Third, this design makes error diagnosis cleaner. If an issue appears in the Chapter 2 results, one can immediately distinguish between an upstream issue in the Chapter 1 solution files and a downstream issue in the dominance pipeline. A monolithic solver would blur this distinction and make the source of numerical errors much harder to identify.

Finally, this architecture preserves comparability. Since the token prices used in Chapter 2 are literally the same numerical solutions analyzed in Chapter 1, any difference in dominance regions or threshold behavior must come from the external-asset comparison itself, not from a hidden change in the token solver.

Summary The numerical implementation of Chapter 2 is a controlled and deliberately conservative extension of the Chapter 1 computational architecture. It does not solve new token-pricing equations. Instead, it imports the equilibrium solutions already obtained for the perpetual benchmark, the deterministic finite-maturity model, and the stochastic-termination model, and processes them through a new external-vs-token comparison layer.

Operationally, the pipeline consists of:

- a centralized configuration module for paths, threshold grids, and tolerance controls;
- a loader and validator that standardizes heterogeneous Chapter 1 outputs without altering them;
- a self-contained external-asset block implementing the risk-neutral valuation identity $V^X(X_t) = X_t$;

- a dominance module defining and classifying $D = P - X$;
- a threshold solver extracting $A^*(X)$ from genuine numerical sign brackets on the imported activity grid;
- a comparative-statics layer organizing threshold families across X, m, and δ;
- a figure-generation driver designed specifically for the economic interpretation of Chapter 2;
- an end-to-end testing routine that validates the entire real-data pipeline.

This implementation strategy is numerically stable, theoretically aligned with the structure of the chapter, and fully transparent about what is and is not being computed. Most importantly, it preserves fidelity to the Chapter 1 solutions while making the Chapter 2 economic message precise: the comparison with an external asset is performed on exactly the same token economies already solved in the thesis, and the threshold results are extracted without numerical embellishment or hidden reinterpretation.

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AI Usage Disclosure

This thesis was developed with the support of AI-based tools for specific tasks, primarily related to code development, debugging, and language refinement.

All modeling choices, theoretical results, numerical implementations, and economic interpretations are entirely the responsibility of the author.

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